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CREDIT CRUNCHES AND THE GREAT STAGFLATION

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ABSTRACT

We argue that severe credit crunches in the banking system contributed to the Great Stagflation of the 1970s. The credit crunches were due to Regulation Q, a banking law that capped deposit rates. Under Reg Q, Fed tightening triggered large deposit outflows that led banks to contract lending. The credit crunches line up closely with stagflation in the time series. To explain this, we add Reg Q to a standard model where firms use bank loans to finance working capital. When Reg Q binds and credit contracts, working capital becomes more expensive, leading firms to raise prices and shrink output. The model implies an augmented Phillips curve where monetary tightening reduces aggregate supply in addition to demand. The impact on supply is increasing in the severity of the credit crunches, firms' external finance dependence, and their working capital intensity. We test all three predictions in the cross section of manufacturing industries. In each case, we find that more exposed industries raise prices and cut output relative to others. Our results imply that under severe financial frictions monetary policy affects aggregate supply and not just demand.

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The Great Stagflation was a watershed in macroeconomics and monetary policy. Between 1965 and 1982, the U.S. economy experienced four severe downturns, each one accompanied by surging inflation. The combination of high inflation and low growth, the “stagflation,” ran counter to the prevailing Keynesian view, according to which inflation and output move hand in hand along a stable Phillips curve. Economists concluded that the Phillips curve is not stable but shifts up and down with inflation expectations. By not raising interest rates aggressively enough, the Federal Reserve had allowed inflation expectations to drift up and become self-fulfilling (Clarida, Gali and Gertler, 1999, 2000). This view underlies the textbook New Keynesian model (Woodford, 2003; Galí, 2007) and the practice of monetary policy today.

It remains difficult, however, to explain the “stag” part of the Great Stagflation under this view. Loose monetary policy can explain bursts of inflation but not why they coincided with recessions. The reason is that in the New Keynesian framework, monetary policy works through aggregate demand, pushing inflation and output in the same direction. Changes in inflation expectations can weaken this relationship but not invert it. To address this, the literature invokes a series of exogenous supply shocks, most prominently the oil shocks, that kept hitting the economy throughout the period. Because these shocks are seen as exogenous “bad luck,” there is nothing monetary policy could do about them. This narrative led to the view that central banks should take supply as given and focus on managing demand.

In this paper, we challenge this view. We argue that the negative supply shocks were in part endogenous, due to monetary tightening in the presence of a large financial friction. The friction was the banking law known as Regulation Q (“Reg Q”), which put a hard ceiling on bank deposit rates. The ceiling became binding whenever the Fed raised rates, leading to large outflows of deposits that triggered severe contractions in lending—credit crunches. Since firms rely on credit to pay for materials and labor up front, the credit crunches disrupted their ability to produce. This negative supply shock led firms to raise prices and cut output, contributing to the stagflation.

We formalize this mechanism by extending the standard New Keynesian model to incorporate deposits, banks, and Reg Q. Households derive liquidity services from deposits, and banks use deposits to make loans. Firms use loans to finance working capital, which enters production as a factor alongside labor. When Reg Q does not bind, banks set deposit rates freely, deposit funding is readily available, lending is unconstrained, and

the economy behaves exactly as in the New Keynesian benchmark.

But when the market interest rate crosses the Reg Q ceiling, the economy's behavior changes. A deposit spread (the difference between the market rate and the Reg Q ceiling) opens up, leading households to reduce their deposit holdings. Faced with a loss of funding, banks cut lending. A credit crunch ensues and the spread on loans rises. Firms that rely on loans to finance working capital see their production costs rise, leading them to raise prices and cut production. This is how monetary tightening can induce an adverse supply shock under Reg Q .

The key analytical result of the model is an augmented Phillips curve. Relative to the New Keynesian benchmark, it includes a term that is only active when Reg Q binds. This term is proportional to the deposit spread and captures how the credit crunches cause inflation by raising firms' marginal costs. It is larger when firms finance more of their working capital externally with loans, when their production process is more working-capital intensive, and when deposits are more sensitive to the deposit spread. These comparative statics give us direct testable predictions.

Since our model nests the New Keynesian benchmark, it incorporates the conventional demand-side effects of monetary policy and the role of inflation expectations. We provide further extensions where the credit crunches themselves have demand effects through household borrowing, as the literature has emphasized. We obtain conditions under which the credit crunches (and monetary policy more broadly) on net reduce aggregate supply as opposed to demand. These conditions boil down to whether the credit crunches primarily affect firms or households, and on whether the financial system becomes more constrained at higher interest rates, as it did under Reg Q .

We test the predictions of the model using aggregate, bank-level, and industry-level data. At the aggregate level, we find patterns that align closely with the model mechanism. When the Fed funds rate rises above the Reg Q ceiling, deposit growth falls sharply. As deposits shrink, banks contract credit. The credit contractions coincide with lower output growth and higher inflation, consistent with a negative supply shock. We confirm these patterns using formal time-series analysis. Credit-tightening shocks predict lower output and higher inflation during the Reg Q period, consistent with the view that the Reg Q credit crunches contributed to the stagflation.

The aggregate time-series alone cannot fully separate our mechanism from other sup-

ply shocks such as the oil shocks.¹ We therefore turn to the cross section, where we can directly test the mechanism of the model. We use detailed data on 459 manufacturing industries at the four-digit SIC level from the NBER-CES database, which is derived from the U.S. Census. Importantly, the data contain information on both prices and quantities, allowing us to test for supply shocks (stagflation) in the cross section.

Our first test exploits industry-level variation in exposure to the credit crunches based on external finance dependence. We measure it as the fraction of production costs (materials and labor) not covered by profits.² We show that this measure identifies the external finance dependence parameter in our model under the assumption that firms generate internal financing by retaining profits. We validate the measure using balance-sheet data. We find that industries with high external finance dependence (“finance-dependent industries”) have more debt, rely more on bank financing, hold less cash, and have lower debt service ratios than industries with low external finance dependence.

The model predicts that when the Fed raises rates above the Reg Q ceiling, a gap opens up between market rates and deposit rates. This leads to deposit outflows and a credit crunch, which cause finance-dependent industries to raise prices and reduce output relative to other industries. We test this prediction in panel regressions of price and output growth on external finance dependence interacted with the Reg Q spread (the Fed funds rate minus the Reg Q ceiling). We find a strong positive effect for prices and a negative one for output. The effects are economically large and robust to controlling for materials costs, wages, energy intensity, and volatility. External finance dependence thus predicts higher prices and lower output, i.e. “stagflation” in the cross section. We also find that it negatively predicts profits, employment, inventories, and investment. All of these results are consistent with the model.

Our second source of industry-level variation is working capital intensity. We measure it as the ratio of inventories to production costs and show how it maps to the model. As the model predicts, we find that when the Fed raises rates above the Reg Q ceiling, industries with high working capital intensity raise prices and cut output relative to other

¹While the oil shocks played an important role, the literature points out that it is unlikely that they were the main driver of the stagflation (DeLong, 1997; Barsky and Kilian, 2001). The main reasons are that they arrived late (the first in 1973, halfway through the Great Stagflation), and that inflation was already high each time they did so (10% in 1973 and 12% in 1979). See Online Appendix A for discussion.

²Rajan and Zingales (1998) use a similar measure. The main difference is that they scale by capital expenditure instead of production costs because they are interested in investment instead of output.

industries over the following year.³ The model further predicts that external finance dependence and working capital intensity reinforce each other. We interact the two to test this prediction and find that it holds.

The third test we run uses variation in Reg Q exposure coming from the bank side. Reg Q imposed different ceilings on different types of deposits, and banks differed in their deposit mix. We use this institutional feature to construct a bank-level Reg Q spread based on each bank's deposit mix. We show that it strongly predicts deposit outflows and contractions in commercial and industries (C&I) loans during the credit crunches. The bank-level Reg Q spread thus captures the severity of the credit crunches at the bank level. We further show that this variation carries over to the regional level: counties whose banks have higher Reg Q spreads experience larger deposit outflows, greater contractions in C&I lending, and larger declines in manufacturing employment.

We map the bank-level Reg Q spread to our industry data using the geographic distribution of each industry's establishments. This is appropriate in our sample because lending was largely local before deregulation in the 1980s. The resulting industry-level Reg Q spread captures each industry's exposure to the credit crunches through the exposure of the banks that lend to it. We find that industries experiencing a higher Reg Q spread (a more severe credit crunch) raise prices and cut output relative to other industries. These results support the model, which further predicts that they should be stronger for finance-dependent industries. We find this to be the case as well.

Our cross-sectional results support the mechanism of the model by showing that firms more exposed to the Reg Q credit crunches experienced a negative supply shock when the Fed raised rates. The model also provides guidance on how to translate these cross-sectional effects into aggregate ones. This can be done through a multiplier that takes into account general equilibrium forces. Using standard parameter values, we find that the multiplier exceeds one, hence the model-implied aggregate effects are economically large. However, these effects are conditional on the model's assumptions and its focus on the supply-side impact of the credit crunches.

To evaluate the broader importance of this mechanism, we compare it to the modern "deposits channel" of monetary policy (Drechsler, Savov and Schnabl, 2017). This is a natural comparison because the deposits channel also centers on a low passthrough of policy rates to deposit rates (due to market power instead of Reg Q). This low passthrough in-

³We look over the following year because these industries have long product cycles by construction.

duces deposit outflows and lending contractions when interest rates rise. The difference is that whereas Reg Q imposed a hard ceiling on deposit rates, under the deposits channel banks are free to set their rates optimally. We therefore expect any effects under the deposits channel to be weaker than under Reg Q.

We find this to be the case empirically. First, the Fed funds rate itself is less volatile post Reg Q. Second, its passthrough to deposit rates, while still low, is much higher. And third, banks facing strong loan demand are able to pay higher rates, dampening the contraction in lending and avoiding credit crunches. Consistent with this, we find qualitatively similar but economically smaller cross-sectional estimates post Reg Q. All in, we estimate that the mechanism was at least three times more powerful during Reg Q than after.

Taken together, our results challenge the conventional narrative of the Great Stagflation, which emphasizes inflation expectations and a run of “bad luck” epitomized by the oil shocks. Our analysis suggests that this narrative is incomplete: monetary tightening under Reg Q triggered credit crunches that raised firms’ costs of production, inducing an endogenous negative supply shock. Reg Q thus emerges as an overlooked contributor to the Great Stagflation, one that implies it due to more than bad luck. Beyond the Great Stagflation, our results imply that monetary policy does not operate solely through aggregate demand. In the presence of severe financial frictions, it also operates partly through aggregate supply. This means that central banks must not only consider the effects of monetary policy on demand, but also on supply.

The rest of the paper is organized as follows: Section 1 reviews the literature; Section 2 looks at the aggregate time series; Section 3 presents the model; Section 4 reports cross-sectional results; and Section 5 concludes.

1 Related literature

The Great Stagflation of the 1970s inspired a large and influential literature. Phelps (1967, 1968) and Friedman (1968, 1977) were the first to argue that the higher inflation was caused by unanchored inflation expectations. While this work went on to transform macro, it does not explain the negative comovement between inflation and output (the stagflation). To address this, Okun (1975), Gordon (1975), and Phelps (1978) incorporate supply shocks into a macro model. Gordon (1977), Perry (1978), and Blinder (1979, 1982)

enumerate such shocks: food shortages in 1972–74 and 1978–79, Nixon’s price controls in 1971–74, and, most famously, the OPEC oil shocks of 1973 and 1979.⁴ What these shocks have in common is that they are exogenous from the perspective of monetary policy. DeLong (1997) calls them “bad luck.” Since bad luck can strike at any time, there is nothing to be done about it. It follows that central banks should take supply as given and steer demand to fall in line with supply. This view is at the heart of the textbook New Keynesian model (Woodford, 2003; Galí, 2007), which underlies modern macro and policy thinking.

The view that supply is exogenous explains why the literature shifted focus from stagflation to inflation, arguing that the Fed had failed to control demand.⁵ Clarida, Gali and Gertler (1999, 2000) find that the Fed’s inflation response coefficient was below one, which violates the Taylor (1993) principle and leads to instability.⁶ The coefficient rose above one under Paul Volcker, leading to the disinflation of the 1980s (Goodfriend and King, 2005). By focusing on demand, this literature again leaves unexplained the negative comovement between inflation and output. As Clarida, Gali and Gertler (2000) write, “To account for the negative comovement that occurred in the 1970s, it appears necessary also to mix in adverse supply shocks.”

Our main contribution is to argue that a different type of supply shock arose endogenously due to a friction in the financial system. The friction, Regulation Q, led to severe credit crunches whenever the Fed raised rates. The credit crunches affected supply by disrupting firms’ ability to finance production. What makes them different from other supply shocks, such as the oil shocks, is that they are not exogenous from the perspective of monetary policy.⁷ It follows that the central bank cannot take supply as given and fo-

⁴The oil shocks are highly debated. DeLong (1997), Clarida, Gali and Gertler (2000), and Barsky and Kilian (2001) point out that inflation was already high when the oil shocks hit, and that oil shocks outside the 1970s were not followed by high inflation. Related, Bernanke et al. (1997) argue that oil shocks affect the economy mainly through the response of the central bank.

⁵Goutsmedt (2021) documents how the name of the period transitioned from “the Great Stagflation” to “the Great Inflation”. Blinder and Rudd (2013), who criticize the shift, use the original name.

⁶There are many explanations for why this happened. Kydland and Prescott (1977) and Barro and Gordon (1983) argue that central bankers have an inflationary bias due to lack of commitment. DeLong (1997) sees the roots of the Great Inflation in the experience of the Great Depression. According to Sargent (1999), policymakers were learning the parameters of the inflation-unemployment tradeoff. Romer and Romer (2002) emphasize shifts in policymakers’ beliefs, while Meltzer (2005) focuses on political factors. Orphanides (2003) and Primiceri (2006) argue that the Fed was working with flawed data.

⁷There are two other differences: Since Reg Q became binding in 1965, it helps to explain the pre-1972 stagflation cycles, for which there is no known exogenous supply shock (Blinder and Rudd, 2013). And since Reg Q was effectively repealed in 1982, it also helps to explain why the stagflation ceased. This does not mean that other supply shocks are not as important, for instance the supply disruptions following the Covid-19 pandemic (Bernanke and Blanchard, 2025).

cus on demand, contrary to conventional wisdom. Moreover, the extent to which supply is affected depends on the structure of the financial system.

Our credit crunch mechanism is closely related to the cost channel of monetary policy of Barth and Ramey (2001), which builds on earlier work by Blinder and Stiglitz (1983) and Blinder (1987). Under the cost channel, firms must pay for inputs before receiving payment for their output, i.e. they require working capital. The cost of working capital, the interest rate, therefore enters the cost of production (Gertler and Gilchrist, 1994; Kashyap, Lamont and Stein, 1994; Chevalier and Scharfstein, 1996). Specifically, the nominal rate enters the nominal cost of production and gets passed through to prices, while the real rate enters the real cost and also gets passed through to output. In particular, changes in real rates induce negative comovement between prices and output (stagflation).

The model of Barth and Ramey (2001) follows Christiano, Eichenbaum and Evans (1997) in assuming that the interest rate firms face is equal to the policy rate (there are no frictions in the financial system). Since the real policy rate did not rise during the Great Stagflation (recall the Taylor coefficient was below one), this version of the cost channel cannot explain the observed stagflation. Nevertheless, Barth and Ramey (2001) find strong cost channel effects during this period. They speculate that this is due to tight credit conditions. Our paper is consistent with this view. It identifies the mechanism (Reg Q) by which monetary policy itself led to tight credit conditions (the credit crunches) and the effect this had on prices and output. Identifying the mechanism is important for assessing the supply effects of monetary policy in a given setting. For instance, the Reg Q mechanism naturally explains why Barth and Ramey (2001) find much weaker cost channel effects after 1982, when Reg Q was repealed.

Credit crunches also negatively affect aggregate demand since households borrow to consume and firms borrow to invest. The question arises whether they are on net inflationary or deflationary. The historical evidence suggests they can be both. Schularick and Taylor (2012) study credit crises in 14 countries from 1870 to 2008 and find that they were on average deflationary pre-war and inflationary post-war. Kaehler and Weber (2023) study 192 countries from 1960 to 2019 and also find inflationary effects. On the other hand, the 2008 Global Financial Crisis was on net deflationary. This could be because it primarily affected household borrowing (Mian, Rao and Sufi, 2013). Nevertheless, the drop in inflation was much smaller than conventional models predict (Hall, 2011; Ball and Mazumder, 2011). Christiano, Eichenbaum and Trabandt (2015) attribute this “miss-

ing deflation” to firms’ higher cost of financing working capital. Consistent with this, Gilchrist et al. (2017) find that liquidity-constrained firms raised prices relative to unconstrained firms (Kim, 2021, finds the opposite for firms tied to Lehman Brothers). The data thus broadly supports the view that credit crunches are inflationary when they primarily affect firms and deflationary when they primarily affect households. We use our model to formalize this idea.

Interestingly, Reg Q also has a positive, inflationary effect on aggregate demand. The reason is that it blocks the passthrough of interest rate increases to deposit rates. This means that households who save in deposits face a low marginal rate, leading them to save less and spend more than they otherwise would. We test this inflationary demand effect in Drechsler, Savov and Schnabl (2022) and find it further helps to explain the high inflation (but not the stagflation) of the 1970s.⁸

Reg Q is a form of “credit control,” a regulation that directly constrains credit supply. During the post-war era, many developing economies used credit controls to channel bank deposits to favored industries.⁹ In seminal work, McKinnon (1973) and Shaw (1973) characterize this as “financial repression” and argue that it lowers growth and raises inflation by distorting the allocation of capital. Jacome and Pienknagura (2022) find consistent evidence. We also argue that Reg Q distorted capital allocation, but our focus is on the near-term impact of the credit crunches on production.

The paper builds on the literature on the bank lending channel of monetary policy (Bernanke and Blinder, 1988, 1992; Kashyap and Stein, 1994, 2000; Bernanke and Gertler, 1995), which studies the impact of monetary policy on bank credit. Closely related is the deposits channel of Drechsler, Savov and Schnabl (2017). In the deposits channel, banks use market power to keep deposit rates low when market rates rise. This leads to deposit outflows and a contraction in credit, qualitatively similar to a Reg Q credit crunch. The difference is that banks in the deposits channel set their rates optimally, trading off higher

⁸There are a few macro papers specifically on Reg Q. Friedman (1970) and Tobin (1970) argue that Reg Q is deflationary because it shrinks deposits, which are part of the money supply. This monetarist perspective explains why the Fed allowed Reg Q to become binding. Mertens (2008) argues that deposit rate ceilings increase the impact of monetary policy on output, but does not consider inflation. Most work on Reg Q is in the banking literature. Burger (1969) and Wojnilower (1980) chronicle the credit crunches and their impact on the economy (see also Bordo and Haubrich, 2010). Gilbert (1986) discusses the repeal of Reg Q. Koch (2015) shows that the Reg Q ceilings negatively impacted bank lending.

⁹Credit controls were also common in advanced economies during the post-war era (Hodgman, 1973). They fell out of favor in the 1980s following their failure to curb inflation in the 1970s (Reinhart and Sbrancia, 2015). We return to this literature when discussing the international evidence on the Great Stagflation.

deposit spreads against unmet credit demand. In contrast, Reg Q forces banks to pay low deposit rates, leading to larger outflows and much more severe contractions in credit. Consistent with this, we find that the deposits channel has a similar but much smaller effect on prices and output than the Reg Q credit crunches.

2 Time-series analysis

Figure 1 summarizes the main pattern of the Great Stagflation. The figure shows that inflation rises in four successive cycles, each peaking higher than the one before. The striking feature is that every inflation cycle is mirrored by a sharp decline in GDP growth. This is the “stag” part of the stagflation. The timing is very close in each cycle: inflation starts rising just as GDP growth begins to fall, and peaks just as GDP growth bottoms out. This tight negative relationship (higher prices, lower quantities) implies that the economy experienced an adverse supply shock in each cycle.

Figure 2 shows the impact of Reg Q on the banking system. Panel A plots the Reg Q ceiling rate on savings deposits (by far the largest category), the Fed funds rate, and real deposit growth. The Fed first raised the funds rate above the ceiling at the onset of the Great Stagflation in late 1965. This led deposit growth to fall sharply. When the Fed then lowered the funds rate in 1967, deposit growth rebounded. This pattern repeats in each subsequent rate cycle, becoming more pronounced over time as higher rates make the ceiling ever more binding. The volatility of deposit growth is large, ranging from about +13% to −9%. The pattern ends when Reg Q is effectively repealed in 1982 (Gilbert, 1986), coinciding with the end of the Great Stagflation.¹⁰

Panel B shows that the declines in deposit growth translate roughly one-for-one into declines in the growth of bank credit. Because bank credit is a stock measure, this implies a much larger contraction in the flow of new lending. For example, when credit growth fell from +6% to −4% during the 1969–70 credit crunch, new bank loans shrank by 51% (see Table OA.1 in the Online Appendix). The close relationship between deposits and bank credit indicates that banks were unable to replace lost deposits, their main source

¹⁰What about other countries? Stagflation was a global phenomenon, yet so were credit controls and “financial repression,” policies aimed at constraining the banking system and restricting the flow of credit to the private sector (Hodgman, 1973; McKinnon, 1973). In addition to deposit rate ceilings, these included loan rate ceilings and outright quantity restrictions, all of which are functionally equivalent from the perspective of our mechanism. Thus, while we focus on the U.S., it is plausible that a similar channel existed in other countries. See Online Appendix B for more details.

of funding, with alternative funding. As a result, deposit outflows repeatedly translated into severe credit contractions or crunches.¹¹

Figure 3 shows that the credit crunches spread beyond banks to the entire financial system. Panel A plots the Fed funds rate and the Tbill rate. The difference between them, the Fed funds-Tbill spread, measures the liquidity premium of Tbills (e.g. Nagel, 2016). Since Tbills are a partial substitute for deposits, they absorb some of the deposit outflows triggered by Reg Q. This drives their yield down toward the Reg Q ceiling, which increases the liquidity premium. The premium peaks at 5.4% in 1974, far higher than the 1.3% reached in 2008, reflecting the acute shortage of safe, liquid assets induced by Reg Q. Panel B plots the Chicago Fed financial conditions index, a measure of tightness across loan, bond, and equity markets.¹² The index has a strong negative relationship with deposit growth during the Reg Q period. Thus, when deposits flowed out, financing became very tight across all markets. Notably, the index spikes higher during Reg Q than the 2008 Global Financial Crisis, highlighting the severity of the Reg Q credit crunches.

Figure 4 overlays the Reg Q credit crunches onto GDP growth and inflation. Deposit growth is strongly positively correlated with GDP growth and strongly negatively correlated with inflation. Whenever deposit growth falls and credit contracts, output growth falls and inflation rises. Then when deposit growth recovers, output growth rebounds and inflation subsides. The timing is very close, with the sharpest contractions in deposit growth occurring at the onset of each stagflationary episode. The strong pattern is consistent with a supply-side mechanism for the transmission of Reg Q.

Formal time-series analysis reveals the same pattern. We estimate impulse response functions following Gilchrist and Zakrajšek (2012), who measure credit supply shocks as innovations in the excess bond premium. Since the excess bond premium is only available from 1973, which is halfway through the Great Stagflation, we replace it with a closely related measure that Gilchrist and Zakrajšek (2012) show is highly correlated with the excess bond premium. The measure is the tightness of credit standards from the Fed's Senior Loan Officer Opinion Survey (SLOOS).¹³ This measure goes back to 1964 and is

¹¹In Online Appendices C and D, we discuss the role of money market funds and corporate bonds in partly absorbing the deposit outflows. In general, deposit outflows lead to disintermediation of the banking system. This disintermediation leads to an overall contraction in credit as well as a reallocation from bank-dependent firms to less financially constrained firms.

¹²Caballero, Caravello and Simsek (2025) argue that monetary policy transmits to the real economy through broad financial conditions rather than the short rate alone.

¹³The tightness of credit standards is measured as the net percentage of banks reporting tighter standards for C&I loans to new customers. We obtain it from the Federal Reserve Bank of St. Louis's digital library

also conceptually appropriate for studying the effects of Reg Q as it is bank-based. We keep the variable ordering unchanged and estimate responses separately for the Reg Q period and post-Reg Q period in order to compare them.

The results are shown in Figure 5, which reproduces Figure 5 in Gilchrist and Zakrajšek (2012).¹⁴ The figure plots impulse response functions of macroeconomic variables to a credit tightening shock. Most variables exhibit directionally similar responses in the two samples. The exception is prices. In the post-Reg Q period, prices fall slightly, consistent with Gilchrist and Zakrajšek (2012).¹⁵ In contrast, during Reg Q they rise substantially. This suggests that the Reg Q credit crunches were inflationary.

We also estimate responses to a Fed funds rate shock using the same framework. The results are discussed in Online Appendix F. We find that a Fed funds rate shock is associated with higher inflation during the Reg Q period, mirroring the result in Figure 5. This is consistent with Fed tightening inducing a negative supply shock through the Reg Q credit crunches.¹⁶

In sum, both the raw time series and impulse response functions are consistent with the hypothesis that the Reg Q credit crunches induced a negative supply shock that contributed to stagflation. To further strengthen our empirical analysis and analyze the mechanism, we turn to the cross section after presenting our model.

3 Model

The model follows the canonical New Keynesian (NK) model (e.g., Galí, 2007). Let $r_t = \log(1 + R_t)$ be the log nominal interest rate and assume it follows

$$r_t = \bar{r} + \phi_\pi \pi_t + \phi_y (\mathbf{y}_t - \bar{\mathbf{y}}) + \varepsilon_t \quad (1)$$

$$\varepsilon_{t+1} = \rho \varepsilon_t \text{ for } t \geq 0, \quad (2)$$

FRASER (1964–1981) and extend it to 1982 using data from Lown and Morgan (2006). The survey did not report credit standards from 1984 to 1990, so the post-Reg Q analysis begins in 1990.

¹⁴Table OA.2 in the Online Appendix reports the associated point estimates.

¹⁵This result can be explained by the nature of the 2008 Global Financial Crisis (GFC), which dominates variation in credit standards post Reg Q. The GFC primarily affected household borrowing (primarily mortgages), a component of aggregate demand (Mian, Rao and Sufi, 2013). Nevertheless, as Christiano, Eichenbaum and Trabandt (2015) emphasize, inflation fell much less than expected, consistent with an inflationary channel also being at work. We discuss the GFC in more detail in Online Appendix E.

¹⁶Previous studies have also found that Fed funds rate shocks were associated with higher inflation during this period, see the discussion in Barth and Ramey (2001). Note also that the effects on prices and output kick in quickly without significant lags.

where $\bar{r}$ is the steady-state nominal rate, ε_0 is a monetary policy shock, $\rho < 1$ is its persistence, π_t is inflation, and $y_t - \bar{y}$ is the output gap. The interest rate follows a Taylor rule with coefficients ϕ_π on inflation and ϕ_y on output.

3.1 Households

Households maximize utility of consumption $\mathbf{C}_t$ net of disutility from labor L_t . They also have a preference for liquidity in the form of real deposit balances D_t :

$$\max \sum_{t=0}^{\infty} \beta^t \left[\frac{\mathbf{C}_t^{1-\gamma}}{1-\gamma} - \frac{L_t^{1+\eta}}{1+\eta} + \phi(D_t) \right], \quad (3)$$

where β is the subjective discount factor, γ is the inverse elasticity of intertemporal substitution (EIS), η is the inverse Frisch elasticity of labor supply, and $\phi(D_t)$ is a liquidity demand function that we specify below. The consumption index, $\mathbf{C}_t$, comprises differentiated goods produced by firms indexed by $i \in [0, 1]$:

$$\mathbf{C}_t = \left(\int_0^1 C_{i,t}^{\frac{\epsilon-1}{\epsilon}} di \right)^{\frac{\epsilon}{\epsilon-1}}, \quad (4)$$

where ϵ is the elasticity of substitution between goods. The associated price index (the price level) is $\mathbf{P}_t$, and inflation is its log growth rate: $\log \mathbf{P}_t / \mathbf{P}_{t-1} = \mathbf{p}_t - \mathbf{p}_{t-1} = \pi_t$.

Households invest in deposits D_t and government bonds G_t . Deposits pay a nominal interest rate R_t^D from date t to $t+1$. Government bonds pay the nominal interest rate R_t . The household budget constraint is

$$\mathbf{P}_t (\mathbf{C}_t + D_t + G_t) = W_t L_t + \mathbf{P}_{t-1} \left[D_{t-1} (1 + R_{t-1}^D) + G_{t-1} (1 + R_{t-1}) \right] + \tau_t, \quad (5)$$

where W_t is the nominal wage and τ_t is government transfers. Transfers include the net proceeds from issuing government bonds as well as taxes on firm and bank profits.

The household first-order condition for labor pins down the real wage:

$$w_t - \mathbf{p}_t = \gamma \mathbf{c}_t + \eta l_t, \quad (6)$$

where $w_t = \log W_t$ is the log nominal wage ($w_t - \mathbf{p}_t$ is the log real wage), $\mathbf{c}_t = \log \mathbf{C}_t$ is

log consumption, and $l_t = \log L_t$ is log hours worked.¹⁷ The real wage is decreasing in the marginal utility of consumption and increasing in the marginal disutility from labor.

The first-order condition for bonds gives us the Euler equation

$$r_t - \pi_{t+1} = -\log \beta + \gamma \Delta \mathbf{c}_{t+1}, \quad (7)$$

where $r_t - \pi_{t+1}$ is the log real rate and $\Delta \mathbf{c}_{t+1}$ is log consumption growth. Real rates are high when future consumption growth is high.

The first-order condition for deposits pins down deposit demand:

$$\phi'(D_t) = \mathbf{C}_t^{-\gamma} (r_t - r_t^D). \quad (8)$$

The marginal value of liquidity equals the marginal utility of foregone consumption from paying a deposit spread. If the deposit spread is zero, liquidity demand is satiated: $\phi'(D_t) = 0$. To capture this case, we parameterize ϕ as follows:

$$\phi(D_t) = \begin{cases} -\frac{\delta_0}{2\bar{D}} (\bar{D} - D_t)^2 & \text{if } D_t \leq \bar{D}, \\ 0 & \text{otherwise.} \end{cases} \quad (9)$$

Liquidity demand is increasing and concave in deposits. The parameter δ_0 controls the marginal value of liquidity up to the satiation point $\bar{D}$. This value is zero past $\bar{D}$, hence households treat deposits and government bonds as perfect substitutes. This implies that banks can raise unlimited deposits when deposit demand is satiated.

Demand for deposits (8) can be log-linearized around the steady state as follows:

$$d_t = \bar{d} - \frac{1}{\delta} [r_t - r_t^D]^+, \quad (10)$$

where $d_t = \log D_t$, $\bar{d} = \log \bar{D}$, $\delta = \delta_0 e^{\gamma \bar{c}}$ is the marginal value of liquidity in consumption units, $\bar{c}$ is steady-state consumption, and $r_t^D = \log(1 + R_t^D)$ is the log deposit rate.

Deposit demand is a decreasing function of the deposit spread.

¹⁷The derivation is in Appendix A, as are all others that follow.

3.2 Banks

Banks issue deposits D_t and invest in firm loans $B_{i,t}$, $i \in [0, 1]$, and government bonds. There is no equity, so loans plus government bonds equal deposits. Banks maximize profits from spreads on loans and deposits:

$$\max \int_0^1 B_{i,t} (r_t^B - r_t - \bar{s}) di + D_t (r_t - r_t^D), \quad (11)$$

where $\bar{s}$ is banks' cost of screening and monitoring loans. We include it to keep the loan spread positive when Reg Q does not bind.¹⁸

We assume deposits are subject to an interest rate ceiling:

$$r_t^D \leq r^Q. \quad (12)$$

When binding, the Reg Q ceiling prevents banks from raising additional deposits. For this to affect lending, banks must be unable to raise non-deposit funding by shorting government bonds. This imposes the funding constraint

$$\int_0^1 B_{i,t} di \leq D_t. \quad (13)$$

The implicit assumption is that deposits are a special form of bank funding, a common assumption in the banking literature (e.g. Stein, 1998). This could be due to deposit insurance, depositor "stickiness" (Hanson et al., 2015), or the low interest-rate sensitivity of deposits (Drechsler, Savov and Schnabl, 2021). These features make deposits non-runnable and stable relative to other forms of short-term debt.

We can characterize the impact of Reg Q on deposit and loan rates as follows:

Proposition 1 (Reg Q). *If Reg Q does not bind strictly, then the deposit spread is zero and the loan spread equals the screening and monitoring cost:*

$$r_t - r_t^D = 0 \text{ and } r_t^B - r_t = \bar{s}. \quad (14)$$

If Reg Q binds strictly, then the deposit spread is positive and the loan spread exceeds the screening

¹⁸We assume loan demand weakly exceeds $\bar{D}$ when the loan spread is $\bar{s}$. We provide parameter restrictions for this assumption and derive (11) in Appendix A.

and monitoring cost:

$$r_t - r_t^D > 0 \text{ and } r_t^B - r_t > \bar{s}. \quad (15)$$

Reg Q binds strictly if and only if the market interest rate exceeds the deposit rate ceiling, $r_t > r^Q$.

Reg Q binds whenever the market rate crosses the ceiling. This creates a sharp distinction between regimes where deposits are scarce and abundant. Proposition 1 shows how the two regimes differ. When Reg Q does not bind, the deposit spread is zero and deposits are abundant. This makes loans also abundant and the loan spread is equal to the cost of screening and monitoring. In contrast, when Reg Q binds the deposit spread opens up and this makes deposits scarce. The funding constraint starts to bind and loans also become scarce, causing the loan spread to widen. Reg Q thus drives a wedge between deposit rates and loan rates. Deposit rates fall below market rates and loan rates rise above them. Deposits become scarce and this makes loans also scarce. This is a Reg Q credit crunch.

3.3 Firms

Firm i produces output $Y_{i,t}$ using labor $L_{i,t}$ and working capital $WC_{i,t}$ according to

$$Y_{i,t} = A WC_{i,t}^\alpha L_{i,t}^{1-\alpha}, \quad (16)$$

where A is a productivity parameter and α is the working capital share of output. The working capital share is high if firms hold a lot of inventory, for instance because their product cycle is long or because they face volatile demand or supply conditions, or if they extend more trade credit (receivables) to their customers or suppliers.

Working capital has to be financed. This can be done through internal sources like cash from retained earnings, $\mathcal{C}$, or external sources like bank loans, $B_{i,t}$. We take cash to be constant for simplicity (i.e. changes in profits are absorbed by dividends). We model working capital as follows:

$$WC_{i,t} = \mathcal{C}^{1-\varphi} B_{i,t}^\varphi, \quad (17)$$

where φ measures the external financing share of working capital. This share is high

if firms have thin gross margins.¹⁹ This makes them rely on banks for financing. We therefore refer to φ as external finance dependence.

Firms choose their input mix to minimize production costs, which consist of labor costs (wages) and borrowing costs (the loan spread). The optimal input mix sets the ratio of borrowing to labor in proportion to their relative cost:

$$\frac{B_{i,t}}{L_{i,t}} = \varphi \left(\frac{\alpha}{1-\alpha} \right) \frac{W_t / \mathbf{P}_t}{r_t^B - r_t}. \quad (18)$$

Borrowing is increasing (relative to labor) in external finance dependence φ and the working capital share α . It is also increasing in the price of labor (the real wage) and decreasing in its own price, the loan spread.

Firms face a standard Calvo (1983) pricing friction: every period, a share $1 - \theta$ of firms get to change their price. A firm that gets to set its price maximizes the present value of its profits in all future periods that the price remains in effect:

$$\max_{P_{i,t}} \sum_{\tau=0}^{\infty} \theta^{\tau} \Lambda_{t,t+\tau} \left[P_{i,t} Y_{i,t+\tau} - W_{t+\tau} L_{i,t+\tau} - \left(r_{t+\tau}^B - r_{t+\tau} \right) \mathbf{P}_{t+\tau} B_{i,t+\tau} \right], \quad (19)$$

where $\Lambda_{t,t+\tau}$ is the discount factor between t and $t + \tau$. The pricing problem is subject to the production function (16), the loan demand condition (18), and the demand curve implied by (4), taking into account that $C_{i,t} = Y_{i,t}$ (goods clearing). Differentiating with respect to $P_{i,t}$, the optimal price satisfies

$$P_{i,t} = \frac{\epsilon}{\epsilon - 1} \sum_{\tau=0}^{\infty} \omega_{i,t \rightarrow t+\tau} \left(r_{t+\tau}^B - r_{t+\tau} \right) \frac{\partial B_{i,t+\tau}}{\partial Y_{i,t+\tau}} \mathbf{P}_{t+\tau}, \quad (20)$$

where $\omega_{i,t \rightarrow t+\tau}$ are weights given in Appendix A. Firms charge a markup over a weighted average of current and future discounted marginal costs. Marginal costs increase in the loan spread because it makes working capital, and hence output, more costly. This leads firms to raise prices, more so if they borrow more to produce each unit at the margin.

¹⁹We are implicitly assuming that firms cannot retain enough cash to fully undo their need for external financing. This could be due to a standard free cash flow problem.

3.4 Inflation and Reg Q

We solve the model in the standard way, by log-linearizing the pricing condition (20) around the steady state and aggregating across firms. We then impose market clearing in the loan market ($b_t = d_t$) to replace the loan spread with the deposit spread. We characterize the result with the following proposition:

Proposition 2 (Phillips curve). *Inflation follows the Phillips curve*

$$\pi_t = \kappa_y (\mathbf{y}_t - \bar{\mathbf{y}}) + \frac{\varphi}{\delta} \kappa_b \left[r_t - r^Q \right]^+ + \beta \pi_{t+1}, \quad (21)$$

where κ_y and κ_b are constants given in Appendix A. If Reg Q does not bind ($r_t < r^Q$), if external finance dependence is zero ($\varphi = 0$), or if deposit demand is inelastic ($1/\delta = 0$), inflation follows the standard NK Phillips curve.

Inflation follows an augmented Phillips curve that includes a term in the deposit spread. From Proposition 1, the deposit spread is zero when Reg Q does not bind. In this case, the Phillips curve collapses to the usual one and inflation behaves exactly as in the NK model. In contrast, when Reg Q binds, deposits become scarce and this makes loans also scarce. The loan spread widens, and this makes it costlier for firms to finance production. Firms pass on the higher cost by raising prices. This is how Reg Q credit crunches create inflation.

The impact of Reg Q on inflation depends on two key parameters and their interaction. The first is external finance dependence, φ . When external finance dependence is high, firms lack internal resources and need loans to finance working capital. This makes production costs more sensitive to credit crunches, leading to larger price increases and higher inflation. The second key parameter is the elasticity of deposit demand to the deposit spread, $1/\delta$. When deposit demand is more elastic, deposit outflows are larger and the credit crunches are more severe. This again results in a larger increase in production costs and higher inflation.

Since the deposit spread is a function of the policy rate, Proposition 2 shows that raising rates has an inflationary effect when Reg Q binds. The inflationary effect is in addition to the standard deflationary effect through the output gap.²⁰

²⁰Proposition A.2 in the appendix shows that the inflationary effect dominates when $\varphi \kappa_b / \delta > \kappa_y / [\gamma(1 - \rho)]$. Proposition A.1 further shows that Reg Q requires a stricter version of the Taylor principle for monetary policy stability: $\kappa_y (\phi_\pi - 1) + \phi_y [(1 - \beta) - \varphi \kappa_b / \delta] > 0$, that is the Taylor coefficient on inflation ϕ_π must exceed one by more than in the standard model.

3.5 Testing and aggregating the model:

In principle, we can test the model by regressing inflation on the Reg Q spread. This is what our time series analysis effectively does. However, time series analysis is subject to an identification challenge: It could be that other supply shocks that happen to coincide with the credit crunches drive the negative comovement between inflation and output (the stagflation). We use cross-sectional analysis to control for such shocks and identify our mechanism.

Cross-sectional analysis differences out aggregate variables such as the oil shocks and inflation expectations. However, it also differences out general equilibrium effects that are part of the model mechanism. These include effects through wages (from lower labor demand), aggregate demand (from higher inflation), and interest rates. Cross-sectional effects therefore map to partial equilibrium comparative statics that hold these aggregate quantities fixed. We compute these comparative statics and use them to guide our empirical analysis:

Proposition 3 (Cross-sectional effect). *Holding fixed real wages and aggregate output, the sensitivity of firm i 's optimal price $p_{i,t}$ to a permanent increase in r_t when Reg Q is binding is increasing in φ , α , and $1/\delta$. A first-order approximation around $(\alpha, \varphi) = (0, 0)$ gives*

$$\frac{\partial p_{i,t}}{\partial r_t} \approx \frac{\alpha \varphi}{\delta}. \quad (22)$$

The sensitivity of firm i 's output $y_{i,t}$ is $-\epsilon \frac{\partial p_{i,t}}{\partial r_t}$.

Proposition 3 shows that we can test the model in the cross section using variation in external finance dependence φ , working capital intensity α , deposit demand elasticity $1/\delta$, and their interactions. When the Fed raises interest rates, firms with higher external finance dependence or working capital intensity, and firms borrowing from banks facing higher deposit outflows should raise prices and reduce output by more than other firms. In other words, external finance dependence, working capital intensity, and exposure to deposit outflows should predict stagflation in the cross section. These predictions form the basis of our empirical tests.

Since we are also interested in aggregate effects, we show how to obtain them from cross-sectional estimates by combining Propositions 2 and 3:

Proposition 4 (Aggregate effect). *Holding fixed the path of aggregate output, the sensitivity of inflation π_t to an increase in r_t when Reg Q is binding is related to the cross-sectional sensitivity of firm i 's price $p_{i,t}$ by*

$$\frac{\partial \pi_t}{\partial r_t} = \frac{1}{\theta} \left(\frac{1 + \frac{\alpha\varphi}{1-\alpha}}{\frac{1}{1+\eta} + \frac{\alpha\varphi}{1-\alpha}} \right) \frac{\partial p_{i,t}}{\partial r_t} \approx \left(\frac{1+\eta}{\theta} \right) \frac{\alpha\varphi}{\delta}. \quad (23)$$

Proposition 4 shows that the aggregate inflationary effect of Reg Q is related to the cross-sectional one through a multiplier. The multiplier is decreasing in the share of firms that do not set prices each period, θ , and increasing in the inverse Frisch elasticity of labor supply, η . When fewer firms set prices, inflation increases more slowly (but more persistently). And when labor supply is inelastic, substitution from working capital to labor forces a larger increase in wages for a given level of output and this adds to inflation.

Since $\theta < 1$ and $\eta > 0$, the inflation multiplier is greater than one, i.e. the aggregate inflationary effect is larger than the cross-sectional one. Note, however, that this result is conditional on the model's assumptions.²¹

3.6 Demand effects of Reg Q:

A key model assumption is that credit is used for production, which is part of aggregate supply, and not investment or consumption, which are part of aggregate demand.²² While we look at investment directly in our empirical analysis, in Appendix C we extend the model to capture the effects of Reg Q on consumption.

Many households use credit to finance consumption. We model them in reduced form, by incorporating a fraction of households who are net borrowers. Their Euler equation (7) features the loan rate r_t^B instead of the policy rate r_t . Since Reg Q raises r_t^B by more than r_t , it leads these borrowers to reduce their consumption disproportionately. This reduces aggregate demand and inflation.

We also introduce households who do not participate in bond markets and use deposits to save. These households' Euler equation (7) features the deposit rate r_t^D instead of r_t . Since Reg Q lowers r_t^D relative to r_t , it leads deposit-savers to increase their con-

²¹Also note that we are leaving out the impact on inflation from changes in output (through aggregate demand) and interest rates (through the Taylor rule). We do so because we are interested in the direct effect of higher rates on inflation through the Phillips curve shown in Proposition 2.

²²Caballero and Simsek (2024) argue that monetary policy affects aggregate demand through its effect on borrowing rates and asset prices.

sumption. This raises aggregate demand and inflation. The net impact of Reg Q on inflation via aggregate demand therefore depends on the relative shares of borrowers and deposit-savers.²³

4 Cross-sectional analysis

Proposition 3 shows that we can test our model in the cross section using variation in external finance dependence φ , deposit demand elasticity $1/\delta$, and working capital intensity α . We examine all three and their interactions. We begin with external finance dependence, which we call finance dependence for short.

4.1 Data

We provide a brief description of our data sources; the details are in Appendix E. Aggregate data (interest rates, inflation, deposits) are from the St. Louis Fed’s FRED database. Deposit rates are from the Fed’s “Monthly Survey of Selected Deposits and Other Accounts” and Reg Q ceiling rates are from Gilbert (1986) and Santomero and Siegel (1986). Industry-level data are from the NBER-CES Manufacturing Industry Database. Bank-level data are from the U.S. Call Reports, which we obtained back to 1959 through a Freedom of Information Act (FOIA) request. Finally, establishment data at the county-industry level are from the Census Bureau’s County Business Patterns.

4.2 Finance dependence

We measure finance dependence at the industry level, using i subscripts.²⁴ Since the production function is Cobb-Douglas, φ_i equals industry i ’s expenditure share on external financing out of total working capital. This is the case when both internal resources (cash) $\mathcal{C}_i$ and external financing B_i are optimally chosen. We assume this is the case in steady state, which we map to the period before Reg Q becomes binding in 1965. If we further assume that the steady-state cost of internal and external financing is the same ($\bar{s}$), then

²³The 1983 Survey of Consumer Finances shows that 94% of households in the median (5th) income decile had deposit accounts and only 4% had money market funds. These households held 76% of their liquid assets in deposits versus 4% in money market funds, 5% in debt securities, and 9% in stocks.

²⁴Industries are arguably the appropriate level of aggregation since individual firms are less likely to face downward-sloping demand curves. We also only observe prices at the industry level.

the expenditure share of external financing is equal to its dollar share:

$$\varphi_i = \frac{\bar{B}_i}{\mathcal{C}_i + \bar{B}_i} = 1 - \frac{\mathcal{C}_i}{\mathcal{C}_i + \bar{B}_i}. \quad (24)$$

We proxy for the external financing share using production data since we lack financial data.²⁵ Companies generate internal resources $\mathcal{C}_i$ by retaining profits. We thus estimate an industry's internal resources by the gross profit ($Sales - COGS$) that it generates from one production cycle. Since our data are annual, we observe gross profits over $Turn_i$ production cycles, where $Turn_i$ is industry turnover. This gives us $Sales_i - COGS_i = Turn_i \times \mathcal{C}_i$, which we can use to construct the numerator in (24).

We estimate working capital, $\mathcal{C}_i + B_i$, using inventories INV_i (our data does not contain other forms of working capital such as receivables). We observe inventories both directly and over $Turn_i$ production cycles since $COGS_i = Turn_i \times INV_i$ by the definition of turnover. We therefore use

$$FinDep_i \equiv 1 - \frac{Sales_i - COGS_i}{COGS_i} = 1 - \frac{Turn_i \times \mathcal{C}_i}{Turn_i \times INV_i} = 1 - \frac{\mathcal{C}_i}{INV_i} \quad (25)$$

as our proxy for finance dependence φ_i as in (24). Intuitively, $FinDep_i$ measures the external (non-cash) share of working capital (inventory) financing.²⁶

Our finance dependence measure is closely related to that of Rajan and Zingales (1998), which they define as “the amount of desired investment that cannot be financed through internal cash flows generated by the same business.” The idea is the same as our measure, except we are interested in working capital rather than investment. Following Rajan and Zingales (1998), we cumulate each component of (25) over time to dampen transitory and cyclical fluctuations. We do so from 1958 (the start of the NBER-CES data) to 1965, when Reg Q first binds. We think of this relatively quiet period as mapping to the steady state of our model. Another advantage is that our measure is pre-determined with respect to the Reg Q period, which avoids reverse causality issues. Finally, we winsorize at the 5% level to reduce the effect of outliers.

²⁵We validate our measure in Section 4.2.2 using financial data available at a more aggregated level. Rajan and Zingales (1998) argue in favor of a production-based measure, noting it is more likely to be invariant to financial conditions.

²⁶Since the model takes $\mathcal{C}_i$ as given, it is silent on whether internal resources are earned per production cycle or per year. We use production cycle because it aligns with the use of working capital. If instead we use a year, then $Sales_i - COGS_i = \mathcal{C}_i$ and our measure becomes $1 - \frac{\mathcal{C}_i}{Turn_i \times INV_i}$, i.e. it incorporates information about working capital intensity α , which is inversely related to turnover, in addition to φ .

4.2.1 Summary statistics

Table 1 shows summary statistics for finance dependence and other variables in the NBER-CES data. The average level of finance dependence is 0.5, i.e., firms can cover up to half their working capital with one cycle's gross profit. The standard deviation of finance dependence is 0.2, implying substantial cross-sectional variation across the 459 industries. To give a sense of this variation, Table OA.4 in the Online Appendix lists the average finance dependence of each sector (two-digit SIC level). It ranges from 0.26 for "Instruments and Related Products" to 0.70 for "Textile Mill Products".

Table 1 further splits the sample into high (above-median) and low (below-median) finance dependence industries. As of 1965, which is pre-Reg Q, high finance dependence industries are on average slightly larger in terms of employment, output, and capital. They are as productive as low finance dependence industries. They are slightly more volatile, both in terms of prices (2.62% versus 1.89%) and output (10.44% versus 9.48%). They have a similar labor share but a larger materials share (0.58 versus 0.42). They have slightly lower energy intensity (energy cost over output), but energy intensity is quite low for both groups, suggesting that energy shocks are unlikely to play a large role. High finance dependence industries have similar inventory-output ratios (0.15 and 0.17) and lower capital-output ratios (0.31 versus 0.46) than low finance dependence industries. Their investment rate (investment over capital) is about the same.

The last six rows of Table 1 look at industry outcomes after Reg Q becomes binding. High finance dependence industries have marginally higher average price growth (5.84% versus 5.83% per year) despite having lower materials price growth (6.30% versus 6.77%). They also have significantly lower output growth (2.75% versus 3.72%), lower employment growth (0.52% versus 1.32%), and lower inventory and capital growth. This combination of lower output growth and higher price growth (relative to materials cost) suggests that these industries experienced a negative supply shock during the Reg Q period. Our main tests ask if this occurred during the credit crunches.

4.2.2 Firm balance sheets

We validate our finance dependence measure by relating it to financial data available at a more aggregated level. We use the Quarterly Financial Reports (QFR), which contain balance sheet information at the sector (two-digit SIC code) level, a much smaller cross

section than our main sample (19 sectors versus 459 industries).

Table 2 relates finance dependence to standard financial ratios. As in Table 1, we focus on the period from 1958 to 1965, which is before Reg Q becomes binding. Panel A shows summary statistics and Panel B regresses each financial ratio on finance dependence. The average level of finance dependence in Table 2 is slightly higher than in Table 1 because the QFR data do not break out overhead (SG&A) from production costs. The higher level of aggregation shrinks the cross-sectional variation substantially: the standard deviation of finance dependence is 0.04 across sectors versus 0.2 across industries.

Despite the lower variation, clear differences emerge. Finance dependent industries have much higher leverage. From Panel B, the relationship is highly significant despite the small cross section. A one standard deviation increase in finance dependence across sectors is associated with 5.8 percentage points higher leverage. If instead we apply the industry-level standard deviation, the increase is 29 percentage points, which is very large relative to the average (31%). This result provides direct evidence that finance dependent industries use more external financing, validating our measure.

The difference in leverage is concentrated in short-term debt. A one industry-standard-deviation increase in finance dependence raises the short-term share of debt by 32.5 percentage points. Note that short-term debt is primarily used to finance operations, while long-term debt is more suited to investment. Thus, finance dependent industries appear to be using external financing for production, as we intended.

Finance dependent industries also have a significantly higher bank share of debt. A one industry-standard-deviation increase in finance dependence raises the bank share of debt by as much as 40 percentage points. Finance dependent industries are therefore not only external finance dependent but also bank dependent. This further increases their exposure to the Reg Q credit crunches.

Next, we look at cash ratios (cash over current liabilities), which capture internal liquidity. Finance dependent industries have significantly lower cash ratios: a one industry-standard-deviation increase in finance dependence lowers cash ratios by 35 percentage points, which is very large compared to the average (36%). Finance dependent industries thus lack internal resources, again validating our measure.

Another measure of internal liquidity is the debt service ratio (operating income over debt due in one year). It measures a firm's ability to repay its debt if it cannot roll it over. Finance dependent industries have much lower debt service ratios: raising finance depen-

dence by one industry standard deviation lowers the debt service ratio by 8.3. Finance dependent industries are thus much more vulnerable to credit supply disruptions.²⁷

Overall, our validation tests support using finance dependence as a measure of exposure to the Reg Q credit crunches.

4.2.3 Yearly cross-sectional regressions

We now test the model using the prediction of Proposition 3 that finance dependent industries (higher φ) raise prices and cut output by more than other industries when interest rates rise under a binding Reg Q.

We begin with simple yearly cross-sectional regressions of the form:

$$y_{i,t} = \alpha_t + \beta_t \text{FinDep}_i + \delta_t X_{i,t} + \epsilon_{i,t}, \quad (26)$$

where $y_{i,t}$ is either price growth (the percentage change in the sales deflator) or output growth (the percentage change in real output) of industry i in year t . The control $X_{i,t}$ is the growth in industry i 's materials prices. We include it to make sure firms are not just passing through higher materials costs. Estimating the coefficients β_t separately by year allows us to see if the effect of finance dependence on prices and output coincides with the rate hikes that induced the Reg Q credit crunches. We thus expect β_t to be larger when the credit crunches are more severe.

Figure 6 plots the results. The red line is the yearly coefficient for prices (ΔPrices) and the blue line is for output (ΔOutput). The light shading around these coefficients shows 95% confidence bands. The dashed line, aggregate deposit growth ($\Delta\text{Deposits}$), captures the timing and severity of the credit crunches (see Figure 2). Finally, vertical lines in 1965 and 1982 mark the start and end of the binding Reg Q period.

The figure shows a clear pattern. The coefficients for predicting prices and output are highly negatively related: finance dependent industries raise prices at the same time as they cut output relative to other industries. This pattern appears in four distinct cycles

²⁷We also validate our measure using Compustat data, which focuses on public firms and has fewer industries represented. As with the QFR data, we find that finance dependent firms have higher leverage and lower internal liquidity. We also find that our measure is positively correlated with that of Rajan and Zingales (1998), which captures finance dependence for investment. We further examine the two most popular measures of financial constraints: the KZ index of Kaplan and Zingales (1997) and the HP index of Hadlock and Pierce (2010). Note that external finance dependence and financial constraints are fundamentally different: one is a sensitivity and the other is a constraint. We find a positive correlation with the KZ index and no correlation with the HP index. The results are reported in Online Appendix H.

during the Reg Q period: in 1965-66, 1969-70, 1973-74, and 1978-79. These are precisely the Reg Q credit crunches as captured by deposit growth (dashed line). Thus, finance dependent industries raised prices and cut output by more than other industries during the credit crunches, consistent with the model (Proposition 3).

4.2.4 Panel regressions

We run a formal test of Proposition 3 using panel regressions that interact finance dependence and the Reg Q deposit spread. The Reg Q spread is the difference between the Fed funds rate and the Reg Q ceiling rate if that difference is positive, and zero otherwise ($[r_t - r^Q]^+$ in the model). It is the right way to test the model because finance dependence (φ) only matters when Reg Q binds. Since Reg Q binds throughout most of 1965–82, and since the Reg Q ceilings change relatively little, most of the variation in the Reg Q spread is driven by the Fed funds rate. This allows us to interpret the coefficient on the Reg Q spread as the impact of raising rates under a binding Reg Q.

We measure the Reg Q deposit spread by combining the spreads on savings, small time, and large time deposits. The spread on large time deposits is zero because they were not subject to Reg Q, while the spreads on savings and small time deposits are shown in Panel A of Figure OA.1.²⁸ The overall Reg Q spread averages these spreads, weighting by their shares, and is shown in Panel B of Figure OA.1. It rises sharply during each of the four credit crunches, from 0% to a peak of 3.5%.

We incorporate the Reg Q spread in panel regressions of the form

$$y_{i,t} = \alpha_t + \gamma_i + \beta \text{RegQSpread}_t \times \text{FinDep}_i + \delta X_{i,t} + \epsilon_{i,t}. \quad (27)$$

The controls $X_{i,t}$ include the fixed characteristics energy intensity (energy costs over output), productivity (TFP), and the volatility of prices and output. We measure them pre-1965 and interact them with the Reg Q spread as we do finance dependence. We also control for contemporaneous wage growth (production payroll divided by hours). Year fixed effects α_t absorb aggregate economic conditions such as inflation expectations, while

²⁸Large time deposits were exempted in June 1970 but their ceilings had been close to the Fed funds rate. Eurodollar deposits, which are also a form of large time deposits, were never subject to Reg Q. The spread on savings deposits is the Fed funds rate minus the ceiling on passbook savings accounts until the deregulated Money Market Deposit Accounts (MMDAs) were introduced in December 1982, and zero after. The spread on small time deposits is the six-month TBill rate minus the ceiling on six-month CDs until the deregulated Money Market Certificates (MMCs) were introduced in July 1978, and zero after.

industry fixed effects γ_i absorb industry trends. The price regressions also control for materials price growth. In addition to prices and output, we examine employment, inventories, and investment as outcome variables $y_{i,t}$. We run the regressions during the Reg Q period, from 1965 to 1982. We cluster standard errors at the industry level.

The coefficient of interest, β , tests the prediction of Proposition 3 that raising interest rates leads finance dependent industries to raise prices and cut output relative to other industries when Reg Q is binding. The controls help rule out other negative supply shocks. Most prominent are the oil shocks of 1973 and 1979 (Blinder and Rudd, 2013), which overlap with two of the Reg Q credit crunches. We capture oil shock exposure with energy intensity and interact it with the Reg Q spread to allow for arbitrary correlation between credit crunches and the oil shocks. Controlling for TFP (interacted with the Reg Q spread) helps rule out that finance dependent industries are more sensitive because they are less productive. The volatility of prices and output rule out that their prices are more flexible or their output is more cyclical in general. Controlling for wages and materials prices ensures that our estimates are over and above changes in input costs. The identifying assumption is that finance dependence is uncorrelated with exposure to any remaining unobserved negative supply shocks that coincide with the Reg Q credit crunches.

4.2.5 Prices

Table 3 shows the regression results for prices. Each column includes a different set of controls, with the last one including all the controls. We include materials prices in all columns because we want to know if output prices rise *relative* to inputs.

Column (1) shows a positive and significant coefficient on the interaction of finance dependence and the Reg Q spread. This means that finance dependent industries raise prices relative to non-finance dependent industries when the Reg Q spread widens. The result is consistent Proposition (3) and the mechanism of the model. It therefore supports the hypothesis that the Reg Q credit crunches led firms to raise prices.

In terms of magnitudes, the coefficient implies that a 1% increase in the Fed funds rate when Reg Q is binding leads industries with a finance dependence of one (no internal resources) to raise prices by 1.85% relative to industries with a finance dependence of zero (can finance production from one year's worth of profits). Since the Reg Q spread rose by 3–4% during each credit crunch, this is an economically large effect.²⁹

²⁹Through the lens of the model (Proposition (3)), the coefficient identifies the product of working capital

Column (2) controls for energy intensity and TFP. Energy intensity has no impact, consistent with the finding in Table 1 that energy expenditures are quite small for these firms (2% on average). The oil shocks are thus unlikely to explain our results. TFP enters with a positive sign, implying that less productive industries cut prices relative to more productive ones during the credit crunches. The coefficient on finance dependence is unaffected. Column (3) controls for the volatility of prices and output. Price volatility has a positive impact, while output volatility has no impact. The coefficient on finance dependence drops slightly but remains similar. Our results are thus robust to controlling for price stickiness and output volatility. Column (4) controls for wage growth, which ensures that firms are not simply passing through higher labor costs. The coefficient on finance dependence returns to the value in column (1). Finally, column (5) adds all controls simultaneously. None of the coefficients change much. The coefficient on finance dependence remains large and significant at 1.637. The impact of finance dependence on prices during the credit crunches is thus robust to all the controls and consistent with the predictions of the model.

4.2.6 Output

Table 4 shows the results for output. From column (1), finance dependent industries have lower output growth than other industries when the Reg Q spread is high. The coefficient is -3.760 and highly significant, indicating an economically large effect.³⁰

The opposite signs of the coefficients for prices and output show that finance dependence predicts shifts in supply across industries (prices and quantities move in opposite directions). The supply shifts are negative when the Reg Q spread is high, hence finance dependence predicts stagflation during the credit crunches. This again supports Proposition 3 and the mechanism of the model.

Column (2) controls for energy intensity and productivity. Both are insignificant and have almost no impact on the coefficient on finance dependence (-3.715). This helps to rule out the oil shocks. Column (3) controls for the volatility of prices and output, which are also insignificant and leave the coefficient on finance dependence at -3.703 . Column (4) controls for wages and materials prices. These have large and significant coefficients,

intensity α and the deposit demand elasticity $1/\delta$. Using $\alpha = 0.5$, we get $1/\delta = 1.85/0.5 = 3.7$. This elasticity is consistent with the observed sensitivity of deposits to the Reg Q spread in Figure 2.

³⁰From Proposition 3, the ratio of the coefficients in Tables 4 and 3 reveals ϵ , the elasticity of substitution across industries. Our numbers imply $\epsilon = 3.760/1.850 = 2.032$, a moderate level of substitutability.

as one would expect, but again almost no impact on finance dependence (-3.670). Finally, column (5) includes all the controls. The coefficient on finance dependence settles at -3.694 and remains highly significant.³¹

Taken together, Tables 3 and 4 show that finance dependence explains stagflation in the cross section of industries during the credit crunches. This result supports Proposition 3 and serves as a test of the mechanism of the model.

4.2.7 Employment, inventories, and investment

We turn to employment next. Table 5 Panel A runs panel regressions as in (27) but with employment growth as the dependent variable. From Column (1), the coefficient on finance dependence is negative and significant at -1.978 . Hence, finance dependent industries cut employment by about 2% relative to non-finance dependent industries when the Reg Q spread is high. This is smaller than the impact on output but still substantial. Columns (2) to (5) show that the impact of finance dependence on employment is robust to controlling for energy intensity, productivity, volatility, wage growth, and materials prices. Broadly, the results for employment look similar to those for output in Table 4, showing that output and employment go hand in hand.

Panel B of Table 5 runs the same regressions with inventory growth as the dependent variable. The coefficient on finance dependence is negative and highly significant at -5.738 . Thus, finance dependent industries see a decline in inventories when the Reg Q spread is high. The result is robust to the controls we add in columns (2) to (5). The magnitude of the effect is larger than that for output, consistent with working capital being disproportionately affected.

The decline in inventories is further evidence of a negative supply shock. Even though finance dependent industries charge higher prices, their inventories get depleted. This points to a disruption in their ability to produce. Inventories are themselves a factor of

³¹We explore additional robustness in the Online Appendix. In Online Appendix I.1, we re-estimate Tables 3 and 4 using lagged time-varying controls (instead of pre-period averages) and their interactions with the Reg Q spread. We also include the lagged dependent variable as a control. The results are qualitatively and quantitatively similar. In Online Appendix I.2, we analyze gross profits (revenues minus operating costs) as the outcome variable. A possible alternative interpretation of finance dependence is that it is an inverse proxy for market power. Theory predicts that firms with high market power reduce markups in response to an input cost shock (De Loecker, Eeckhout and Unger, 2020). We find that, if anything, the relationship goes the other way, which is consistent with our model. In Online Appendix I.3, we examine demand-side forces. We proxy for industry-level variation in demand elasticities using industry-level output growth during a large federal funds rate increase in the pre-period (1958–1960). Including this proxy, interacted with the Reg Q spread, leaves our results effectively unchanged.

production (Ramey, 1989), and since they require financing, a natural explanation for the production shock is a lack of credit.³²

Panel C of Table 5 looks at investment. While investment is not a focus of the paper, we expect finance dependent firms to also cut investment during the credit crunches. This is indeed what we find: from column (1), the coefficient on finance dependence is negative and significant.

The large coefficient implies that investment is more sensitive than output. At the same time, firms spend much less on investment than production. As a result, the output effects we uncover are much larger than the investment effects in dollar terms. In our data, investment is on average 3% of output; hence the implied dollar decline of output is about twenty times larger than that of investment ($= 3.760/6.124 \times 1/0.03$). Thus, although the credit crunches decrease investment demand, the main effect we uncover in the cross section is a decrease in supply.³³

4.3 Working capital intensity

Proposition 3 shows that exposure to the Reg Q credit crunches is increasing in working capital intensity α in addition to finance dependence φ . This means that we can also test the model using variation in working capital intensity. We do so in Online Appendix G, where we provide a full set of results. We summarize them here.

We measure working capital intensity analogously to finance dependence as the ratio of working capital (using inventories as a proxy) to production costs. We do so pre-Reg Q just like finance dependence. Intuitively, working capital intensity measures the length of an industry's production cycle. The longer the production cycle, the more working capital financing an industry needs. Note, however, that industries with a long production cycle are likely to experience the effects of a credit crunch with a lag. We account for this by leading the outcome variables (price growth and output growth) by one year. We keep the same sample and controls as in Tables 3 and 4.

We find that when the Reg Q spread widens, industries with high working capital intensity raise prices and shrink output over the following year relative to industries with a

³²Panel B provides further evidence that our results are not driven by market power. If low finance dependence firms have market power leading them to reduce markups, we would expect their inventories to shrink by more. However, we find the opposite.

³³Reg Q can have other demand effects that our cross section is not designed to uncover. See our discussion of these effects in Section 3.6.

low working capital intensity in the following year. The coefficients are economically and statistically significant and robust across specifications. This result provides independent support for Proposition 3 and the mechanism of the model.

Proposition 3 further predicts that working capital intensity and finance dependence should complement each other. That is, exposure to the credit crunches should be increasing in their interaction. This is indeed what we find. The interaction of working capital intensity and finance dependence predicts higher prices and lower output during the Reg Q credit crunches. The two measures thus go hand in hand, as predicted.

Finally, we study the relationship between finance dependence and working capital intensity. We find that the two are negatively related. In other words, industries with higher working capital needs tend to have more internal resources to finance those needs. As a result, we find that finance dependence is even more strongly correlated with leverage (using QFR data) holding working capital intensity fixed. Working capital intensity itself is only weakly related to leverage. This result motivates us to focus on finance dependence as our main measure.

4.4 The Reg Q spread across banks, counties, and industries

The third key parameter in the model (Proposition 3) is the elasticity of deposit demand δ , which determines the severity of the Reg Q credit crunches. We do not observe δ directly but we do observe other sources of variation that play the same role. In particular, Reg Q imposed different ceilings on different types of deposits, and banks differ in their deposit mix. This makes the severity of the credit crunches vary at the bank level. We measure it with a bank-level version of the Reg Q spread.

We construct the bank-level Reg Q spread similarly to the aggregate spread in Section 4.2 as a weighted average of the spreads on different types of deposits using quarterly bank-level data. The key difference is that we weight by each bank's own deposit shares instead of the aggregate ones:

$$\text{Bank RegQSpread}_{b,t} = \sum_d w_{b,t}^d \text{RegQSpread}_t^d, \quad (28)$$

where $w_{b,t}^d$ is bank b 's share of deposits of type d at the start of period t , and RegQSpread_t^d is their Reg Q spread in period t . The deposit types d are savings deposits (other than MMDAs), small time deposits (other than MMCs), large time deposits, MMDAs, and

MMCs.³⁴ The Reg Q spreads for non-MMDA savings and non-MMC small time deposits are taken against their Reg Q ceilings (shown in Appendix Figure OA.1). The spreads on the deregulated products, MMDAs, MMCs, and large time deposits, are zero.

Next, we aggregate the bank-level Reg Q spread to the county level. We weight each bank by its lending share:

$$\text{County RegQSpread}_{c,t} = \sum_b w_{b,c,t} \text{Bank RegQSpread}_{b,t}, \quad (29)$$

where $w_{b,c,t}$ is the C&I loan share of bank b at the start of period t in the county c .³⁵ A county's RegQ spread captures the extent to which Reg Q binds for banks lending to firms in county c . Since county data is annual, we compute the annual county RegQSpread by taking the average of quarterly RegQSpread over the year.

Finally, we construct the industry-level Reg Q by mapping the county-level Reg Q spread to industries by using the locations of an industry's establishments (from the Census' County Business Patterns dataset). We calculate each county's share of an industry's total employment and average it over time so our results are not driven by industry relocations. We then calculate the industry-level Reg Q spread as

$$\text{Industry RegQSpread}_{i,t} = \sum_c w_{i,c} \text{County RegQSpread}_{c,t}, \quad (30)$$

where $w_{i,c}$ is county c 's share of industry i 's employment. An industry's Reg Q spread captures the extent to which Reg Q binds for banks that lend in the areas where the industry is concentrated. It implicitly assumes that firms obtain bank funding locally, a common assumption in the literature (e.g., Petersen and Rajan, 2002). This assumption is plausible in our sample because it predates banking deregulation in the 1980s (Kroszner and Strahan, 2014).

The model (Proposition 3) predicts that industries with a higher Reg Q spread (higher

³⁴Our analysis focuses on deposits because they make up 97% of average bank liabilities. Our measure of bank-level Reg Q spread uses more detail on banks' deposit mix than the aggregate Reg Q spread due to the richness of the call reports. MMCs and MMDAs are the two main types of deposits exempt from Reg Q. MMCs are six-month Money Market Certificates, a type of small time deposit introduced in July 1978. MDDAs are Money Market Deposit Accounts, a type of savings account introduced in December 1982. We first observe large time deposits in 1974, hence we use this date to back-fill their share for each bank. We exclude demand deposits because their rates were zero both before and after Reg Q.

³⁵In Online Appendix I.4, we also construct the Reg Q spread using pre-determined C&I weights and average C&I weights over the analysis period. The results are qualitatively and quantitatively unchanged.

$1/\delta$) should experience higher price growth and lower output growth than other industries. The industry Reg Q spread thus gives us an alternative way to test the main mechanism of the model. The advantage is that it uses variation coming from the deposit side of bank balance sheets as opposed to from firm balance sheets. It is therefore unlikely that this strategy picks up unobserved industry-level supply shocks correlated with firm balance sheets.

4.4.1 RegQ spread and the severity of credit crunches

We validate the Reg Q spread as a measure of the severity of credit crunches by relating it to bank- and county-level outcomes. We expect banks with a higher Reg Q spread to experience larger deposit outflows and lending contractions. For this to affect firms, it must also aggregate to the county level. We therefore also check if counties with a higher Reg Q spread experience contractions in lending and, as a real effect, employment.

We start with bank-level panel regressions of the form

$$y_{b,t} = \alpha_b + \gamma_t + \beta \text{Bank RegQSpread}_{b,t} + \delta X_{b,t} + \epsilon_{b,t}, \quad (31)$$

where $y_{b,t}$ is bank b 's quarterly real deposit growth or C&I lending growth (winsorized at 5%), $\text{RegQSpread}_{b,t}$ is its Reg Q spread over the same period, $X_{b,t}$ are lagged bank characteristics (log of assets, equity to assets, deposits to liabilities, C&I loans to assets, and return on assets), and α_b and γ_t are bank and time fixed effects. The sample includes all banks with non-missing controls during the Reg Q period from 1965 to 1982.³⁶ Standard errors are clustered at the bank level.

Panel A of Table 6 reports the results. Column 1 shows that a 1% increase in the Reg Q spread leads to a 1.9% reduction in deposits, which is statistically significant. The effect rises to 2.8% when we add the controls in column 2. Columns 3 and 4 show similar effects for C&I lending, which falls by 1.6% and 3.2%, respectively. This shows that banks for whom Reg Q binds more strongly lose deposits and are unable to replace them with alternative sources of funding.

For county-level outcomes, we use the County Business Patterns dataset, which is

³⁶There are 15,658 banks in the sample. The average Reg Q spread is 0.02 with a standard deviation of 0.02. Average annualized deposit and C&I lending growth rates are 0.10 and 0.11, respectively. Average assets total \$30 million (average log assets of 9.61). The average bank is financed through equity and deposits, with an equity ratio of 8% and deposits representing 97% of liabilities.

reported yearly in March and goes back to 1971. We restrict the sample to counties with at least 100 manufacturing workers (2,530 out of 3,152 counties). Our outcome variables are the log growth rates of manufacturing employment, total employment, and total payrolls. To connect lending to county outcomes, we aggregate deposits and C&I lending to the county level and match the March reporting date.

We run the following panel regression:

$$y_{c,t} = \alpha_c + \gamma_t + \beta \text{County Reg Q Spread}_{c,t} + \epsilon_{c,t}, \quad (32)$$

where $y_{c,t}$ is manufacturing employment growth, total employment growth, or real payroll growth; $\text{Reg Q Spread}_{c,t}$ is the county-level Reg Q spread; and α_c and γ_t are county and time fixed effects. Standard errors are clustered at the county level.

Panel B of Table 6 presents the results for deposits and lending. Columns (1) and (3) report specifications without county fixed effects and show that a 1% increase in the county-level Reg Q spread shrinks deposits by 2.9% and C&I lending by 3.6%. These magnitudes are similar to the bank-level ones. In specifications with county fixed effects (columns 2 and 4), the coefficients remain significant but are somewhat smaller: 1.7% and 2.7%, respectively. Since C&I loans are typically fixed-term, this is an economically large effect: with an average maturity of 3 years, banks would need to cut new lending by 8.1-10.8% for every 1% increase in the Reg Q spread.

Panel C of Table 6 presents results for county-level real outcomes. In the specification without county fixed effects, we find a statistically significant coefficient of -1.045 for manufacturing employment. The effects on total employment (-0.710) and payrolls (-1.311) are of similar magnitude. With county fixed effects, the coefficients decline to -0.805 for manufacturing employment, -0.513 for total employment, and -0.737 for payroll. All remain statistically significant. These findings suggest that differences in the severity of credit crunches (as captured by the Reg Q spread) aggregate to the county level and affect real outcomes.

4.4.2 The industry Reg Q spread, prices and output

We now test model's prediction (Proposition 3) that industries experiencing more severe credit crunches raise prices and shrink output relative to other industries. We do so by regressing price and output growth on the industry Reg Q spread in panel regressions of

the form

$$y_{i,t} = \alpha_t + \gamma_i + \beta \text{Industry RegQSpread}_{i,t} + \delta X_{i,t} + \epsilon_{i,t}, \quad (33)$$

where $y_{i,t}$ is either price growth or output growth of industry i in year t .³⁷ The time fixed effects α_t absorb aggregate economic conditions and the industry fixed effects γ_i absorb fixed industry characteristics. We use the same set of controls $X_{i,t}$ as in Eq. (27). This includes interacting fixed industry characteristics (energy intensity, TFP, volatility of prices and output) with the average Industry Reg Q spread to rule out other industry supply shocks that might have coincided with the credit crunches. We also use the same sample period (1965 to 1982) and industry clustering.³⁸

The coefficient of interest in Eq. (33) is β . It captures the extent to which industries whose local lenders are experiencing more severe credit crunches raise prices or cut output relative to industries whose lenders are less constrained. The identifying assumption is that these industries are not differentially affected by unobserved supply shocks that coincide with increases in their Reg Q spread.

Table 7 presents the results for prices. From column (1), the coefficient on the Reg Q spread is positive and significant at 1.567. This means that a 1% increase in the Reg Q spread leads to about 1.6% higher price growth, an economically large impact. This result supports Proposition (3) and the mechanism of the model.

To compare the estimates in Tables 3 and 7, we have to take into account that Eq. (33) does not separate industries by finance dependence, hence it holds for its average level. From Table 1, this average is 0.5, so we have to multiply the coefficient in Table 7 by 2 to compare it to Table 3. This gives 3.13, which is somewhat larger but within a standard error. The results in Tables 3 and 7 are therefore broadly consistent.

Turning to column (2), energy intensity and productivity (TFP) have no explanatory power, nor do they affect the coefficient on the Reg Q spread. In column (3), the volatilities of prices and output have insignificant positive coefficients and also do not impact the main coefficient. The same is true for wage growth in column (4). Finally, column (5)

³⁷The industry Reg Q spread varies both in the time series and the cross section, hence it plays the role of the interaction of the treatment and exposure variables in Equation (27).

³⁸A potential concern with this clustering is that it might not account for correlated error terms arising from a “shift-share” setting (recall our measure is constructed using employment shares). We address this by grouping industries based on their Reg Q spread, which has the most potential to affect our coefficient of interest. Online Appendix I.5 reports the results and shows that our findings remain statistically significant under this alternative clustering approach.

includes all controls. The coefficient on the Reg Q spread is unchanged at 1.610 and remains significant. The results in Table 7 thus show that the Reg Q spread has a robust positive impact on price growth, consistent with the model.

Table 8 shows the results for output. From column 1, the coefficient on the Reg Q spread is -4.184 and significant. Thus, industries facing a higher Reg Q spread shrink output relative to other industries. Combined with Table 7, the industry Reg Q spread predicts high prices and low output, implying a negative supply shock.

The estimated coefficient on the industry Reg Q spread is economically large: a 1% increase leads to a 4.2% decline in output. Multiplying by 2 to compare to Table 4, the coefficient is again somewhat larger but not significantly so. As with finance dependence, the negative impact of the Reg Q spread on output is larger than the positive impact on prices, hence revenues decline.

Column 2 controls for energy intensity and productivity. The coefficient on energy intensity is positive and insignificant, hence exposure to the energy shocks does not explain the decline in output (recall the shocks hit in 1973 and 1979 when the aggregate Reg Q spread was high). Productivity has a positive impact on output but does not affect the coefficient on the Reg Q spread. Neither does controlling for the volatility of prices and output in column 3 or wages and materials prices in column 4. Finally, column 5 includes all controls. The coefficient on the Reg Q spread is -4.768 and remains significant. The negative impact of the Reg Q spread on output is thus robust to all the controls.³⁹

Overall, the industry Reg Q spread predicts negative supply shifts (stagflation) across industries similar to finance dependence in Section 4.2. This supports the mechanism of the model (Proposition 3), where finance dependence (φ) and the severity of credit crunches (δ) play a key role.

4.4.3 The Industry Reg Q spread and finance dependence

Our final test combines finance dependence and the industry Reg Q spread. Finance dependence measures exposure to the credit crunches and the Reg Q spread measures their severity. The two are therefore complementary. This is easily seen in Proposition 3, where the sensitivity of prices to interest rates depends on the interaction of finance

³⁹In Online Appendix I.1, we re-estimate Tables 7 and 8 using lagged time-varying controls (instead of pre-period averages) and their interactions with the Reg Q spread. We also include the lag of the dependent variable as a control. The results are qualitatively and quantitatively the same, which provides additional robustness for our results.

dependence (φ) and credit crunch severity ($1/\delta$). We test this prediction by running panel regressions of the form

$$y_{i,t} = \alpha_t + \gamma_i + \beta \text{Industry RegQSpread}_{i,t} + \gamma \text{FinDep}_i \times \text{Industry RegQSpread}_{i,t} + \delta X_{i,t} + \epsilon_{i,t}. \quad (34)$$

The coefficient of interest is γ . It measures the additional impact of the Reg Q spread for industries with high versus low finance dependence. All other terms, the sample period, and clustering of standard errors are the same as in our earlier tests.

The identifying assumption in Eq. (34) is that there are no unobserved negative supply shocks affecting finance dependent industries with a high Reg Q spread more than ones with a low Reg Q spread. Violating this assumption requires a negative supply shock that is correlated with (i) finance dependence across industries, and (ii) banks' deposit mix across geographical areas. The shock must also coincide with the credit crunches in the time series. Since this is a highly specific set of conditions, the identifying assumption in Eq. (34) is weaker than in Eqs. (27) and (33), giving us a stronger test of the model.

The results for prices and output are presented in Tables 9 and 10, respectively. From Table 9 column (1), the estimated coefficient on the interaction between the Reg Q spread and finance dependence is 0.758 and highly significant. The coefficient changes little as we add controls in columns (2) to (4). It settles at 0.628 and remains significant conditional on all controls in column (5). The result is thus robust to the controls. This shows that the effect of the Reg Q spread on prices is increasing in finance dependence.

Turning to output in Table 10, column (1) shows a large negative coefficient on the interaction between the Reg Q spread and finance dependence. The point estimate is -2.972 and highly significant. The result is again stable and robust as we add in controls, settling in at -3.257 in the most saturated specification in column (5). The impact of the Reg Q spread on output is thus increasing in finance dependence, as predicted.

Taken together, the results in Tables 9 and 10 show that credit crunch exposure (finance dependence) and severity (the Reg Q spread) reinforce each other in explaining prices and output. This provides further evidence for the model's prediction that the Reg Q credit crunches led to stagflation (higher prices and lower output) across industries.

4.5 Deposit flows post Reg Q

We test whether the impact of deposit outflows on prices and output persists after the Great Stagflation. Even though there is no Reg Q, deposits remain sensitive to changes in the Fed funds rate, consistent with the deposits channel of monetary policy (Drechsler, Savov and Schnabl, 2017). However, we expect smaller effects due to three key differences: First, the Fed funds rate itself is less volatile in the post-Reg Q period, which reduces the volatility of deposits. Second, the passthrough of interest rates to deposit rates (the deposit “beta”) is higher, which reduces the sensitivity of deposits to the Fed funds rate. And third, banks set deposit rates optimally, taking loan demand into account, which reduces the impact on lending. All three differences affect the impact of the deposits channel on inflation and output. The first two affect only the aggregate impact, while the third also affects the cross-sectional one. We therefore estimate the first two in the time series and the third in the cross section and then combine them.⁴⁰

Panel A of Table 11 compares the variation in the Fed funds rate during the Reg Q period 1965–1982 and the post-Reg Q period 1985–2018.⁴¹ The interquartile range of one-year fed funds rate changes is 3.29% during Reg Q versus 1.72% post Reg Q. The standard deviations are 2.49% and 1.31%, respectively. Variation in the Fed funds range is thus about half as large post Reg Q.

Next we examine deposit spreads during Reg Q and post Reg Q. There are no deposit rate ceilings after Reg Q, so we cannot use the Reg Q spread from the original regressions. We instead use the deposit spread from Drechsler, Savov and Schnabl (2017), which can be consistently measured over the entire period.⁴²

Panel B of Table 11 calculates the passthrough of the Fed funds rate to the deposit spread during Reg Q and post Reg Q by regressing deposit spread changes on Fed funds rate changes at a yearly frequency. During Reg Q, the deposit spread rises by 84 bps per 100 bps increase in the Fed funds rate. This is close to one-for-one as expected given deposit rates were bound by rate ceilings.⁴³ Post Reg Q, deposit spreads rise by 63 bps

⁴⁰We extend the model to account for these differences in Appendix D.

⁴¹We leave out 1983–1984 because Reg Q effectively ended with the introduction of MMDAs in December 1982. This led to a surge in deposits in 1983 that creates base effects in 1984 growth rates. We end in 2018 with the NBER-CES Manufacturing Industry Database.

⁴²Specifically, we use the FDIC’s Historical Bank Data, which provides aggregate deposits and deposit interest expense since 1934. We calculate a deposit rate by dividing deposit interest expense by deposits. We then calculate a deposit spread by subtracting the deposit rate from the fed funds rate.

⁴³It is slightly less than one-for-one because, as explained in Section 4.4, large time deposits did not have a ceiling after 1970.

per 100, i.e. their sensitivity declines by about a quarter.

Finally, we use the cross section to see if the impact of the deposits channel post Reg Q is dampened as we expect due to endogenous deposit betas. Panel C of Table 11 repeats our main cross-sectional analysis from Table 3 for prices (columns 1-4) and Table 4 for output (columns 5-8). The only difference is that we replace the Reg Q spread with the deposit spread to make the two periods comparable. Columns 1, 2, 5, and 6 show that our results for Reg Q are robust to this change: this is indeed the case. The magnitude of the coefficients for prices (0.472 without controls and 0.412 with controls) and output (-1.480 without controls and -1.599 with controls) is somewhat smaller than using the Reg Q spread but robust and highly significant.

Columns 3, 4, 7, and 8 of Panel C look at the post Reg Q sample. As in columns for the Reg Q period, the treatment variable is the deposit spread (since the Reg Q spread is not available). We have also updated the finance dependence measure and controls to the start of the post Reg Q sample.

We find remarkably similar cross-sectional results post Reg Q and during Reg Q. The coefficients for prices, 0.334 without controls and 0.339 with controls, are 30% and 20% smaller than the corresponding coefficients during Reg Q. The coefficients for output, -1.620 without controls and -1.355 with controls, are 9% larger and 15% smaller. All are robust and significant. Thus, we find slightly smaller cross-sectional effects post Reg Q, consistent with our framework.

Overall, we find that the deposits channel continues to affect firm prices and output post Reg Q. The cross sectional effects are smaller, consistent with endogenous betas. Deposit spreads are also less sensitive to the Fed funds rate, and the Fed funds rate itself is less volatile. Chaining them together, the variation in inflation and output induced by the deposits channel is between three and four times smaller than under Reg Q.⁴⁴

4.6 Additional discussion

The Online Appendix provides in-depth discussions of several questions that are not covered in the main text due to space constraints, including the pervasiveness of credit controls in other countries during this time period, the role of money market funds as a

⁴⁴Other important changes in the financial system are also likely to affect this estimate. Firms have become less reliant on banks due to the growth of nonbank intermediaries and the bond market. At the same time, banks are the dominant investor in mortgages, potentially opening a real estate channel.

substitute for deposits, the role of corporate bonds and commercial paper as substitutes for bank loans, and a comparison with the 2008 Global Financial Crisis.

5 Conclusion

The Great Stagflation brought about a sea change in economic policy and thinking. Its central feature was a negative relation between inflation and output. To explain it, the literature evokes a series of exogenous adverse supply shocks, e.g. oil. Because these shocks are exogenous, there is little to be done about them. This led to the view that monetary policy should take supply as given and focus on demand.

We argue that the Great Stagflation was due, in part, to severe credit crunches triggered by the banking law known as Regulation Q. Reg Q placed a ceiling on deposit rates that became binding whenever the Fed raised rates. This led to large outflows of deposits, triggering a series of credit crunches. The credit crunches made it hard for firms to finance working capital, raising production costs and resulting in higher prices and lower output. We formalize this mechanism in an otherwise standard model and find consistent evidence in aggregate and cross-sectional tests.

Our findings imply that the adverse supply shocks that hit the economy in the 1970s were not just due to “bad luck”. They were partly due to the Fed’s own efforts to reduce inflation under Reg Q. This does not mean that the Fed should not have raised rates. Rather, it helps explain why the Fed found its job so difficult. Fed chair Arthur Burns lamented this in “The Anguish of Central Banking” (Burns, 1979), a speech much derided as an exercise in excuse-making. But perhaps some sympathy is due, as Burns was thwarted by a backfiring financial system. The lesson for us is to fix it; to ensure that raising rates does not choke off credit to firms. Only then would monetary policy transmit solely through demand—as conventional wisdom would have it—and the supply side can safely be taken as given.

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Table 1: Industry summary statistics

Summary statistics at the industry level. Finance dependence is one minus industry i 's gross profit (sales minus production costs) divided by production costs (materials and labor), averaged over 1958 to 1965 and winsorized at the 5% level. Employment is in thousands. Output is sales plus the change in inventories in millions of 1965 dollars. TFP is five-factor total factor productivity. The volatilities of output and prices are the standard deviations of real output growth and the growth of the shipments deflator from 1958 to 1965. The labor share is production labor costs divided by output. The materials share and energy intensity are calculated analogously. The investment rate is real investment over real capital. The bottom rows are averages of yearly growth rates during the Reg Q period from 1965 to 1982. The data are from the NBER-CES manufacturing database, which covers 459 manufacturing industries at the four-digit SIC level.

	Finance dependence					
	All		Low		High	
	Mean	St. Dev.	Mean	St. Dev.	Mean	St. Dev.
Finance dependence	0.50	(0.20)	0.34	(0.15)	0.66	(0.08)
<i>Levels, 1958–1965:</i>						
Employment	35.75	(50.61)	32.15	(41.43)	39.33	(58.21)
Output (1965 \$)	946.03	(1,854.81)	754.79	(923.41)	1,136.44	(2,441.05)
Capital stock	1,234.94	(3,806.29)	1,063.79	(1,765.33)	1,405.35	(5,080.80)
TFP	0.90	(0.28)	0.90	(0.29)	0.90	(0.28)
$\sigma(\text{Prices})$	2.25	(1.55)	1.89	(1.15)	2.62	(1.79)
$\sigma(\text{Output})$	9.96	(5.78)	9.48	(5.65)	10.44	(5.88)
Labor share	0.17	(0.07)	0.18	(0.07)	0.17	(0.07)
Materials share	0.50	(0.13)	0.42	(0.08)	0.58	(0.11)
Energy intensity	0.02	(0.02)	0.02	(0.02)	0.01	(0.01)
Inventory/Output	0.16	(0.08)	0.17	(0.08)	0.15	(0.08)
Capital/Output	0.39	(0.27)	0.46	(0.31)	0.31	(0.21)
Investment rate	0.08	(0.04)	0.08	(0.03)	0.08	(0.04)
<i>Growth rates, 1965–1982:</i>						
Δ Prices	5.83	(2.22)	5.83	(2.04)	5.84	(2.39)
Δ Materials prices	6.53	(1.48)	6.77	(1.32)	6.30	(1.59)
Δ Output	3.24	(4.25)	3.72	(4.37)	2.75	(4.08)
Δ Employment	0.92	(2.86)	1.32	(2.92)	0.52	(2.74)
Δ Inventories	4.34	(6.16)	4.87	(4.47)	3.83	(7.44)
Δ Capital	4.11	(2.73)	4.23	(2.73)	3.99	(2.73)
# Industries	459		229		230	

Table 2: Finance dependence and balance sheet characteristics

This table uses the Quarterly Financial Reports (QFR) to relate finance dependence to financial characteristics. The QFR reports are available at the two-digit SIC level starting in 1947. We calculate finance dependence in the QFR data in the same way as in the NBER-CES dataset (and over the same period from 1958 to 1965). The QFR measure of costs includes SG&A (selling, general, and administrative expenses), which makes the measure higher on average. The balance sheet characteristics are measured in 1965. Leverage is the ratio of debt over equity. The short-term share of debt is short-term debt over debt. The bank share of debt is bank debt over debt. The cash ratio is cash and securities over current liabilities. The debt service ratio is operating income divided by debt due in one year. Panel A shows average and standard deviations for all sectors and sectors with below- and above-median finance dependence. Panel B regresses each balance sheet characteristic on finance dependence.

Panel A:						
	All		Finance dependence			
			Low		High	
	Mean	St. Dev.	Mean	St. Dev.	Mean	St. Dev.
Finance dependence	0.91	(0.04)	0.88	(0.02)	0.94	(0.02)
Leverage	0.31	(0.09)	0.26	(0.06)	0.37	(0.08)
Short-term share of debt	0.23	(0.12)	0.18	(0.11)	0.28	(0.11)
Bank share of debt	0.38	(0.11)	0.31	(0.10)	0.45	(0.09)
Cash ratio	0.36	(0.13)	0.41	(0.15)	0.30	(0.09)
Debt service ratio	3.39	(2.03)	4.76	(1.85)	1.87	(0.67)
# Sectors	19		10		9	

Panel B:					
	Leverage	Short-term share of debt	Bank share of debt	Cash ratio	Debt service ratio
	(1)	(2)	(3)	(4)	(5)
Fin. dep.	1.450*** (0.460)	1.625** (0.726)	2.008*** (0.598)	−1.753** (0.790)	−41.440*** (9.362)
Constant	−1.012** (0.419)	−1.255* (0.662)	−1.451** (0.546)	1.957** (0.721)	41.169*** (8.541)
Obs.	19	19	19	19	19
R ²	0.369	0.228	0.399	0.225	0.535

Table 3: Finance dependence and prices

Panel regressions of prices on finance dependence and the Reg Q spread:

$$\Delta\text{Prices}_{i,t} = \alpha_t + \gamma_i + \beta\text{RegQSpread}_t \times \text{FinDep}_i + X_{i,t} + \epsilon_{i,t}.$$

Price growth is the percentage change of industry i 's shipments deflator in year t . Finance dependence is one minus industry i 's gross profit (sales minus production costs) divided by production costs (materials and labor), averaged over 1958 to 1965 and winsorized at the 5% level. The Reg Q spread is the weighted average spread between market rates and the Reg Q ceiling rates on savings, small time, and large time deposits (using their shares as weights). Energy intensity is energy costs divided by output. TFP is total five-factor productivity in 1965. The standard deviations of price growth and output growth are from 1958 to 1965 (output is shipments plus the change in inventories deflated by the shipments deflator). Wage growth is the percentage change of hourly production wages (production wages divided by hours worked). Materials price growth is the percentage change of industry i 's cost of materials deflator. Standard errors are clustered by industry. The data are from the NBER-CES manufacturing database, which covers 459 manufacturing industries. The sample is the Reg Q period from 1965 to 1982.

	ΔPrices				
	(1)	(2)	(3)	(4)	(5)
Reg Q spread $\times$ Fin. dep.	1.850*** (0.367)	1.991*** (0.377)	1.473*** (0.344)	1.855*** (0.366)	1.637*** (0.357)
Reg Q spread $\times$ Energy int.		0.105** (0.047)			0.100** (0.045)
Reg Q spread $\times$ TFP		0.595* (0.325)			0.497 (0.335)
Reg Q spread $\times \sigma(\Delta\text{Prices})$			15.893** (6.599)		14.454** (6.539)
Reg Q spread $\times \sigma(\Delta\text{Output})$			0.047 (0.808)		0.394 (0.799)
ΔWage				0.019 (0.015)	0.018 (0.015)
$\Delta\text{Materials price}$	0.855*** (0.069)	0.852*** (0.069)	0.848*** (0.067)	0.855*** (0.069)	0.846*** (0.067)
Time FE	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes
Obs.	8,262	8,262	8,262	8,262	8,262
R^2	0.587	0.588	0.588	0.587	0.589

Table 4: Finance dependence and output

Panel regressions of output on finance dependence and the Reg Q spread:

$$\Delta\text{Output}_{i,t} = \alpha_t + \gamma_i + \beta\text{RegQSpread}_t \times \text{FinDep}_i + X_{i,t} + \epsilon_{i,t}.$$

Output growth is the percentage change of industry i 's output (shipments plus the change in inventories deflated by the shipments deflator) in year t . Finance dependence is one minus industry i 's gross profit (sales minus production costs) divided by production costs (materials and labor), averaged over 1958 to 1965 and winsorized at the 5% level. The Reg Q spread is the weighted average spread between market rates and the Reg Q ceiling rates on savings, small time, and large time deposits (using their shares as weights). Energy intensity is energy costs divided by output. TFP is total five-factor productivity in 1965. The standard deviations of price growth and real output growth are from 1958 to 1965. Wage growth is the percentage change of hourly production wages (production wages divided by hours worked). Materials price growth is the percentage change of industry i 's cost of materials deflator. Standard errors are clustered by industry. The data are from the NBER-CES manufacturing database, which covers 459 manufacturing industries. The sample is the Reg Q period from 1965 to 1982.

	ΔOutput				
	(1)	(2)	(3)	(4)	(5)
Reg Q spread $\times$ Fin. dep.	-3.760*** (0.817)	-3.715*** (0.817)	-3.703*** (0.853)	-3.670*** (0.838)	-3.694*** (0.871)
Reg Q spread $\times$ Energy int.		0.065 (0.094)			0.108 (0.093)
Reg Q spread $\times$ TFP		1.446 (0.884)			1.453* (0.848)
Reg Q spread $\times \sigma(\Delta\text{Prices})$			0.226 (10.326)		8.523 (10.020)
Reg Q spread $\times \sigma(\Delta\text{Output})$			-1.886 (3.126)		-1.955 (3.062)
ΔWage				0.145*** (0.043)	0.147*** (0.043)
$\Delta\text{Materials price}$				-0.290*** (0.048)	-0.298*** (0.047)
Time FE	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes
Obs.	8,262	8,262	8,262	8,262	8,262
R^2	0.249	0.249	0.249	0.260	0.261

Table 5: Finance dependence, employment, inventory, and investment

Panel regressions of employment on finance dependence and the Reg Q spread:

$$\Delta Y_{i,t} = \alpha_t + \gamma_i + \beta \text{RegQSpread}_t \times \text{FinDep}_i + X_{i,t} + \epsilon_{i,t}.$$

where $\Delta Y_{i,t}$ is the percentage change of industry i 's total employment, inventory, or investment in year t . Finance dependence is one minus industry i 's gross profit (sales minus production costs) divided by production costs (materials and labor), averaged over 1958 to 1965 and winsorized at the 5% level. The Reg Q spread is the weighted average spread between market rates and the Reg Q ceiling rates on savings, small time, and large time deposits (using their shares as weights). Control variables for each column are hidden but the same as in Tables 3 and 4. Standard errors are clustered by industry. The data are from the NBER-CES manufacturing database, which covers 459 manufacturing industries. The sample is the Reg Q period from 1965 to 1982.

Panel A: Δ Employment

	Δ Employment				
	(1)	(2)	(3)	(4)	(5)
Reg Q spread $\times$ Fin. dep.	-1.978*** (0.580)	-1.890*** (0.575)	-2.029*** (0.596)	-2.045*** (0.578)	-1.996*** (0.587)
Time FE	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes
Obs.	8,262	8,262	8,262	8,262	8,262
R^2	0.235	0.236	0.235	0.244	0.244

Panel B: Δ Inventory

	Δ Inventory				
	(1)	(2)	(3)	(4)	(5)
Reg Q spread $\times$ Fin. dep.	-5.738*** (1.111)	-5.794*** (1.139)	-4.791*** (1.045)	-5.680*** (1.102)	-4.870*** (1.030)
Time FE	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes
Obs.	8,262	8,262	8,262	8,262	8,262
R^2	0.105	0.105	0.106	0.110	0.111

Panel C: Δ Investment

	Δ Investment				
	(1)	(2)	(3)	(4)	(5)
Reg Q spread $\times$ Fin. dep.	-6.124** (2.445)	-5.729** (2.501)	-6.289** (2.573)	-6.062** (2.438)	-5.696** (2.642)
Time FE	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes
Obs.	8,262	8,262	8,262	8,262	8,262
R^2	0.088	0.089	0.088	0.089	0.089

Table 6: Reg Q spread, bank lending, and county-level outcomes

Panel A shows results from bank-level regressions of deposit growth and C&I loan growth on bank Reg Q spread. Deposit growth and C&I loan growth are annualized quarterly growth rates of non-demand deposits and C&I loans, respectively. Bank Reg Q spread is constructed by averaging aggregate Reg-Q spreads across deposit types using bank-level deposit-type shares. The control variables are log assets, equity to assets, deposits to liabilities, C&I loans to assets, and return on assets, all lagged by one quarter. Standard errors are clustered at the bank level. The data are bank call reports and the sample is from 1965q1 to 1982q4. Panels B and C show results from county-level regressions of non-demand deposit growth and C&I loan growth on county Reg Q spread, and of growth in manufacturing employment, total employment, and total payroll on county Reg Q spread, respectively. County Reg Q spread is constructed by aggregating bank Reg Q spreads weighted by each bank's share of C&I loans in the county. Standard errors are clustered at the county level. The data are from bank call reports (Panel B) and County Business Patterns (Panel C). The sample covers counties with at least 100 employees in manufacturing from 1971 to 1982.

Panel A: Bank-Level Deposit Growth and Lending

	Δ Deposits		Δ C&I Loans	
	(1)	(2)	(3)	(4)
Bank Reg Q Spread	-1.866*** (0.059)	-2.830*** (0.062)	-1.638*** (0.129)	-3.186*** (0.143)
Quarter FE	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes
Obs.	947,230	947,230	947,230	947,230
R ²	0.184	0.224	0.036	0.070
Controls	No	Yes	No	Yes

Panel B: County-Level Deposit Growth and Lending

	Δ Deposits		Δ C&I Loans	
	(1)	(2)	(3)	(4)
County Reg Q Spread	-2.939*** (0.139)	-1.726*** (0.162)	-3.643*** (0.243)	-2.740*** (0.349)
Year FE	Yes	Yes	Yes	Yes
County FE	No	Yes	No	Yes
Obs.	22,685	22,685	22,685	22,685
R ²	0.262	0.451	0.053	0.172

Panel C: County-level Real Outcomes

	Δ Emp. Manuf.		Δ Emp.		Δ Payrolls	
	(1)	(2)	(3)	(4)	(5)	(6)
County Reg Q Spread	-1.045*** (0.220)	-0.805** (0.336)	-0.710*** (0.106)	-0.513*** (0.138)	-1.311*** (0.136)	-0.737*** (0.171)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
County FE	No	Yes	No	Yes	No	Yes
Obs.	22,685	22,685	22,685	22,685	22,685	22,685
R ²	0.257	0.307	0.353	0.431	0.138	0.240

Table 7: Industry Reg Q spread and prices

Panel regressions of prices on the industry Reg Q spread:

$$\Delta \text{Prices}_{i,t} = \alpha_t + \gamma_i + \beta \text{Industry Reg Q Spread}_{i,t} + X_{i,t} + \epsilon_{i,t}.$$

Price growth is the percentage change of industry i 's shipments deflator in year t . The industry Reg Q spread is the average Reg Q spread of banks in the counties where an industry operates (we weight banks by C&I loans within a county and counties by industry employment within an industry). A bank's Reg Q spread is the average spread on savings, small time and large time deposits using the bank's shares of each type as weights. Ind. Reg Q spread averages the spread across counties using total manufacturing employment as weights. Energy intensity is energy costs divided by output. TFP is total five-factor productivity in 1965. The standard deviations of price growth and real output growth are from 1958 to 1965. Wage growth is the percentage change of hourly production wages (production wages divided by hours worked). Materials price growth is the percentage change of industry i 's cost of materials deflator. Standard errors are clustered by industry. The data are from the NBER-CES manufacturing database, which covers 459 manufacturing industries. The sample is the Reg Q period from 1965 to 1982.

	ΔPrices				
	(1)	(2)	(3)	(4)	(5)
Industry Reg Q spread	1.567** (0.719)	1.610** (0.708)	1.589** (0.726)	1.552** (0.718)	1.610** (0.711)
Ind. Reg Q spread $\times$ Energy int.		-0.019 (0.029)			-0.015 (0.028)
Ind. Reg Q spread $\times$ TFP		0.267 (0.255)			0.274 (0.260)
Ind. Reg Q spread $\times \sigma(\Delta \text{Prices})$			5.182 (4.913)		4.879 (4.903)
Ind. Reg Q spread $\times \sigma(\text{Output})$			1.102* (0.663)		1.145* (0.665)
ΔWage				0.023 (0.018)	0.023 (0.018)
$\Delta \text{Materials price}$	0.853*** (0.076)	0.853*** (0.076)	0.851*** (0.075)	0.853*** (0.076)	0.851*** (0.075)
Time FE	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes
Obs.	6,477	6,477	6,477	6,477	6,477
R^2	0.580	0.580	0.580	0.580	0.581

Table 8: Industry Reg Q spread and output

Panel regressions of output on the industry Reg Q spread:

$$\Delta\text{Output}_{i,t} = \alpha_t + \gamma_i + \beta\text{Industry RegQSpread}_{i,t} + X_{i,t} + \epsilon_{i,t}.$$

Output growth is the percentage change of industry i 's output (shipments plus the change in inventories deflated by the shipments deflator) in year t . The industry Reg Q spread is the average Reg Q spread of banks in the counties where an industry operates (we weight banks by C&I loans within a county and counties by industry employment within an industry). A bank's Reg Q spread is the average spread on savings, small time and large time deposits using the bank's shares of each type as weights. Ind. Reg Q spread averages the spread across counties using total manufacturing employment as weights. Energy intensity is energy costs divided by output. TFP is total five-factor productivity in 1965. The standard deviations of price growth and real output growth are from 1958 to 1965. Wage growth is the percentage change of hourly production wages (production wages divided by hours worked). Materials price growth is the percentage change of industry i 's cost of materials deflator. Standard errors are clustered by industry. The data are from the NBER-CES manufacturing database, which covers 459 manufacturing industries. The sample is the Reg Q period from 1965 to 1982.

	ΔOutput				
	(1)	(2)	(3)	(4)	(5)
Industry Reg Q spread	-4.184** (1.656)	-4.373*** (1.562)	-4.370*** (1.624)	-4.303*** (1.656)	-4.768*** (1.531)
Ind. Reg Q spread $\times$ Energy int.		0.035 (0.060)			0.061 (0.059)
Ind. Reg Q spread $\times$ TFP		1.263** (0.616)			1.148* (0.606)
Ind. Reg Q spread $\times \sigma(\Delta\text{Prices})$			2.064 (6.464)		4.988 (6.241)
Ind. Reg Q spread $\times \sigma(\text{Output})$			-3.410* (1.909)		-3.318* (1.896)
ΔWage				0.125** (0.049)	0.126** (0.049)
$\Delta\text{Materials price}$				-0.258*** (0.051)	-0.264*** (0.051)
Time FE	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes
Obs.	6,477	6,477	6,477	6,477	6,477
R^2	0.226	0.227	0.227	0.236	0.238

Table 9: Industry Reg Q spread, finance dependence, and prices

Panel regressions of prices on the industry Reg Q spread and finance dependence:

$$\Delta \text{Prices}_{i,t} = \alpha_t + \gamma_i + \beta \text{Industry Reg Q Spread}_{i,t} \times \text{FinDep}_i + X_{i,t} + \epsilon_{i,t}.$$

Price growth is the percentage change of industry i 's shipments deflator in year t . Finance dependence is one minus industry i 's gross profit (sales minus production costs) divided by production costs (materials and labor), averaged over 1958 to 1965 and winsorized at the 5% level. The industry Reg Q spread is the average Reg Q spread of banks in the counties where an industry operates (we weight banks by C&I loans within a county and counties by industry employment within an industry). A bank's Reg Q spread is the average spread on savings, small time and large time deposits using the bank's shares of each type as weights. $\overline{\text{Ind. Reg Q spread}}$ averages the spread across counties using total manufacturing employment as weights. Energy intensity, TFP, and the standard deviations of prices and real output growth are measured as of 1965. Wage growth is the percentage change of hourly production wages (production wages divided by hours worked). Materials price growth is the percentage change of industry i 's cost of materials deflator. Standard errors are clustered by industry. The data are from the NBER-CES manufacturing database, which covers 459 manufacturing industries. The sample is the Reg Q period from 1965 to 1982.

	ΔPrices				
	(1)	(2)	(3)	(4)	(5)
Ind. Reg Q spread $\times$ Fin. dep.	0.758*** (0.282)	0.745** (0.291)	0.624** (0.255)	0.768*** (0.281)	0.628** (0.271)
Ind. Reg Q spread	1.130 (0.745)	1.128 (0.743)	1.234* (0.739)	1.109 (0.745)	1.210 (0.734)
$\overline{\text{Ind. Reg Q spread}} \times \text{Energy int.}$		-0.002 (0.031)			-0.001 (0.031)
$\overline{\text{Ind. Reg Q spread}} \times \text{TFP}$		0.253 (0.254)			0.268 (0.258)
$\overline{\text{Ind. Reg Q spread}} \times \sigma(\Delta \text{Prices})$			3.688 (4.939)		3.382 (4.954)
$\overline{\text{Ind. Reg Q spread}} \times \sigma(\text{Output})$			0.996 (0.658)		1.056 (0.660)
ΔWage				0.025 (0.018)	0.024 (0.018)
$\Delta \text{Materials price}$	0.853*** (0.076)	0.853*** (0.076)	0.852*** (0.075)	0.853*** (0.076)	0.852*** (0.075)
Time FE	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	53	Yes	Yes	Yes
Obs.	6,477	6,477	6,477	6,477	6,477
R^2	0.580	0.580	0.581	0.580	0.581

Table 10: Industry Reg Q spread, finance dependence, and output

Panel regressions of output on the industry Reg Q spread and finance dependence:

$$\Delta \text{Output}_{i,t} = \alpha_t + \gamma_i + \beta \text{Industry Reg Q Spread}_{i,t} \times \text{FinDep}_i + X_{i,t} + \epsilon_{i,t}.$$

Output growth is the percentage change of industry i 's real output in year t . Finance dependence is one minus industry i 's gross profit (sales minus production costs) divided by production costs (materials and labor), averaged over 1958 to 1965 and winsorized at the 5% level. The industry Reg Q spread is the average Reg Q spread of banks in the counties where an industry operates (we weight banks by C&I loans within a county and counties by industry employment within an industry). A bank's Reg Q spread is the average spread on savings, small time and large time deposits using the bank's shares of each type as weights. $\overline{\text{Ind. Reg Q spread}}$ averages the spread across counties using total manufacturing employment as weights. Energy intensity, TFP, and the standard deviations of prices and real output growth are measured as of 1965. Wage growth is the percentage change of hourly production wages (production wages divided by hours worked). Materials price growth is the percentage change of industry i 's cost of materials deflator. Standard errors are clustered by industry. The data are from the NBER-CES manufacturing database, which covers 459 manufacturing industries. The sample is the Reg Q period from 1965 to 1982.

	ΔOutput				
	(1)	(2)	(3)	(4)	(5)
Ind. Reg Q spread $\times$ Fin. dep.	-2.972*** (0.662)	-3.085*** (0.658)	-3.098*** (0.697)	-2.982*** (0.666)	-3.257*** (0.696)
Ind. Reg Q spread	-2.472 (1.649)	-2.379 (1.558)	-2.606 (1.622)	-2.582 (1.663)	-2.689* (1.552)
$\overline{\text{Ind. Reg Q spread}} \times \text{Energy int.}$		-0.035 (0.062)			-0.011 (0.060)
$\overline{\text{Ind. Reg Q spread}} \times \text{TFP}$		1.319** (0.599)			1.178** (0.594)
$\overline{\text{Ind. Reg Q spread}} \times \sigma(\Delta \text{Prices})$			9.401 (6.726)		12.755** (6.449)
$\overline{\text{Ind. Reg Q spread}} \times \sigma(\text{Output})$			-2.877 (1.821)		-2.852 (1.805)
ΔWage				0.120** (0.049)	0.120** (0.049)
$\Delta \text{Materials price}$				-0.260*** (0.050)	-0.267*** (0.050)
Time FE	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	54 Yes	Yes	Yes	Yes
Obs.	6,477	6,477	6,477	6,477	6,477
R^2	0.229	0.230	0.230	0.239	0.241

Table 11: Post Reg Q

Panel A presents the standard deviation and interquartile range of the Fed funds rate during Reg Q (1965-1982) and post Reg Q (1985-2018). Panel B displays coefficients from regressing the yearly change in the deposit spread on the yearly change in the Fed funds rate: $\Delta \text{Deposit spread}_t = \alpha + \beta \Delta \text{Fed funds}_t + \epsilon_t$ during Reg Q and post Reg Q. Panel C presents results from panel regressions of prices or output on finance dependence and the deposits spread, following the specifications in Tables 3 and 4. The deposit spread is calculated by subtracting from the Fed funds rate the deposits interest rate, which is interest expense on domestic deposits divided by total domestic deposits, using data from FDIC's Annual Historical Bank Data.

Panel A: Variation in the Fed funds rate

	Reg Q	Post Reg Q
SD	2.490	1.314
IQR	3.285	1.724

Panel B: Fed funds rate passthrough to the deposit spread

	Reg Q	Post Reg Q
	(1)	(2)
Δ Fed funds rate	0.841*** (0.038)	0.633*** (0.032)
Constant	-0.002*** (0.001)	0.001* (0.000)
Obs.	18	34
R^2	0.965	0.919

Panel C: Deposit spread, finance dependence, prices and output

	Δ Prices				Δ Output			
	Reg Q		Post Reg Q		Reg Q		Post Reg Q	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dep. spr. $\times$ Fin. dep.	0.472*** (0.152)	0.412*** (0.150)	0.334** (0.145)	0.339** (0.134)	-1.480*** (0.343)	-1.599*** (0.367)	-1.620*** (0.403)	-1.355*** (0.406)
Obs.	8,262	8,262	15,447	15,440	8,262	8,262	15,447	15,440
Time, Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	No	Yes	No	Yes	No	Yes	No	Yes

Figure 1: The Great Stagflation

The figure shows the Fed funds rate, inflation, and real GDP growth (ΔGDP). Gray shading covers the Great Stagflation using conventional dating (e.g., Bryan, 2013). Inflation and ΔGDP are year-over-year. Inflation is based on the Consumer Price Index (CPI). The data are monthly from 1962 to 1986 except for GDP, which is quarterly.

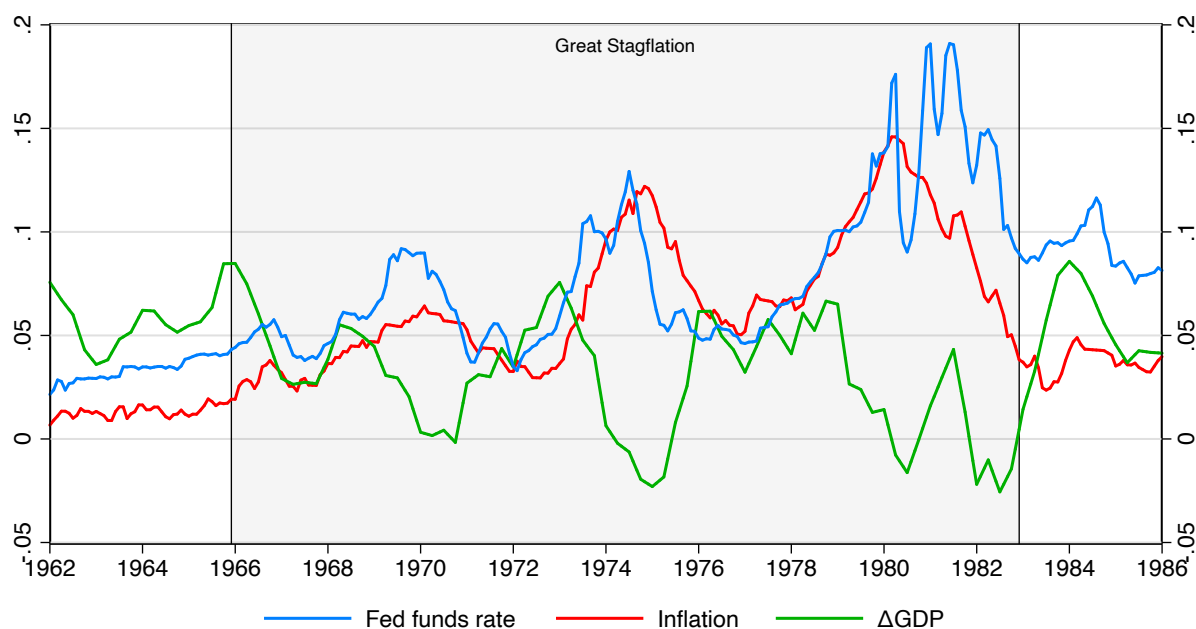


Figure 2: The Reg Q Credit Crunches

Panel A shows the Reg Q ceiling rate on savings deposits, which becomes the rate on MMDAs (a type of deregulated savings account) in December 1982. Also shown is deposit growth (savings and small time deposits). Gray shading covers the period when Reg Q is binding: from December 1965 when the ceiling first binds to December 1982 when MMDAs are introduced. Panel B shows the growth in bank credit (loans and securities), from the Federal Reserve's H.8 release. Deposits and bank credit are deflated by the CPI. The data are monthly from 1962 to 1986.

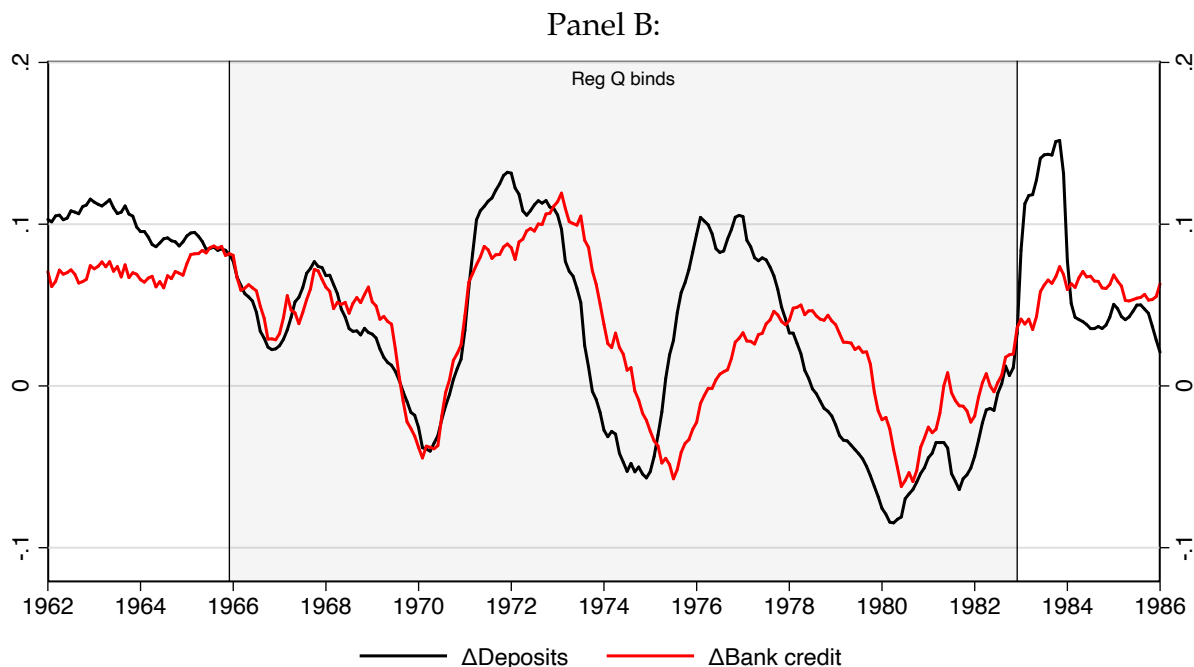
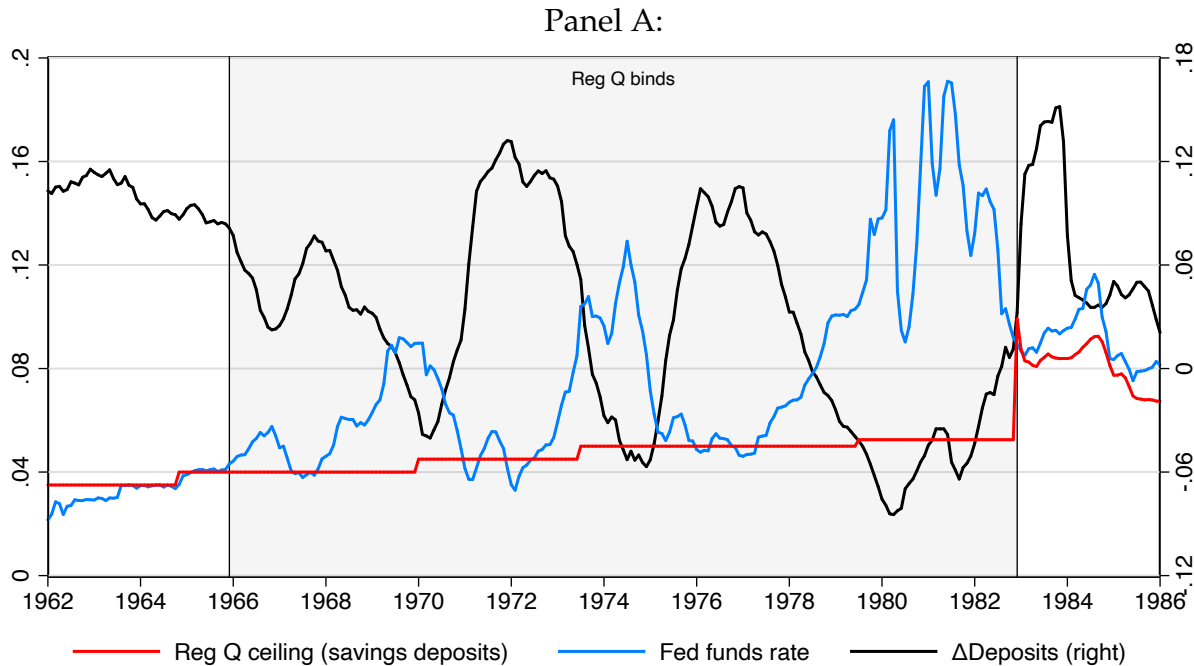


Figure 3: Credit Crunches and Financial Conditions

Panel A plots real deposit growth (savings and small time deposits) against the spread between the Fed funds rate and the three-month Treasury Bill rate, a measure of the liquidity premium in treasury bill prices. The data are monthly from 1962 to 1986. Panel B plots real deposit growth against the Chicago Fed Adjusted National Financial Conditions Index. The index measures the tightness of financial conditions across U.S. debt, equity, and loan markets. The adjusted version controls for economic conditions. The data are monthly from 1971 to 2021. Gray shading covers the period when Reg Q is binding.

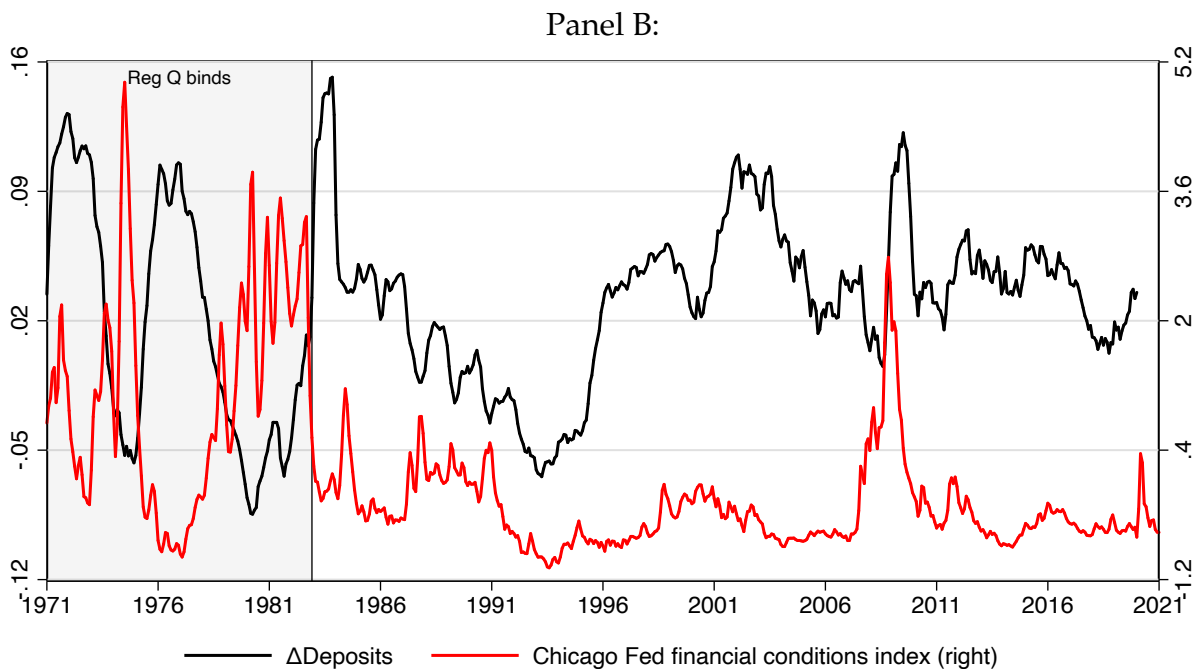
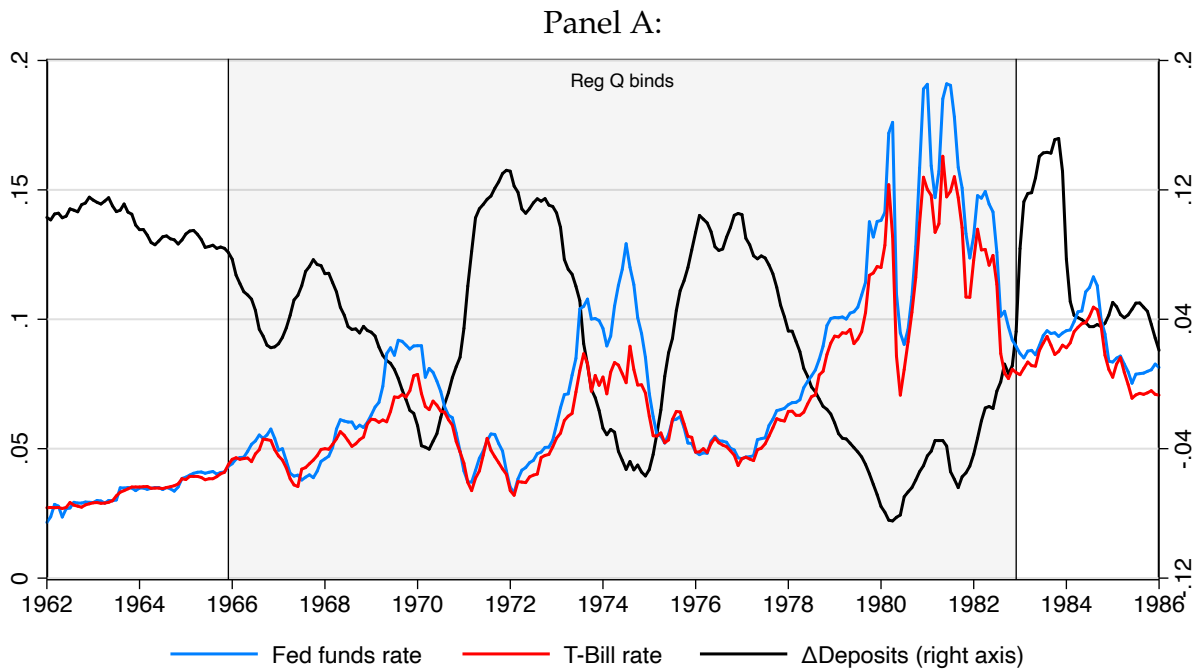


Figure 4: Credit Crunches and the Great Stagflation

The figure plots real deposit growth (savings and small time deposits), inflation, and GDP growth. Inflation and GDP growth are the year-over-year percentage changes in the Consumer Price Index (CPI) and real GDP, respectively. Gray shading covers the period when Reg Q is binding. The data are monthly from 1962 to 1986.

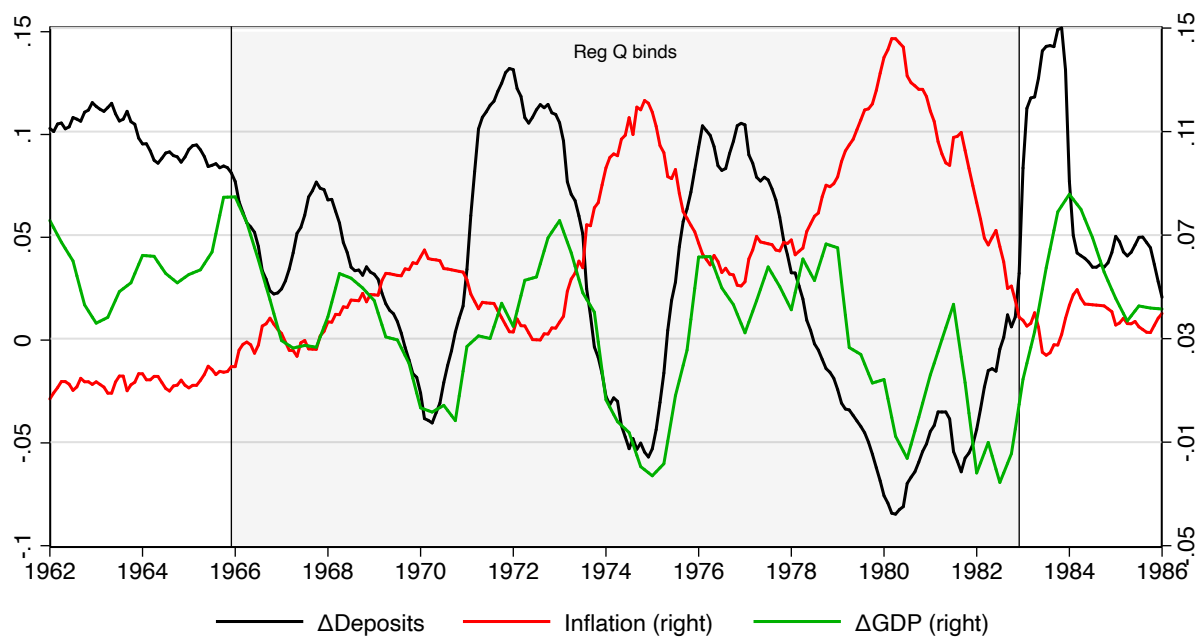


Figure 5: Impulse response from shock to credit standards

The figure plots impulse responses to an orthogonalized shock to credit standards based on an estimation of a vector autoregression model (VAR) based on Gilchrist and Zakrajsek (2012) but replacing the excess bond premium with credit standard tightening from the Fed's Senior Loan Office Survey. The responses of consumption, investment, output, and price growth and that of the excess market return have been accumulated. The vertical axis is in percentage point. Shaded bands are 95-percent confidence intervals based on 2,000 bootstrap replications. Binding Reg-Q period is 1965Q1-1982Q4 and post Reg-Q period is 1990Q2-2018Q4.

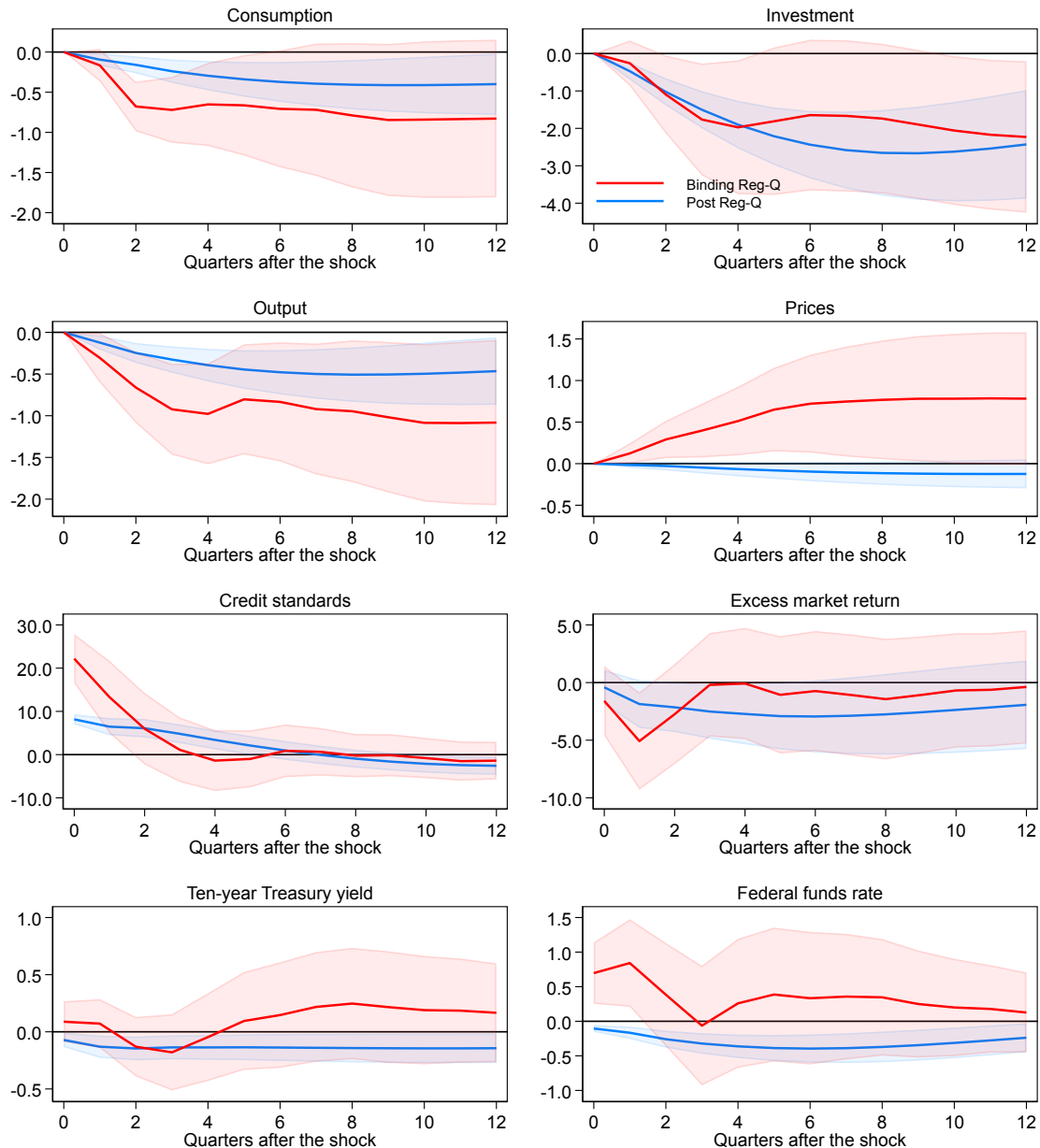
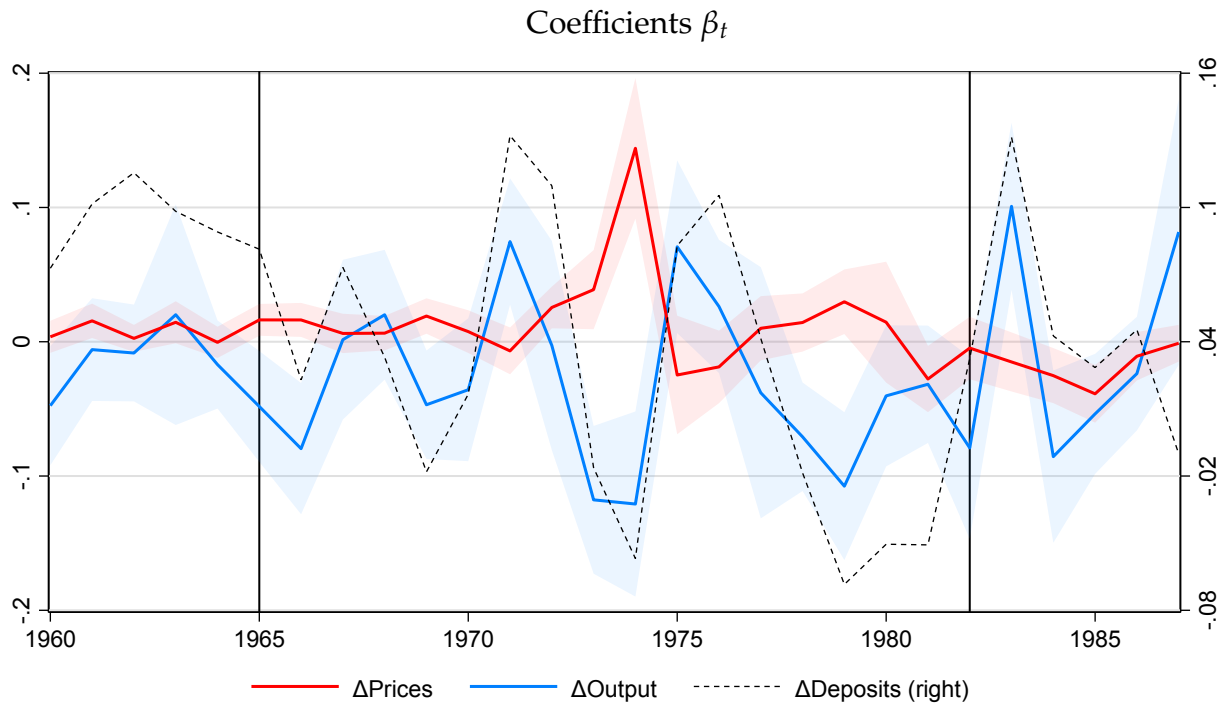


Figure 6: Finance dependence, prices, and output

The figure plots the coefficients β_t from yearly cross-sectional regressions of price growth (red) and output growth (blue) on finance dependence at the industry level:

$$y_{i,t} = \alpha_t + \beta_t \text{FinDep}_i + X_{i,t} + \epsilon_{i,t}.$$

Price growth is the percentage change of industry i 's shipments deflator in year t . Output growth is the percentage change of industry i 's real output (shipments plus the change in inventories deflated by the shipments deflator) in year t . Finance dependence is one minus industry i 's gross profit (sales minus production costs) divided by production costs (materials and labor), averaged over 1958 to 1965 and winsorized at the 5% level. The price regressions control for materials price growth, the percentage change of industry i 's cost of materials deflator. Shading denotes 95% confidence intervals. The dashed line is real deposit growth with values on the right axis. The data are from the NBER manufacturing database, which covers 459 manufacturing industries.



Appendix

A Model derivations

Derivation of (6): The household first-order condition with respect to labor L_t is:

$$L_t^\eta = \mathbf{C}_t^{-\gamma} \left(\frac{W_t}{\mathbf{P}_t} \right). \quad (\text{A.1})$$

Taking logs gives (6).

Derivation of (7): The household first-order condition with respect to bonds G_t is:

$$1 = \beta \left(\frac{\mathbf{C}_{t+1}}{\mathbf{C}_t} \right)^{-\gamma} \frac{\mathbf{P}_t}{\mathbf{P}_{t+1}} (1 + R_t). \quad (\text{A.2})$$

Taking logs gives (7).

Derivation of (8): The first-order condition with respect to deposits D_t is:

$$\phi'(D_t) = \mathbf{C}_t^{-\gamma} \left(1 - \frac{1 + R_t^D}{1 + R_t} \right). \quad (\text{A.3})$$

The deposit spread is approximated by

$$1 - \frac{1 + R_t^D}{1 + R_t} = 1 - e^{r_t^D - r_t} \approx r_t - r_t^D. \quad (\text{A.4})$$

Substituting into (A.3) gives (8).

Derivation of (11): The bank's objective in levels is:

$$\max \int_0^1 B_{i,t} \left[\frac{1 + R_t^B}{1 + R_t} - (1 + \bar{S}) \right] di + D_t \left(1 - \frac{1 + R_t^D}{1 + R_t} \right), \quad (\text{A.5})$$

where $\bar{S} = e^{\bar{s}} - 1$. From (A.4) above, the deposit spread is approximated by $r_t - r_t^D$, while the loan spread is approximated by

$$\frac{1 + R_t^B}{1 + R_t} - (1 + \bar{S}) = \left(e^{r_t^B - r_t} - 1 \right) - \left(e^{\bar{s}} - 1 \right) \approx r_t^B - r_t - \bar{s}. \quad (\text{A.6})$$

Plugging into (A.5) gives (11).

Proof of Proposition 1. Let $B_t = \int_0^1 B_{i,t} di$ be total loans at date t and let D_t^Q be the maximum amount of deposits banks can raise by paying the reg Q ceiling rate r^Q given r_t . From (11) and (13), the representative bank's Lagrangian is

$$\max_{B_t, D_t} B_t \left(r_t^B - r_t - \bar{s} \right) + D_t \left(r_t - r_t^D \right) + \mu_1 (D_t - B_t) + \mu_2 \left(D_t^Q - D_t \right), \quad (\text{A.7})$$

where μ_1 and μ_2 are the Lagrange multipliers on the funding and Reg Q constraints, respectively. The first-order conditions are

$$r_t^B - r_t = \bar{s} + \mu_1 \quad (\text{A.8})$$

$$r_t - r_t^D = \mu_2 - \mu_1. \quad (\text{A.9})$$

Since $\mu_1, \mu_2 \geq 0$ and $r_t - r_t^D \geq 0$ (the latter follows from (8) and $\phi' \geq 0$), if $\mu_2 = 0$ (Reg Q does not bind strictly), then by (A.9) $r_t - r_t^D = 0$ and $\mu_1 = 0$ (the deposit spread is zero and the funding constraint does not bind strictly). Then from (A.8), $r_t^B - r_t = \bar{s}$ (the loan spread equals the screening and monitoring cost).

If $\mu_2 > 0$ (Reg Q binds strictly), then $\mu_1 < \mu_2$ because if $\mu_1 = \mu_2$, then from (A.9) $r_t - r_t^D = 0$ and from (8) deposit demand is satiated ($\phi' = 0$). But if deposit demand is satiated, households treat deposits and government bonds as perfect substitutes and provide deposits elastically. This implies that the funding constraint cannot bind, $\mu_1 = 0$, which contradicts $\mu_1 = \mu_2$ since $\mu_2 > 0$.

Since $\mu_1 < \mu_2$, by (A.9) $r_t - r_t^D > 0$ and $D_t < \bar{D}$. Recall we assumed that loan demand weakly exceeds $\bar{D}$ at a loan spread of $\bar{s}$. This implies that the funding constraint binds

and $\mu_1 > 0$. From (A.8), $r_t^B - r_t > \bar{s}$.

Reg Q binds strictly if and only if $\mu_2 > 0$. We showed that if $\mu_2 > 0$, then $\mu_1 < \mu_2$, hence from (A.9) $r_t - r_t^D > 0$. Conversely, if $r_t - r_t^D > 0$, then from (A.9) $\mu_2 > 0$ and Reg Q binds strictly. $\square$

Derivation of (18): Firms choose their borrowing to minimize production costs for a given level of output according to (16). Their cost minimization problem is

$$\min_{B_{i,t}} \quad \frac{W_t}{\mathbf{P}_t} L_{i,t} + B_{i,t} \left(\frac{1 + R_t^B}{1 + R_t} - 1 \right) \quad (\text{A.10})$$

Production costs are the sum of labor costs (the wage bill) and borrowing costs (the loan spread). Internal financing (cash) does not enter because it is fixed or, more generally, because it earns a fair rate of return R_t , i.e. it is invested in government bonds.

Solving (16) and (17) for $L_{i,t}$ and substituting into (A.10),

$$\min_{B_{i,t}} \quad \frac{W_t}{\mathbf{P}_t} \left(\frac{Y_{i,t}}{A} \right)^{\frac{1}{1-\alpha}} C_i^{-(1-\varphi)\frac{\alpha}{1-\alpha}} B_{i,t}^{-\varphi\frac{\alpha}{1-\alpha}} + B_{i,t} \left(\frac{1 + R_t^B}{1 + R_t} - 1 \right). \quad (\text{A.11})$$

The first-order condition with respect to $B_{i,t}$ is

$$\varphi \frac{\alpha}{1-\alpha} \frac{W_t}{\mathbf{P}_t} \frac{L_{i,t}}{B_{i,t}} = \frac{1 + R_t^B}{1 + R_t} - 1. \quad (\text{A.12})$$

Substituting the loan spread in logs from (A.6) and re-arranging gives (18).

Derivation of (A.16): Utility maximization implies that household demand for good i depends on its relative price:

$$C_{i,t} = \left(\frac{P_{i,t}}{\mathbf{P}_t} \right)^{-\epsilon} \mathbf{C}_t. \quad (\text{A.13})$$

where $P_{i,t}$ is the price good i at t . The aggregate price index, which weighs individual goods' prices by their consumption shares, can be written as

$$\mathbf{P}_t = \int_0^1 P_{i,t} \left(\frac{C_{i,t}}{\mathbf{C}_t} \right) di = \left(\int_0^1 P_{i,t}^{1-\epsilon} di \right)^{\frac{1}{1-\epsilon}}. \quad (\text{A.14})$$

Initial prices are $P_{i,-1} = 1$ for all goods on date -1 . Then, starting on date 0 and on every date t after that, a fraction $1 - \theta$ of firms get to change their price while the remaining firms do not. Let $P_{i,t}^*$ be the optimal price chosen by a firm i that gets to change its price on date t . The remaining firms charge $\mathbf{P}_{t-1}$ on average (this ignores price dispersion, which is second-order). The price index (A.14) can therefore be written as

$$\mathbf{P}_t = \left[(1 - \theta) (P_{i,t}^*)^{1-\epsilon} + \theta \mathbf{P}_{t-1}^{1-\epsilon} \right]^{\frac{1}{1-\epsilon}}. \quad (\text{A.15})$$

Let $p_{i,t}^* = \log P_{i,t}^*$ and $\mathbf{p}_{t-1} = \log \mathbf{P}_{t-1}$. Log-linearizing around $p_{i,t}^* = \mathbf{p}_{t-1}$ gives

$$\pi_t = (1 - \theta) (p_{i,t}^* - \mathbf{p}_{t-1}). \quad (\text{A.16})$$

Inflation is high when many firms change their price by a large amount.

Derivation of (20): By the envelope theorem, the optimality condition for $P_{i,t}$ keeps $B_{i,t+\tau}$, $\tau \geq 0$ fixed. Therefore, we can drop the last term from the maximization problem. Substituting the production and working capital functions (16) and (17), and the demand function (A.13) into (19),

$$\max_{P_{i,t}} \sum_{\tau=0}^{\infty} \theta^{\tau} \Lambda_{t,t+\tau} \left[P_{i,t} \left(\frac{P_{i,t}}{\mathbf{P}_{t+\tau}} \right)^{-\epsilon} \mathbf{Y}_{t+\tau} - W_{t+\tau} B_{i,t}^{-\varphi \frac{\alpha}{1-\alpha}} \left(\frac{P_{i,t}}{\mathbf{P}_{t+\tau}} \right)^{-\frac{\epsilon}{1-\alpha}} \left(\frac{\mathbf{Y}_{t+\tau}}{\mathcal{A}} \right)^{\frac{1}{1-\alpha}} \right], \quad (\text{A.17})$$

where $\mathcal{A} = A\mathcal{C}^{\alpha(1-\varphi)}$ is the firm's cash-augmented productivity. Differentiating with respect to $P_{i,t}$ gives the first-order condition

$$0 = \sum_{\tau=0}^{\infty} \theta^{\tau} \Lambda_{t,t+\tau} \left[(1-\epsilon) \left(\frac{P_{i,t}}{\mathbf{P}_{t+\tau}} \right)^{-\epsilon} \mathbf{Y}_{t+\tau} + \frac{\epsilon}{1-\alpha} \frac{W_{t+\tau}}{\mathbf{P}_{t+\tau}} B_{i,t}^{-\varphi \frac{\alpha}{1-\alpha}} \left(\frac{P_{i,t}}{\mathbf{P}_{t+\tau}} \right)^{-\frac{\epsilon}{1-\alpha}-1} \left(\frac{\mathbf{Y}_{t+\tau}}{\mathcal{A}} \right)^{\frac{1}{1-\alpha}} \right]. \quad (\text{A.18})$$

Simplifying using (16) and (A.13) and re-arranging,

$$0 = \sum_{\tau=0}^{\infty} \theta^{\tau} \Lambda_{t,t+\tau} Y_{i,t+\tau} \left[\frac{1}{1-\alpha} \frac{\epsilon}{\epsilon-1} \frac{W_{t+\tau}}{\mathbf{P}_{t+\tau}} \frac{L_{i,t+\tau}}{Y_{i,t+\tau}} \left(\frac{P_{i,t}}{\mathbf{P}_{t+\tau}} \right)^{-1} - 1 \right]. \quad (\text{A.19})$$

Substituting for labor using (18),

$$0 = \sum_{\tau=0}^{\infty} \theta^{\tau} \Lambda_{t,t+\tau} Y_{i,t+\tau} \left[\frac{1}{\varphi \alpha} \frac{\epsilon}{\epsilon-1} (r_{t+\tau}^B - r_{t+\tau}) \frac{B_{i,t+\tau}}{Y_{i,t+\tau}} \left(\frac{P_{i,t}}{\mathbf{P}_{t+\tau}} \right)^{-1} - 1 \right]. \quad (\text{A.20})$$

Solving for $P_{i,t}$ gives

$$P_{i,t} = \frac{\epsilon}{\epsilon-1} \sum_{\tau=0}^{\infty} \omega_{i,t \rightarrow t+\tau} (\alpha \varphi)^{-1} (r_{t+\tau}^B - r_{t+\tau}) \frac{B_{i,t+\tau}}{Y_{i,t+\tau}} \mathbf{P}_{t+\tau}, \quad (\text{A.21})$$

where $\omega_{i,t \rightarrow t+\tau} = (\theta^{\tau} \Lambda_{t,t+\tau} Y_{i,t+\tau}) / \sum_{s=0}^{\infty} (\theta^s \Lambda_{t,t+s} Y_{i,t+s})$. From (16) and (17),

$$\frac{\partial Y_{i,t} / Y_{i,t}}{\partial B_{i,t} / B_{i,t}} = \alpha \varphi. \quad (\text{A.22})$$

Plugging into (A.21) and simplifying gives (20).

Note that from (A.2), the τ -period nominal discount factor starting on date t is

$$\Lambda_{t,t+\tau} = \beta^{\tau} \left(\frac{\mathbf{C}_{t+\tau}}{\mathbf{C}_t} \right)^{-\gamma} \frac{\mathbf{P}_t}{\mathbf{P}_{t+\tau}} = \frac{1}{\Pi_{s=0}^{\tau-1} (1 + R_{t+s})}. \quad (\text{A.23})$$

Proof of Proposition 2. We first evaluate (A.19) in steady state, where $P_{i,t} = \mathbf{P}_t$, $\mathbf{P}_{t+\tau} = \mathbf{P}_t$, and $\Lambda_{t,t+\tau} = \beta^{\tau}$:

$$1 = \frac{1}{1-\alpha} \frac{\epsilon}{\epsilon-1} \overline{W / \mathbf{P}} \frac{\overline{L}}{\overline{\mathbf{Y}}}. \quad (\text{A.24})$$

In logs,

$$0 = -\log(1 - \alpha) + \log \frac{\epsilon}{\epsilon - 1} + \overline{w - p} + \bar{l} - \bar{y}. \quad (\text{A.25})$$

Using the production function (16), labor supply (6) and loan demand (18) conditions,

$$\bar{y} = \log A + \alpha \varphi \bar{b} + (1 - \alpha) \bar{l} \quad (\text{A.26})$$

$$\overline{w - p} = \eta \bar{l} + \gamma \bar{y} \quad (\text{A.27})$$

$$\bar{b} - \bar{l} = \log \varphi \alpha - \log(1 - \alpha) + \overline{w - p} - \log \bar{s} \quad (\text{A.28})$$

Solving the system (A.25)–(A.28), steady-state output is

$$\bar{y} = \frac{\log A + \alpha \varphi \log \frac{\alpha \varphi}{\bar{s}} - \left(\alpha \varphi + \frac{1 - \alpha}{1 + \eta} \right) \log \frac{\epsilon}{\epsilon - 1} + \frac{1 - \alpha}{1 + \eta} \log(1 - \alpha)}{1 - \alpha \varphi + \frac{1 - \alpha}{1 + \eta} (\gamma - 1)}. \quad (\text{A.29})$$

Steady-state borrowing is then

$$\bar{b} = \frac{\log \frac{\alpha \varphi}{1 - \alpha} - \frac{1 + \eta}{1 - \alpha} \log A + \left(\gamma + \frac{1 + \eta}{1 - \alpha} \right) \bar{y} - \log \bar{s}}{1 + \alpha \varphi \frac{1 + \eta}{1 - \alpha}}. \quad (\text{A.30})$$

For loan demand to weakly exceed deposits when the loan spread is $\bar{s}$, we need $\bar{d} \leq \bar{b}$, with $\bar{b}$ as in (A.30). This is the condition needed to satisfy footnote 18.

A log-linear approximation of (A.19) around $p_{i,t} = \mathbf{p}_t$ gives

$$\begin{aligned} p_{i,t} - \mathbf{p}_t &= -\log(1 - \alpha) + \log \frac{\epsilon}{\epsilon - 1} \\ &+ (1 - \theta \beta) \sum_{\tau=0}^{\infty} \theta^{\tau} \beta^{\tau} [w_{t+\tau} - \mathbf{p}_{t+\tau} + l_{i,t+\tau} - y_{i,t+\tau} + \pi_{t,t+\tau}], \end{aligned} \quad (\text{A.31})$$

where $\pi_{t,t+\tau} = \mathbf{p}_{t+\tau} - \mathbf{p}_t$ is cumulative inflation from t to $t + \tau$. Using (A.25), we can

write this in terms of deviations from steady state as

$$p_{i,t} - \mathbf{p}_t = (1 - \theta\beta) \sum_{\tau=0}^{\infty} \theta^\tau \beta^\tau [w_{t+\tau} - \mathbf{p}_{t+\tau} - \overline{w - p} + l_{i,t+\tau} - y_{i,t+\tau} - \overline{l - y} + \pi_{t,t+\tau}]. \quad (\text{A.32})$$

Next, using the production function (16),

$$l_{i,t+\tau} - \bar{l} = \frac{1}{1 - \alpha} \left[(y_{i,t+\tau} - \bar{y}) - \alpha \varphi (b_{i,t+\tau} - \bar{b}) \right] \quad (\text{A.33})$$

$$l_{i,t} - y_{i,t} - \overline{l - y} = \frac{\alpha}{1 - \alpha} \left[(y_{i,t+\tau} - \bar{y}) - \varphi (b_{i,t+\tau} - \bar{b}) \right]. \quad (\text{A.34})$$

Plugging into (A.32),

$$p_{i,t} - \mathbf{p}_t = (1 - \theta\beta) \sum_{\tau=0}^{\infty} \theta^\tau \beta^\tau [w_{t+\tau} - \mathbf{p}_{t+\tau} - \overline{w - p} + \frac{\alpha}{1 - \alpha} \left[(y_{i,t+\tau} - \bar{y}) - \varphi (b_{i,t+\tau} - \bar{b}) \right] + \pi_{t,t+\tau}]. \quad (\text{A.35})$$

Also, plugging (A.33) applied to aggregate labor (this ignores dispersion across firms, which is second-order) into the labor supply condition (6),

$$w_{t+\tau} - \mathbf{p}_{t+\tau} - \overline{w - p} = \gamma (\mathbf{y}_{t+\tau} - \bar{y}) + \eta (l_{t+\tau} - \bar{l}) \quad (\text{A.36})$$

$$= \left(\gamma + \frac{\eta}{1 - \alpha} \right) (\mathbf{y}_{t+\tau} - \bar{y}) - \eta \frac{\alpha \varphi}{1 - \alpha} (b_{t+\tau} - \bar{b}), \quad (\text{A.37})$$

Plugging into (A.35),

$$p_{i,t} - \mathbf{p}_t = (1 - \theta\beta) \sum_{\tau=0}^{\infty} \theta^\tau \beta^\tau \left[\left(\gamma + \frac{\eta}{1 - \alpha} \right) (\mathbf{y}_{t+\tau} - \bar{y}) - \eta \frac{\alpha \varphi}{1 - \alpha} (b_{t+\tau} - \bar{b}) + \frac{\alpha}{1 - \alpha} \left[(y_{i,t+\tau} - \bar{y}) - \varphi (b_{i,t+\tau} - \bar{b}) \right] + \pi_{t,t+\tau} \right]. \quad (\text{A.38})$$

Using the production function (16), loan demand condition (18), and (A.37),

$$y_{i,t} - \bar{y} = \alpha\varphi(b_{i,t} - \bar{b}) + (1 - \alpha)(l_{i,t} - \bar{l}) \quad (\text{A.39})$$

$$b_{i,t} - l_{i,t} - \overline{b - l} = \left(\gamma + \frac{\eta}{1 - \alpha}\right)(\mathbf{y}_t - \bar{\mathbf{y}}) - \eta \frac{\alpha\varphi}{1 - \alpha}(b_t - \bar{b}) - \left[\log(r_t^B - r_t) - \log \bar{s}\right]. \quad (\text{A.40})$$

Aggregating across firms (this again ignores dispersion) and solving for the loan spread,

$$\log(r_t^B - r_t) - \log \bar{s} = \left(\gamma + \frac{1 + \eta}{1 - \alpha}\right)(\mathbf{y}_t - \bar{\mathbf{y}}) - \left(1 + \alpha\varphi \frac{1 + \eta}{1 - \alpha}\right)(b_t - \bar{b}). \quad (\text{A.41})$$

Solving for $b_t - \bar{b}$,

$$b_t - \bar{b} = \left(\frac{\gamma + \frac{1 + \eta}{1 - \alpha}}{1 + \alpha\varphi \frac{1 + \eta}{1 - \alpha}}\right)(\mathbf{y}_t - \bar{\mathbf{y}}) - \left(\frac{1}{1 + \alpha\varphi \frac{1 + \eta}{1 - \alpha}}\right)\log\left(\frac{r_t^B - r_t}{\bar{s}}\right). \quad (\text{A.42})$$

Using $\log x \approx 1 + x$ when x is small, this can be approximated as

$$b_t - \bar{b} = \left(\frac{\gamma + \frac{1 + \eta}{1 - \alpha}}{1 + \alpha\varphi \frac{1 + \eta}{1 - \alpha}}\right)(\mathbf{y}_t - \bar{\mathbf{y}}) - \left(\frac{1/\bar{s}}{1 + \alpha\varphi \frac{1 + \eta}{1 - \alpha}}\right)(r_t^B - r_t - \bar{s}). \quad (\text{A.43})$$

Plugging (A.41) into (A.39) and (A.40) and solving for $b_{i,t} - b_t$,

$$b_{i,t} - \bar{b} = (b_t - \bar{b}) + \frac{\frac{1}{1 - \alpha}}{1 + \frac{\alpha\varphi}{1 - \alpha}}[(y_{i,t} - \bar{\mathbf{y}}) - (\mathbf{y}_t - \bar{\mathbf{y}})]. \quad (\text{A.44})$$

This holds for all periods. Plugging into (A.38),

$$\begin{aligned} p_{i,t} - \mathbf{p}_t &= (1 - \theta\beta) \sum_{\tau=0}^{\infty} \theta^\tau \beta^\tau \left[\left(\gamma + \frac{1}{1 - \alpha} \left(\eta + \frac{\frac{\alpha\varphi}{1 - \alpha}}{1 + \frac{\alpha\varphi}{1 - \alpha}} \right) \right) (\mathbf{y}_{t+\tau} - \bar{\mathbf{y}}) \right. \\ &\quad \left. - \alpha\varphi \frac{1 + \eta}{1 - \alpha} (b_{t+\tau} - \bar{b}) + \frac{\alpha}{1 - \alpha} \left(\frac{1 - \varphi}{1 + \frac{\alpha\varphi}{1 - \alpha}} \right) (y_{i,t+\tau} - \bar{\mathbf{y}}) + \pi_{t,t+\tau} \right]. \end{aligned} \quad (\text{A.45})$$

Plugging in the demand curve (A.13) evaluated in logs and with $c_{i,t} = y_{i,t}$,

$$p_{i,t} - \mathbf{p}_t = (1 - \theta\beta) \sum_{\tau=0}^{\infty} \theta^\tau \beta^\tau \left[\left(\gamma + \frac{1}{1-\alpha} \left(\eta + \frac{\frac{\alpha\varphi}{1-\alpha}}{1 + \frac{\alpha\varphi}{1-\alpha}} \right) \right) (\mathbf{y}_{t+\tau} - \bar{\mathbf{y}}) \right. \quad (\text{A.46})$$

$$\left. - \alpha\varphi \frac{1+\eta}{1-\alpha} (b_{t+\tau} - \bar{b}) - \frac{\epsilon\alpha}{1-\alpha} \left(\frac{1-\varphi}{\eta + \frac{\alpha\varphi}{1-\alpha}} \right) (p_{i,t} - \mathbf{p}_{t+\tau}) + \pi_{t,t+\tau} \right]$$

$$= (1 - \theta\beta) \sum_{\tau=0}^{\infty} \theta^\tau \beta^\tau \left[\left(\gamma + \frac{1}{1-\alpha} \left(\eta + \frac{\frac{\alpha\varphi}{1-\alpha}}{1 + \frac{\alpha\varphi}{1-\alpha}} \right) \right) (\mathbf{y}_{t+\tau} - \bar{\mathbf{y}}) \right. \quad (\text{A.47})$$

$$\left. - \alpha\varphi \frac{1+\eta}{1-\alpha} (b_{t+\tau} - \bar{b}) - \frac{\epsilon\alpha}{1-\alpha} \left(\frac{1-\varphi}{1 + \frac{\alpha\varphi}{1-\alpha}} \right) (p_{i,t} - \mathbf{p}_t) \right. \\ \left. + \left(1 + \frac{\epsilon\alpha}{1-\alpha} \left(\frac{1-\varphi}{1 + \frac{\alpha\varphi}{1-\alpha}} \right) \right) \pi_{t,t+\tau} \right]$$

$$= (1 - \theta\beta) \sum_{\tau=0}^{\infty} \theta^\tau \beta^\tau \left[\left(\frac{\gamma + \frac{1}{1-\alpha} \left(\eta + \frac{\frac{\alpha\varphi}{1-\alpha}}{1 + \frac{\alpha\varphi}{1-\alpha}} \right)}{1 + \frac{\epsilon\alpha}{1-\alpha} \frac{1-\varphi}{1 + \frac{\alpha\varphi}{1-\alpha}}} \right) (\mathbf{y}_{t+\tau} - \bar{\mathbf{y}}) \right. \quad (\text{A.48})$$

$$\left. - \left(\frac{\alpha\varphi \frac{1+\eta}{1-\alpha}}{1 + \frac{\epsilon\alpha}{1-\alpha} \frac{1-\varphi}{1 + \frac{\alpha\varphi}{1-\alpha}}} \right) (b_{t+\tau} - \bar{b}) + \pi_{t,t+\tau} \right].$$

Note that we can re-write (A.16) as

$$\pi_t = (1 - \theta) (p_{i,t}^* - \mathbf{p}_t + \pi_t) \quad (\text{A.49})$$

$$= \frac{1 - \theta}{\theta} (p_{i,t}^* - \mathbf{p}_t). \quad (\text{A.50})$$

Plugging into (A.48),

$$\pi_t = (1 - \theta\beta) \frac{1 - \theta}{\theta} \sum_{\tau=0}^{\infty} \theta^\tau \beta^\tau \left[\left(\frac{\gamma + \frac{1}{1-\alpha} \left(\eta + \frac{\frac{\alpha\varphi}{1-\alpha}}{1 + \frac{\alpha\varphi}{1-\alpha}} \right)}{1 + \frac{\epsilon\alpha}{1-\alpha} \frac{1-\varphi}{1 + \frac{\alpha\varphi}{1-\alpha}}} \right) (\mathbf{y}_{t+\tau} - \bar{\mathbf{y}}) \right. \quad (\text{A.51})$$

$$\left. - \varphi \left(\frac{\frac{\alpha}{1-\alpha} (1 + \eta)}{1 + \frac{\epsilon\alpha}{1-\alpha} \frac{1-\varphi}{1 + \frac{\alpha\varphi}{1-\alpha}}} \right) (b_{t+\tau} - \bar{b}) + \pi_{t,t+\tau} \right].$$

Plugging in (A.43) applied to $t + \tau$,

$$\begin{aligned} \pi_t = & (1 - \theta\beta) \frac{1 - \theta}{\theta} \sum_{\tau=0}^{\infty} \theta^{\tau} \beta^{\tau} \left[\left(\frac{\frac{\gamma + \frac{\eta}{1-\alpha}}{1 + \alpha\varphi^{\frac{1+\eta}{1-\alpha}}} + \frac{1}{1-\alpha} \frac{\frac{\alpha\varphi}{1 + \frac{\alpha\varphi}{1-\alpha}}}{1 + \frac{\alpha\varphi}{1-\alpha}} - \frac{\frac{\alpha\varphi}{1-\alpha} \frac{1+\eta}{1-\alpha}}{1 + \alpha\varphi^{\frac{1+\eta}{1-\alpha}}} \right) (\mathbf{y}_{t+\tau} - \bar{\mathbf{y}}) \right. \\ & \left. + \frac{\varphi}{\bar{s}} \left(\frac{\frac{\frac{\alpha}{1-\alpha}(1+\eta)}{1 + \alpha\varphi^{\frac{1+\eta}{1-\alpha}}}}{1 + \frac{\epsilon\alpha}{1-\alpha} \frac{1-\varphi}{1 + \frac{\alpha\varphi}{1-\alpha}}} \right) (r_{t+\tau}^B - r_{t+\tau} - \bar{s}) + \pi_{t,t+\tau} \right]. \end{aligned} \quad (\text{A.52})$$

Let $\kappa_{\theta} = (1 - \theta\beta) \frac{1-\theta}{\theta}$ and

$$\kappa_y^0 = \kappa_{\theta} \left(\frac{\frac{\gamma + \frac{\eta}{1-\alpha}}{1 + \alpha\varphi^{\frac{1+\eta}{1-\alpha}}} + \frac{1}{1-\alpha} \frac{\frac{\alpha\varphi}{1 + \frac{\alpha\varphi}{1-\alpha}}}{1 + \frac{\alpha\varphi}{1-\alpha}} - \frac{\frac{\alpha\varphi}{1-\alpha} \frac{1+\eta}{1-\alpha}}{1 + \alpha\varphi^{\frac{1+\eta}{1-\alpha}}} \right) \quad (\text{A.53})$$

$$\kappa_b^0 = \frac{\kappa_{\theta}}{\bar{s}} \left(\frac{\frac{\frac{\alpha}{1-\alpha}(1+\eta)}{1 + \alpha\varphi^{\frac{1+\eta}{1-\alpha}}}}{1 + \frac{\epsilon\alpha}{1-\alpha} \frac{1-\varphi}{1 + \frac{\alpha\varphi}{1-\alpha}}} \right). \quad (\text{A.54})$$

Plugging in,

$$\pi_t = \sum_{\tau=0}^{\infty} \theta^{\tau} \beta^{\tau} \left[\kappa_y^0 (\mathbf{y}_{t+\tau} - \bar{\mathbf{y}}) + \varphi \kappa_b^0 (r_{t+\tau}^B - r_{t+\tau} - \bar{s}) + \kappa_{\theta} \pi_{t,t+\tau} \right].$$

We can re-write this recursively to derive the Phillips curve:

$$\pi_t = \kappa_y^0 (\mathbf{y}_t - \bar{\mathbf{y}}) + \varphi \kappa_b^0 (r_t^B - r_t - \bar{s}) \quad (\text{A.55})$$

$$+ \sum_{\tau=1}^{\infty} \theta^\tau \beta^\tau \left[\kappa_y^0 (\mathbf{y}_{t+\tau} - \bar{\mathbf{y}}) + \varphi \kappa_b^0 (r_{t+\tau}^B - r_{t+\tau} - \bar{s}) + \kappa_\theta \pi_{t,t+\tau} \right] \\ = \kappa_y^0 (\mathbf{y}_t - \bar{\mathbf{y}}) + \varphi \kappa_b^0 (r_t^B - r_t - \bar{s}) \quad (\text{A.56})$$

$$+ \theta \beta \sum_{s=0}^{\infty} \theta^s \beta^s \left[\kappa_y^0 (\mathbf{y}_{t+1+s} - \bar{\mathbf{y}}) + \varphi \kappa_b^0 (r_{t+1+s}^B - r_{t+1+s} - \bar{s}) + \kappa_\theta \pi_{t,t+1+s} \right] \\ = \kappa_y^0 (\mathbf{y}_t - \bar{\mathbf{y}}) + \varphi \kappa_b^0 (r_t^B - r_t - \bar{s}) \quad (\text{A.57})$$

$$+ \theta \beta \sum_{s=0}^{\infty} \theta^s \beta^s \left[\kappa_y^0 (\mathbf{y}_{t+1+s} - \bar{\mathbf{y}}) + \varphi \kappa_b^0 (r_{t+1+s}^B - r_{t+1+s} - \bar{s}) + \kappa_\theta \pi_{t+1} + \kappa_\theta \pi_{t+1,t+1+s} \right] \\ = \kappa_y^0 (\mathbf{y}_t - \bar{\mathbf{y}}) + \varphi \kappa_b^0 (r_t^B - r_t - \bar{s}) + \beta (1 - \theta) \pi_{t+1} \quad (\text{A.58})$$

$$+ \theta \beta \sum_{s=0}^{\infty} \theta^s \beta^s \left[\kappa_y^0 (\mathbf{y}_{t+1+s} - \bar{\mathbf{y}}) + \varphi \kappa_b^0 (r_{t+1+s}^B - r_{t+1+s} - \bar{s}) + \kappa_\theta \pi_{t+1,t+1+s} \right] \\ = \kappa_y^0 (\mathbf{y}_t - \bar{\mathbf{y}}) + \varphi \kappa_b^0 (r_t^B - r_t - \bar{s}) + \beta \pi_{t+1}. \quad (\text{A.59})$$

When Reg Q does not bind strictly or $\varphi = 0$, the second term drops out and inflation follows the standard New Keynesian Phillips curve (not necessarily with the same κ_y^0).

From (A.43),

$$r_t^B - r_t - \bar{s} = \bar{s} \left(\gamma + \frac{1+\eta}{1-\alpha} \right) (\mathbf{y}_t - \bar{\mathbf{y}}) - \bar{s} \left(1 + \alpha \varphi \frac{1+\eta}{1-\alpha} \right) (b_t - \bar{b}). \quad (\text{A.60})$$

Plugging in to (A.59) and simplifying,

$$\pi_t = \left[\kappa_y^0 + \varphi \bar{s} \kappa_b^0 \left(\gamma + \frac{1+\eta}{1-\alpha} \right) \right] (\mathbf{y}_t - \bar{\mathbf{y}}) - \varphi \bar{s} \kappa_b^0 \left(1 + \alpha \varphi \frac{1+\eta}{1-\alpha} \right) (b_t - \bar{b}) + \beta \pi_{t+1}. \quad (\text{A.61})$$

The loan market clearing condition is $b_t = d_t$. Since it also holds in steady state, $\bar{b} = \bar{d}$.

Therefore, using (10)

$$\pi_t = \left[\kappa_y^0 + \varphi \bar{s} \kappa_b^0 \left(\gamma + \frac{1+\eta}{1-\alpha} \right) \right] (\mathbf{y}_t - \bar{\mathbf{y}}) + \frac{\varphi}{\delta} \bar{s} \kappa_b^0 \left(1 + \alpha \varphi \frac{1+\eta}{1-\alpha} \right) [r_t - r_t^D]^+ + \beta \pi_{t+1}. \quad (\text{A.62})$$

Let

$$\kappa_y = \kappa_y^0 + \varphi \bar{s} \kappa_b^0 \left(\gamma + \frac{1+\eta}{1-\alpha} \right) = \kappa_\theta \left[\frac{\gamma + \frac{1}{1-\alpha} \left(\eta + \frac{\frac{\alpha\varphi}{1-\alpha}}{1 + \frac{\frac{\alpha\varphi}{1-\alpha}}{1-\alpha}} \right)}{1 + \frac{\epsilon\alpha}{1-\alpha} \frac{1-\varphi}{1 + \frac{\frac{\alpha\varphi}{1-\alpha}}{1-\alpha}}} \right] \quad (\text{A.63})$$

$$\kappa_b = \bar{s} \kappa_b^0 \left(1 + \alpha \varphi \frac{1+\eta}{1-\alpha} \right) = \kappa_\theta \left(\frac{\frac{\alpha}{1-\alpha} (1+\eta)}{1 + \frac{\epsilon\alpha}{1-\alpha} \frac{1-\varphi}{1 + \frac{\frac{\alpha\varphi}{1-\alpha}}{1-\alpha}}} \right). \quad (\text{A.64})$$

Plugging into (A.62) and using Proposition 1 to replace r_t^D with r^Q gives (21).

□

Solving the model: We simulate the interest rate shock ε_t following (2) up to some large T such that ε_T is arbitrarily small. We then solve the model backwards from T with the initial condition that inflation and output are at their steady states: $\mathbf{y}_T = \bar{\mathbf{y}}$, where $\bar{\mathbf{y}}$ is as in (A.29), and $\pi_T = 0$. This also gives $r_T = \bar{r} = -\log \beta$.

Since Reg Q does not bind in steady state ($\bar{r} < r^Q$), the state $(r_t, \mathbf{y}_t, \pi_t)$ initially follows the dynamics given by (2), (7), and (21) with $[r_t - r^Q]^+ = 0$:

$$r_t = \bar{r} + \phi_\pi \pi_t + \phi_y (\mathbf{y}_t - \bar{\mathbf{y}}) + \varepsilon_t \quad (\text{A.65})$$

$$r_t - \pi_{t+1} = \bar{r} + \gamma (\mathbf{y}_{t+1} - \mathbf{y}_t) \quad (\text{A.66})$$

$$\pi_t = \kappa_y (\mathbf{y}_t - \bar{\mathbf{y}}) + \beta \pi_{t+1}. \quad (\text{A.67})$$

Solving the system for output and inflation gives the recursive dynamics

$$\begin{bmatrix} \mathbf{y}_t - \bar{\mathbf{y}} \\ \pi_t \end{bmatrix} = \begin{bmatrix} \gamma + \phi_y & \phi_\pi \\ -\kappa_y & 1 \end{bmatrix}^{-1} \left\{ \begin{bmatrix} \gamma & 1 \\ 0 & \beta \end{bmatrix} \begin{bmatrix} \mathbf{y}_{t+1} - \bar{\mathbf{y}} \\ \pi_{t+1} \end{bmatrix} + \begin{bmatrix} -1 \\ 0 \end{bmatrix} \varepsilon_t \right\} \quad (\text{A.68})$$

$$= A_{NK}^{-1} \left(B_{NK} \begin{bmatrix} \mathbf{y}_{t+1} - \bar{\mathbf{y}} \\ \pi_{t+1} \end{bmatrix} + C_{NK} \varepsilon_t \right). \quad (\text{A.69})$$

The solution is unique for any persistence $\rho \leq 1$ if the eigenvalues of $A_{NK}^{-1} B_{NK}$ lie inside the unit circle, where

$$A_{NK}^{-1} B_{NK} = \frac{1}{\gamma + \phi_y + \phi_\pi \kappa_y} \begin{bmatrix} \gamma & 1 - \phi_\pi \beta \\ \gamma \kappa_y & (\gamma + \phi_y) \beta + \kappa_y \end{bmatrix}. \quad (\text{A.70})$$

This requires that its trace be less than one plus its determinant:

$$\frac{\gamma + (\gamma + \phi_y) \beta + \kappa_y}{\gamma + \phi_y + \phi_\pi \kappa_y} < 1 + \frac{\gamma [(\gamma + \phi_y) \beta + \kappa_y] - \gamma \kappa_y (1 - \phi_\pi \beta)}{(\gamma + \phi_y + \phi_\pi \kappa_y)^2}. \quad (\text{A.71})$$

Simplifying and rearranging,

$$\kappa_y (\phi_\pi - 1) + \phi_y (1 - \beta) > 0. \quad (\text{A.72})$$

This is the same condition in Galí (2007).⁴⁵

Iterating backwards, as the epsilons get bigger, if the interest rate never crosses the Reg Q ceiling, then the dynamics continue as above. If it does cross the ceiling, the dynamics follow (2), (7), and (21) with $r_t - r^Q > 0$:

$$r_t = \bar{r} + \phi_\pi \pi_t + \phi_y (\mathbf{y}_t - \bar{\mathbf{y}}) + \varepsilon_t \quad (\text{A.73})$$

$$r_t - \pi_{t+1} = \bar{r} + \gamma (\mathbf{y}_{t+1} - \mathbf{y}_t) \quad (\text{A.74})$$

$$\pi_t = \kappa_y (\mathbf{y}_t - \bar{\mathbf{y}}) + \frac{\phi}{\delta} \kappa_b (r_t - r^Q) + \beta \pi_{t+1}. \quad (\text{A.75})$$

⁴⁵For a given ρ , the condition is $\gamma (1 - \rho) (1 - \rho \beta) + \kappa_y (\phi_\pi - \rho) + \phi_y (1 - \rho \beta) > 0$.

Solving the system for output and inflation gives the recursive dynamics

$$\begin{bmatrix} \mathbf{y}_t - \bar{\mathbf{y}} \\ \pi_t \end{bmatrix} = \begin{bmatrix} \gamma + \phi_y & \phi_\pi \\ -\kappa_y - \phi_y \frac{\varphi}{\delta} \kappa_b & 1 - \phi_\pi \frac{\varphi}{\delta} \kappa_b \end{bmatrix}^{-1} \left\{ \begin{bmatrix} 0 \\ -\frac{\varphi}{\delta} \kappa_b (r^Q - \bar{r}) \end{bmatrix} \right. \quad (\text{A.76})$$

$$\left. + \begin{bmatrix} \gamma & 1 \\ 0 & \beta \end{bmatrix} \begin{bmatrix} \mathbf{y}_{t+1} - \bar{\mathbf{y}} \\ \pi_{t+1} \end{bmatrix} + \begin{bmatrix} -1 \\ \frac{\varphi}{\delta} \kappa_b \end{bmatrix} \varepsilon_t \right\} \\ = A_{RegQ}^{-1} \left(\begin{bmatrix} 0 \\ -\frac{\varphi}{\delta} \kappa_b (r^Q - \bar{r}) \end{bmatrix} + B_{RegQ} \begin{bmatrix} \mathbf{y}_{t+1} - \bar{\mathbf{y}} \\ \pi_{t+1} \end{bmatrix} + C_{RegQ} \varepsilon_t \right) \quad (\text{A.77})$$

The stability condition is that the eigenvalues of $A_{RegQ}^{-1} B_{RegQ}$ must be lie inside the unit circle. This requires that $\left| \text{tr} \left(A_{RegQ}^{-1} B_{RegQ} \right) \right| < 1 + \text{det} \left(A_{RegQ}^{-1} B_{RegQ} \right)$. Plugging in and simplifying,

Proposition A.1 (Stability condition). *The solution to (A.73)-(A.75) is unique if*

$$\kappa_y (\phi_\pi - 1) + \phi_y \left[(1 - \beta) - \frac{\varphi}{\delta} \kappa_b \right] > 0. \quad (\text{A.78})$$

This is the analog of the NK stability condition (A.72) under the Reg Q regime. It is a stricter condition because the PC is effectively flatter or even inverted.

Always binding Reg Q: Consider the limit $r^Q \rightarrow \bar{r}$, which implies that $T \rightarrow \infty$, where T is the last period in which Reg Q binds. In other words, Reg Q binds arbitrarily long.

Using the evolution of ε_t from (2), the solution to (A.77) in period 0 is

$$\begin{bmatrix} \mathbf{y}_0 - \bar{\mathbf{y}} \\ \pi_0 \end{bmatrix} = (A_{RegQ} - \rho B_{RegQ})^{-1} C_{RegQ} \varepsilon_0. \quad (\text{A.79})$$

Stability for a given ρ (assumes away sunspots) requires that the determinant of $A_{RegQ} - \rho B_{RegQ}$ is positive:

$$\gamma (1 - \rho) (1 - \rho \beta) + \left[\kappa_y (\phi_\pi - \rho) - \gamma (1 - \rho) \phi_\pi \frac{\varphi}{\delta} \kappa_b \right] + \phi_y \left[(1 - \rho \beta) - \rho \frac{\varphi}{\delta} \kappa_b \right] > 0. \quad (\text{A.80})$$

Note that (A.78) is a special case of (A.80) when $\rho = 1$, which ensures stability even with sunspot shocks.

Substituting into (A.79) using the matrices as they appear in (A.76),

$$\begin{bmatrix} \mathbf{y}_0 - \bar{\mathbf{y}} \\ \pi_0 \end{bmatrix} = \left\{ \begin{bmatrix} \gamma + \phi_y & \phi_\pi \\ -(\kappa_y + \phi_y \frac{\varphi}{\delta} \kappa_b) & 1 - \phi_\pi \frac{\varphi}{\delta} \kappa_b \end{bmatrix} - \rho \begin{bmatrix} \gamma & 1 \\ 0 & \beta \end{bmatrix} \right\}^{-1} \begin{bmatrix} -1 \\ \frac{\varphi}{\delta} \kappa_b \end{bmatrix} \varepsilon_0 \quad (\text{A.81})$$

$$= \begin{bmatrix} \gamma(1 - \rho) + \phi_y & \phi_\pi - \rho \\ -(\kappa_y + \phi_y \frac{\varphi}{\delta} \kappa_b) & (1 - \rho\beta) - \phi_\pi \frac{\varphi}{\delta} \kappa_b \end{bmatrix}^{-1} \begin{bmatrix} -1 \\ \frac{\varphi}{\delta} \kappa_b \end{bmatrix} \varepsilon_0 \quad (\text{A.82})$$

$$= \frac{1}{\det(I - \rho A_{RegQ}^{-1} B_{RegQ})} \begin{bmatrix} (1 - \rho\beta) - \phi_\pi \frac{\varphi}{\delta} \kappa_b & -(\phi_\pi - \rho) \\ \kappa_y + \phi_y \frac{\varphi}{\delta} \kappa_b & \gamma(1 - \rho) + \phi_y \end{bmatrix} \begin{bmatrix} -1 \\ \frac{\varphi}{\delta} \kappa_b \end{bmatrix} \varepsilon_0 \quad (\text{A.83})$$

$$= \frac{1}{\det(I - \rho A_{RegQ}^{-1} B_{RegQ})} \begin{bmatrix} -(1 - \rho\beta) + \rho \frac{\varphi}{\delta} \kappa_b \\ -\kappa_y + \gamma(1 - \rho) \frac{\varphi}{\delta} \kappa_b \end{bmatrix} \varepsilon_0. \quad (\text{A.84})$$

Since the determinant is positive, we have the following result:

Proposition A.2 (Net effect of monetary policy). *Consider a tightening monetary policy shock $\varepsilon_0 > 0$ and suppose $\bar{r}^Q = \bar{r}$ so that Reg Q is always binding. If*

$$\frac{\varphi}{\delta} \kappa_b > \frac{1}{\gamma(1 - \rho)} \kappa_y, \quad (\text{A.85})$$

then the shock is inflationary, $\pi_0 > 0$.

Under Reg Q, monetary policy tightening is on net inflationary if the aggregate supply effect of the credit crunches dominates the standard aggregate demand effect through the household Euler equation.

Proof of Proposition 3. We start with (A.35) from the proof of Proposition 2:

$$\begin{aligned} p_{i,t} - \mathbf{p}_t &= (1 - \theta\beta) \sum_{\tau=0}^{\infty} \theta^\tau \beta^\tau [w_{t+\tau} - \mathbf{p}_{t+\tau} - \overline{w} - \overline{\mathbf{p}} \\ &\quad + \frac{\alpha}{1 - \alpha} [(y_{i,t+\tau} - \bar{y}) - \varphi(b_{i,t+\tau} - \bar{b})] + \pi_{t,t+\tau}]. \end{aligned} \quad (\text{A.86})$$

Taking logs of the production function (16) and loan demand (18) conditions,

$$y_{i,t} - \bar{y} = \alpha \varphi (b_{i,t} - \bar{b}) + (1 - \alpha) (l_{i,t} - \bar{l}) \quad (\text{A.87})$$

$$b_{i,t} - l_{i,t} - \bar{b} - \bar{l} = w_t - \mathbf{p}_t - \overline{w - \mathbf{p}} - \left[\log (r_t^B - r_t) - \log \bar{s} \right]. \quad (\text{A.88})$$

Solving for $l_{i,t} - \bar{l}$ from (A.87), into (A.88), and re-arranging,

$$b_{i,t} - \bar{b} = \frac{\frac{1}{1-\alpha}}{1 + \frac{\alpha\varphi}{1-\alpha}} (y_{i,t} - \bar{y}) + \frac{1}{1 + \frac{\alpha\varphi}{1-\alpha}} \left(w_t - \mathbf{p}_t - \overline{w - \mathbf{p}} - \left[\log (r_t^B - r_t) - \log \bar{s} \right] \right). \quad (\text{A.89})$$

From (A.42)

$$\log (r_t^B - r_t) - \log \bar{s} = \left(\gamma + \frac{1+\eta}{1-\alpha} \right) (\mathbf{y}_t - \bar{\mathbf{y}}) - \left(1 + \alpha \varphi \frac{1+\eta}{1-\alpha} \right) (b_t - \bar{b}). \quad (\text{A.90})$$

Plugging in,

$$\begin{aligned} b_{i,t} - \bar{b} &= \frac{\frac{1}{1-\alpha}}{1 + \frac{\alpha\varphi}{1-\alpha}} (y_{i,t} - \bar{y}) + \frac{1}{1 + \frac{\alpha\varphi}{1-\alpha}} (w_t - \mathbf{p}_t - \overline{w - \mathbf{p}}) \\ &\quad - \frac{\gamma + \frac{1+\eta}{1-\alpha}}{1 + \frac{\alpha\varphi}{1-\alpha}} (\mathbf{y}_t - \bar{\mathbf{y}}) + \frac{1 + \alpha \varphi \frac{1+\eta}{1-\alpha}}{1 + \frac{\alpha\varphi}{1-\alpha}} (b_t - \bar{b}). \end{aligned} \quad (\text{A.91})$$

Applying to period $t + \tau$ and plugging into (A.86),

$$\begin{aligned} p_{i,t} - \mathbf{p}_t &= (1 - \theta\beta) \sum_{\tau=0}^{\infty} \theta^\tau \beta^\tau \left[\frac{1}{1 + \frac{\alpha\varphi}{1-\alpha}} (w_{t+\tau} - \mathbf{p}_{t+\tau} - \overline{w - \mathbf{p}}) + \frac{\frac{\alpha}{1-\alpha} - \frac{\alpha\varphi}{1-\alpha}}{1 + \frac{\alpha\varphi}{1-\alpha}} (y_{i,t+\tau} - \bar{y}) \right. \\ &\quad \left. + \frac{\alpha\varphi}{1-\alpha} \frac{\gamma + \frac{1+\eta}{1-\alpha}}{1 + \frac{\alpha\varphi}{1-\alpha}} (\mathbf{y}_{t+\tau} - \bar{\mathbf{y}}) - \frac{\alpha\varphi}{1-\alpha} \frac{1 + \alpha \varphi \frac{1+\eta}{1-\alpha}}{1 + \frac{\alpha\varphi}{1-\alpha}} (b_{t+\tau} - \bar{b}) + \pi_{t,t+\tau} \right]. \end{aligned} \quad (\text{A.92})$$

Plugging in the demand curve (A.13) evaluated in logs and with $c_{i,t} = y_{i,t}$ and re-arranging,

$$\begin{aligned}
(p_{i,t} - \mathbf{p}_t) = & (1 - \theta\beta) \sum_{\tau=0}^{\infty} \theta^{\tau} \beta^{\tau} \left[\frac{1}{1 + \frac{\alpha\varphi}{1-\alpha} + \epsilon \left(\frac{\alpha}{1-\alpha} - \frac{\alpha\varphi}{1-\alpha} \right)} (w_{t+\tau} - \mathbf{p}_{t+\tau} - \overline{w} - \overline{\mathbf{p}}) \right. \\
& + \frac{\frac{\alpha}{1-\alpha} - \frac{\alpha\varphi}{1-\alpha} + \frac{\alpha\varphi}{1-\alpha} \left(\gamma + \frac{1+\eta}{1-\alpha} \right)}{1 + \frac{\alpha\varphi}{1-\alpha} + \epsilon \left(\frac{\alpha}{1-\alpha} - \frac{\alpha\varphi}{1-\alpha} \right)} (\mathbf{y}_{t+\tau} - \overline{\mathbf{y}}) \\
& \left. - \frac{\frac{\alpha\varphi}{1-\alpha} \left(1 + \alpha\varphi \frac{1+\eta}{1-\alpha} \right)}{1 + \frac{\alpha\varphi}{1-\alpha} + \epsilon \left(\frac{\alpha}{1-\alpha} - \frac{\alpha\varphi}{1-\alpha} \right)} (b_{t+\tau} - \bar{b}) + \pi_{t,t+\tau} \right]. \quad (\text{A.93})
\end{aligned}$$

Using market-clearing in the loans market, $b_t = d_t$ and the deposit demand equation (8),

$$\begin{aligned}
(p_{i,t} - \mathbf{p}_t) = & (1 - \theta\beta) \sum_{\tau=0}^{\infty} \theta^{\tau} \beta^{\tau} \left[\frac{1}{1 + \frac{\alpha\varphi}{1-\alpha} + \epsilon \left(\frac{\alpha}{1-\alpha} - \frac{\alpha\varphi}{1-\alpha} \right)} (w_{t+\tau} - \mathbf{p}_{t+\tau} - \overline{w} - \overline{\mathbf{p}}) \right. \\
& + \frac{\frac{\alpha}{1-\alpha} - \frac{\alpha\varphi}{1-\alpha} + \frac{\alpha\varphi}{1-\alpha} \left(\gamma + \frac{1+\eta}{1-\alpha} \right)}{1 + \frac{\alpha\varphi}{1-\alpha} + \epsilon \left(\frac{\alpha}{1-\alpha} - \frac{\alpha\varphi}{1-\alpha} \right)} (\mathbf{y}_{t+\tau} - \overline{\mathbf{y}}) \\
& \left. + \frac{1}{\delta} \frac{\frac{\alpha\varphi}{1-\alpha} \left(1 + \alpha\varphi \frac{1+\eta}{1-\alpha} \right)}{1 + \frac{\alpha\varphi}{1-\alpha} + \epsilon \left(\frac{\alpha}{1-\alpha} - \frac{\alpha\varphi}{1-\alpha} \right)} [r_{t+\tau} - r^Q]^+ + \pi_{t,t+\tau} \right]. \quad (\text{A.94})
\end{aligned}$$

The sensitivity of the optimal price to a permanent shock to the deposit spread holding constant the real wage and aggregate output is

$$\frac{\partial p_{i,t}}{\partial r_t} = \frac{1}{\delta} \frac{1 + \frac{\alpha\varphi}{1-\alpha} (1 + \eta)}{1 + \frac{1-\alpha}{\alpha\varphi} + \epsilon \left(\frac{1}{\varphi} - 1 \right)}. \quad (\text{A.95})$$

The expression is monotonically increasing in φ and α . A first-order approximation around $(\alpha, \varphi) = (0, 0)$ gives

$$\frac{\partial p_{i,t}}{\partial r_t} \approx \frac{\alpha\varphi}{\delta}. \quad (\text{A.96})$$

The sensitivity of output is obtained by plugging back into the demand curve (A.13). $\square$

Proof of Proposition 4. Note that if instead of permanent, the increase in the deposit

spread has persistence ρ , the sensitivities in Proposition 3 are

$$\frac{\partial p_{i,t}}{\partial r_t} = \left(\frac{1 - \theta\beta}{1 - \theta\beta\rho} \right) \frac{1}{\delta} \frac{1 + \frac{\alpha\varphi}{1-\alpha} (1 + \eta)}{1 + \frac{1-\alpha}{\alpha\varphi} + \epsilon \left(\frac{1}{\varphi} - 1 \right)}. \quad (\text{A.97})$$

and similarly for output. Moreover, averaging over all firms to include the ones that do not set prices in period t ,

$$\frac{\partial p_{i,t}}{\partial r_t} \approx (1 - \theta) \left(\frac{1 - \theta\beta}{1 - \theta\beta\rho} \right) \frac{1}{\delta} \frac{1 + \frac{\alpha\varphi}{1-\alpha} (1 + \eta)}{1 + \frac{1-\alpha}{\alpha\varphi} + \epsilon \left(\frac{1}{\varphi} - 1 \right)}. \quad (\text{A.98})$$

Iterating forward the Phillips curve in Proposition 2, the aggregate effect on inflation holding the path of aggregate output fixed is

$$\frac{\partial \pi_t}{\partial r_t} = \frac{1}{\delta} \left(\frac{1 - \theta\beta}{1 - \theta\beta\rho} \right) \frac{1 - \theta}{\theta} \left(\frac{(1 + \eta) \left(1 + \frac{\alpha\varphi}{1-\alpha} \right)}{1 + \frac{1-\alpha}{\alpha\varphi} + \epsilon \left(\frac{1}{\varphi} - 1 \right)} \right). \quad (\text{A.99})$$

Plugging in (A.98),

$$\frac{\partial \pi_t}{\partial r_t} = \frac{1}{\theta} \left(\frac{1 + \frac{\alpha\varphi}{1-\alpha}}{\frac{1}{1+\eta} + \frac{\alpha\varphi}{1-\alpha}} \right) \frac{\partial p_{i,t}}{\partial r_t} \approx \left(\frac{1 + \eta}{\theta} \right) \frac{\partial p_{i,t}}{\partial r_t}. \quad (\text{A.100})$$

□

B Numerical illustration

We generate impulse responses to a monetary policy shock ε_0 under the model, separately when Reg Q binds and when it does not bind. For the latter, we solve the linear system (A.69) and for the former we use (A.77). We set the initial shock to $\varepsilon_0 = 0.01/4$, which corresponds to 100 bps annualized. We parameterize the model using standard parameter values from Galí (2007), except for the shock persistence ρ and non-labor share (working capital intensity) α . We set $\rho = 0.4$, slightly lower than the standard value of 0.5 (this ensures stability under Reg Q), and $\alpha = 0.5$, slightly higher than the standard value of 1/3 (this makes working capital more important). For the binding Reg Q scenario, we

set the external finance dependence parameter $\varphi = 2/3$ and the inverse deposit demand elasticity $\delta = 0.2$ (this corresponds to 5% deposit outflows per 1% increase in the deposit spread, somewhat higher but broadly consistent with Figure 2).

Figure A.1 presents the results. The interest rate, measured as a deviation from steady state, shoots up by more than 100 bps on impact under Reg Q, more than twice as much as without Reg Q. This is because inflation rises under Reg Q, and the Taylor rule leads the central bank to offset it. Deposits fall and the loan spread spikes under Reg Q whereas they are unaffected absent Reg Q.

The output gap declines both with and without Reg Q but the impact with Reg Q is twice as large. Most strikingly, the response of inflation diverges between the two regimes. Absent Reg Q, inflation declines by about 40 bps, while under Reg Q it rises by 20 bps. The opposing effects are consistent with the VAR evidence for the Reg Q versus post Reg Q samples in Section F of the Online Appendix.

C Demand effects of Reg Q

In the baseline model, Reg Q affects the economy through firms' working capital, which is part of aggregate supply. Yet there are also channels by which Reg Q can affect aggregate demand. We extend the model to consider two such channels that operate through household consumption. We do so in reduced form by directly modifying the three-equation description of the model (A.73)–(A.75).

Household consumption is determined by the Euler equation (7) (see also (A.74)). In the baseline model, the Euler equation features the market interest rate r_t because all households save in government bonds. Let $\mathbf{y}^S$ denote the consumption of such households and write their Euler equation as

$$r_t - \pi_{t+1} = \bar{r} + \gamma \left(\mathbf{y}_{t+1}^S - \mathbf{y}_t^S \right). \quad (\text{A.101})$$

In practice, many households borrow instead of save. For these households, the Euler

equation features the loan interest rate r_t^B :

$$r_t^B - \pi_{t+1} = \bar{r} + \gamma \left(\mathbf{y}_{t+1}^B - \mathbf{y}_t^B \right), \quad (\text{A.102})$$

where $\mathbf{y}^B$ is borrowers' consumption. When Reg Q binds, borrowers face a higher effective real rate than savers, leading them to consume less. Reg Q therefore has a deflationary impact on aggregate demand via household borrowing.

Another important group are households who do not participate in government bond markets or money market funds. This was the case for most households during the Reg Q period. The relevant rate in their Euler equation is the deposit rate r^D :

$$r_t^D - \pi_{t+1} = \bar{r} + \gamma \left(\mathbf{y}_{t+1}^D - \mathbf{y}_t^D \right), \quad (\text{A.103})$$

where $\mathbf{y}^D$ is deposit-savers' consumption. When Reg Q binds, the deposit rate falls relative to the market interest rate (the deposit spread widens). This leads deposit-savers to consume more than savers in government bonds. Reg Q therefore has a positive impact on aggregate demand via deposit-savers and is inflationary.

Let λ^B be the share of borrowers and λ^D be the share of deposit-savers. The share of households who save in government bonds is then $\lambda^S = 1 - \lambda^B - \lambda^D$. Let their consumption be $\mathbf{y}^S$. Then, log-linearizing around steady state, total output satisfies

$$\Delta \mathbf{y}_{t+1} = \Delta \mathbf{y}_{t+1}^S + \lambda^B \left(\Delta \mathbf{y}_{t+1}^B - \Delta \mathbf{y}_{t+1}^S \right) + \lambda^D \left(\Delta \mathbf{y}_{t+1}^D - \Delta \mathbf{y}_{t+1}^S \right). \quad (\text{A.104})$$

Substituting (A.101), (A.102), and (A.103), into (A.104), we get a modified version of the aggregate Euler equation, which replaces (A.74) in the three-equation system (A.73)–(A.75) with

$$r_t - \pi_{t+1} + \lambda^B \left(r_t^B - \bar{r} \right) - \lambda^D \left(r_t - r_t^D \right) = \bar{r} + \gamma \left(\mathbf{y}_{t+1} - \mathbf{y}_t \right). \quad (\text{A.105})$$

The extended model with borrowers and deposit savers is equivalent to the baseline

model with a modified real interest rate. The modified real rate is higher by $\lambda^B (r_t^B - \bar{r})$ due to borrowers and lower by $\lambda^D (r_t - r_t^D)$ due to deposit savers. The net effect of Reg Q on aggregate demand therefore depends on the relative magnitudes of these two terms. If there are more borrowers, then Reg Q has a deflationary effect on aggregate demand whereas if there are more deposit savers it has an inflationary effect.

To characterize the net effect more fully, we proceed under the assumption that the loan spread household borrowers pay is proportional to the deposit spread, $r_t^B - r_t = \kappa_B (r_t - r_t^D)$. Then, if Reg Q is always binding, $\bar{r} = r^Q$, we get the three-equation system

$$r_t - \bar{r} = \phi_\pi \pi_t + \phi_y (\mathbf{y}_t - \bar{\mathbf{y}}) + \varepsilon_t \quad (\text{A.106})$$

$$(1 + \lambda^B \kappa_B - \lambda_D) (r_t - \bar{r}) - \pi_{t+1} = \gamma (\mathbf{y}_{t+1} - \mathbf{y}_t) \quad (\text{A.107})$$

$$\pi_t = \kappa_y (\mathbf{y}_t - \bar{\mathbf{y}}) + \frac{\varphi}{\delta} \kappa_b (r_t - \bar{r}) + \beta \pi_{t+1}. \quad (\text{A.108})$$

The parameters κ_y and κ_b are generally different than in the baseline model because the market clearing condition for loans now includes loans to borrowing households and savings from deposit savers.

Let $\lambda = 1 + \lambda^B \kappa_B - \lambda_D$. Following the approach we took in the baseline model, this system can be represented in the form

$$\begin{aligned} \begin{bmatrix} \mathbf{y}_t - \bar{\mathbf{y}} \\ \pi_t \end{bmatrix} &= \begin{bmatrix} \gamma + (1 + \lambda) \phi_y & (1 + \lambda) \phi_\pi \\ -\kappa_y - \phi_y \frac{\varphi}{\delta} \kappa_b & 1 - \phi_\pi \frac{\varphi}{\delta} \kappa_b \end{bmatrix}^{-1} \left\{ \begin{bmatrix} \gamma & 1 \\ 0 & \beta \end{bmatrix} \begin{bmatrix} \mathbf{y}_{t+1} - \bar{\mathbf{y}} \\ \pi_{t+1} \end{bmatrix} \right. \\ &\quad \left. + \begin{bmatrix} -(1 + \lambda) \\ \frac{\varphi}{\delta} \kappa_b \end{bmatrix} \varepsilon_t \right\} \\ &= A_{RegQ\lambda}^{-1} \left(B_{RegQ} \begin{bmatrix} \mathbf{y}_{t+1} - \bar{\mathbf{y}} \\ \pi_{t+1} \end{bmatrix} + C_{RegQ\lambda} \varepsilon_t \right). \end{aligned} \quad (\text{A.109})$$

$$(\text{A.110})$$

The stability condition is

$$\kappa_y [\phi_\pi (1 + \lambda) - 1] + \phi_y \left[(1 - \beta) (1 + \lambda) - \frac{\varphi}{\delta} \kappa_b \right] > 0. \quad (\text{A.111})$$

The stability condition is tighter than in the baseline model if $\lambda < 0$, i.e. if the inflationary

demand effect through deposit savers dominates the deflationary demand effect through household borrowers.

Solving the model backward,

$$\begin{bmatrix} \mathbf{y}_0 - \bar{\mathbf{y}} \\ \pi_0 \end{bmatrix} = \frac{1}{\mathbf{det} \left(I - \rho A_{RegQ\lambda}^{-1} B_{RegQ} \right)} \begin{bmatrix} -(1 + \lambda) (1 - \rho \beta) + \rho \frac{\varphi}{\delta} \kappa_b \\ -(1 + \lambda) \kappa_y + \gamma (1 - \rho) \frac{\varphi}{\delta} \kappa_b \end{bmatrix} \varepsilon_0 \quad (\text{A.112})$$

As a result, a tightening monetary policy shock $\varepsilon_0 > 0$ is inflationary if

$$\frac{\varphi}{\delta} \kappa_b > \frac{1 + (\lambda^B \kappa_B - \lambda^D)}{\gamma (1 - \rho)} \kappa_y. \quad (\text{A.113})$$

Thus, the net effect of the demand-based channels depends on the relative share of borrowers, λ^B (and the sensitivity of the loan spread to the deposit spread κ_B), versus deposit savers λ^D . While borrowers make monetary tightening less inflationary, non-participants make it more inflationary. The reason is that monetary tightening does not pass through to the real rate they face, hence it fails to reduce their demand.

Empirically, the 1983 Survey of Consumer Finances shows that 94% of households in the median (50th) income decile had deposit accounts and only 4% had money market funds. These households held 76% of their liquid assets in deposits versus 4% in money market funds, 5% in debt securities, and 9% in stocks. This suggests that λ^D was quite high.

We provide evidence for this inflationary mechanism in a related paper, Drechsler, Savov and Schnabl (2022).

D Relation to the deposits channel

Under the deposits channel of monetary policy (Drechsler, Savov and Schnabl, 2017), deposit rates are not constrained by a ceiling but nevertheless have a low “beta”, which means they increase less than one-for-one with market rates. Consequently, the deposit spread is $(1 - \beta^D) r_t$ instead of $r_t - r^Q$. We can accommodate this in our framework so

that the Phillips curve becomes

$$\pi_t = \kappa_y (\mathbf{y}_t - \bar{\mathbf{y}}) + \frac{\varphi}{\delta} \kappa_b (1 - \beta^D) r_t + \beta \pi_{t+1}.$$

Empirically, the average deposit beta has been approximately 0.4 during the post-Reg Q era (Drechsler, Savov and Schnabl, 2021).

An important difference between the deposits channel and the Reg Q ceiling is that β^D is a choice variable, i.e. banks can pay a higher deposit rate to retain deposits if they face strong loan demand. The fact that β^D is a choice of banks explains why, unlike Reg Q, the deposits channel does not lead to severe credit crunches. We can capture the endogeneity of β^D in our framework by allowing it to depend on φ , i.e. $\beta^D = \beta^D(\varphi)$ with $\partial \beta^D / \partial \varphi > 0$. Such variation likely exists both in the cross section and time series. The inflationary impact of the deposits channel is further reduced by the covariance between loan demand and deposit betas:

$$\pi_t = \kappa_y (\mathbf{y}_t - \bar{\mathbf{y}}) + \frac{\kappa_b}{\delta} \left[E[\varphi] (1 - E[\beta^D]) - Cov(\varphi, \beta^D) \right] r_t + \beta \pi_{t+1}.$$

Thus, due to the higher, endogenous deposit betas, the deposits channel should have a weaker inflationary effect than Reg Q.

E Data appendix

Aggregate data: The Fed funds rate is from the Federal Reserve's H.15 release. Inflation is the year-over-year percentage change of the seasonally-adjusted Consumer Price Index (CPI) from the Bureau of Labor Statistics (BLS). GDP is real Gross Domestic Product from the Bureau of Economic Analysis. Employment is total non-farm employment from the BLS' Current Employment Statistics, as is manufacturing employment. The oil price is the spot price of West Texas Intermediate crude oil, deflated by the CPI. The Chicago Fed Adjusted National Financial Conditions Index is produced by the Federal Reserve Bank of Chicago. Deposits are savings and small time deposits at commercial banks

and thrifts from the Federal Reserve's H.6 release, deflated by the CPI. Bank credit and commercial and industrial loans are from the Federal Reserve's H.8 release, deflated by the CPI. We downloaded all of these series from the Federal Reserve Bank of St. Louis' FRED database. Finally, unfilled orders are from the Census Bureau's Manufacturers' Shipments, Inventories, & Orders release.⁴⁶

Bank lending survey: We use data from the Changes in Bank Lending Practices survey administered by the Federal Reserve quarterly from 1964 to 1981 and available on FRASER, the digital library of the Federal Reserve Bank of St. Louis. The survey is the progenitor of the Senior Loan Officer Opinion Survey (SLOOS) and has the same basic structure. Following the methodology of SLOOS, we compute the net percentage of banks reporting higher loan demand and tighter credit standards for commercial and industrial loans. For loan demand we use the item "Strength of demand for commercial and industrial loans: Anticipated in next 3 months" and for tightness of credit standards we use "Practice concerning review of credit lines or loan applications: New customers." The survey focuses on large banks in financial centers (cities with \$150 million or more in business loans in 1964). It contains 80 banks until 1967 and 125 after that.

Deposit rates and ceilings: We obtain the interest rates on six-month Money Market Certificates (MMCs) and Money Market Deposit Accounts (MMDAs) from the Federal Reserve Board's Monthly Survey of Selected Deposits and Other Accounts.⁴⁷ We take the average rate across banks in the survey. The Reg Q ceiling rates are from Gilbert (1986) and Santomero and Siegel (1986).

Industry data: Our main industry-level data are annual from the NBER-CES Manufacturing Industry Database. The dataset is an industry-level aggregation of the U.S. Census Bureau's Annual Survey of Manufactures (ASM). Firms in the ASM are drawn from the

⁴⁶The release is available at <https://www.census.gov/manufacturing/m3>. We use the seasonally adjusted series for unfilled orders in manufacturing ex defense (MXD). There is a break in the series in January 1968 as defense was not excluded prior to this date. To remove the break we scale the series prior to January 1968 by the average ratio of total (MTM) unfilled orders to MXD unfilled orders during 1968.

⁴⁷The survey is available at <https://www.federalreserve.gov/data/mmttd.htm>.

Census of Manufactures, which covers the universe of manufacturing establishments in the U.S. Importantly, this includes private firms, which are the vast majority.⁴⁸ The NBER-CES database is thus representative of the entire U.S. manufacturing sector. The data are annual from 1958 to 2018 and cover 459 manufacturing industries at the four-digit SIC level. Important for our study, they contain information on both prices and quantities. There are price deflators for shipments (sales), materials, investment, and energy. We calculate price growth as the percentage change of the shipments deflator. Output is shipments plus the change in inventories deflated by the shipments deflator. We also use information on materials prices, wages, employment, inventories, and investment. A second source of industry data are the Quarterly Financial Reports from the Census Bureau, which include detailed information on firm balance sheets.⁴⁹ They are available at the two-digit SIC code (sector) level starting in 1947.

Bank data: Our bank-level data come from the U.S. Commercial Bank Call Reports. The Call Reports are publicly available back to 1976 through Wharton Research Data Services (WRDS). We used a Freedom of Information Act (FOIA) request to obtain Call Reports covering the Reg Q period back to 1959.

Establishment data: We use data on establishments at the county-industry level from the Census Bureau's County Business Patterns database.⁵⁰ The data are annual back to 1967. We use this data to aggregate county-level banking measures to the industry level by weighting across counties by industry employment. Since employment is reported in bins, we use the bin midpoints for the weights.

Rated Firms: Our data on credit ratings is from the S&P Capital IQ Credit Ratings dataset.⁵¹ For each year, we record how many firms had a credit rating and how many firms obtained a rating for the first time. The rating we use is the entity-level long-term

⁴⁸It is available at <https://www.nber.org/research/data/nber-ces-manufacturing-industry-database>.

⁴⁹They are available at <https://www.census.gov/econ/qfr/>.

⁵⁰See <https://www.census.gov/programs-surveys/cbp/data.html>.

⁵¹We use S&P because the vast majority of rated firms had an S&P rating and because it is the only dataset that extends back to the entire Reg Q period.

rating. To ensure the entities are corporations we restrict the sample to entities that appear in Compustat.

Aggregate bank loan and corporate bond issuance: For aggregate bank loan and corporate bond issuance we use Table F.102 of the Financial Accounts of the United States (Federal Reserve Board Z.1 release). The data contains aggregate net new borrowing by nonfinancial businesses from depository institutions (including commercial mortgages), as well as net commercial paper and corporate bond issuance. We also calculate net issuance excluding utilities and oil companies using bond-level issuance data from Mergent Fixed Income Securities Database and SDC Platinum, which we aggregate to the industry level using the issuer's primary SIC classification. For best coverage we use the Mergent data for 1965–1969 and SDC Platinum for 1970–1982.

Figure A.1: Model-implied impulse responses

Impulse responses to a monetary policy shock as implied by the model in Section 3 under two regimes: binding Reg Q (red squares) and non-binding Reg Q (blue circles). We parameterize the model with the following values: $\phi_\pi = 1.5$, $\phi_y = 0.125$, $\rho = 0.4$, $\theta = \frac{2}{3}$, $\alpha = 0.5$, $\gamma = 1$, $\eta = 1$, $\epsilon = 6$, $\beta = 0.99$, $\varphi = \frac{2}{3}$, $\delta = 0.2$, $\kappa_y = 0.26$, $\kappa_b = 0.15$. The interest rate, loan spread, and inflation are annualized in percent, while output and deposits are in percent deviations from steady state. The horizontal axis denotes quarters.

